



Elegant $\text{\LaTeX}$

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Elegant $\text{\LaTeX}$ Program

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1

1.1 (Skewness Coefficient)

$X \quad \mu \quad \sigma \quad \gamma_1$

$$\gamma_1 = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{E[(X - \mu)^3]}{\sigma^3}$$

$x_1, x_2, \dots, x_n \quad g_1$

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{3/2}}$$

$\gamma_1 > 0$

$\gamma_1 < 0$

$\gamma_1 = 0$



tip 1.1

0,

“ ”

“ ”

1.1 $X \quad \mathcal{E}(\lambda)$

$$f(x; \lambda) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, & x > 0; \\ 0, & , \end{cases}$$

$\lambda > 0 \quad (X_1, X_2, \dots, X_n)^T$

$X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim f(x, \lambda)$

$$\prod_{i=1}^n f(x_i; \lambda) = \begin{cases} \frac{1}{\lambda^n} \exp \left(-\frac{1}{\lambda} \sum_{i=1}^n x_i \right), & x_1, x_2, \dots, x_n > 0; \\ 0, & . \end{cases}$$

1.2

$(X_1, X_2, \dots, X_n)^T$

$X \quad f(x; \theta)$

$X_1, X_2, \dots, X_n$

i.i.d.

$$L(x_1, x_2, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta)$$

$x_1, x_2, \dots, x_n \quad \theta$



“ ”

n

•

$\mathcal{E}(\lambda) \quad \lambda$

$x_1 = 1, x_2 = 2, x_3 = 3$

•

• 1 $f(1; \lambda)$

• 2 $f(2; \lambda)$

• 3 $f(3; \lambda)$

“ ”

$$L(\lambda) = f(1; \lambda) \times f(2; \lambda) \times f(3; \lambda)$$

•

• $\lambda \quad \lambda = 1 \quad \lambda = 5$

$\lambda \quad \text{“ ”}$

• $\sum x_i$

• “ ”

2.1

1.3 ()

$X \sim F_X$

$$(X_1, X_2, \dots, X_n)^T,$$

$X \sim F_X$

$$(x_1, x_2, \dots, x_n)$$

n



tip 1.2 ()

1. X

2. X

1.4 ()

$X_1, X_2, \dots, X_n$

• $X_i \sim X$

•

$X_1, X_2, \dots, X_n \sim X$

$X_1, X_2, \dots, X_n \text{ i.i.d.}$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$X_1 \sim F(x) \iff (X_1 \sim f(x)).$$



tip 1.3

$X \sim XI \implies X_1, \dots, X_n \sim XI$

$$X_1, X_2, \dots, X_n.$$

X_i

1.2 $X \sim E(\lambda)$

$$f(x; \lambda) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda}, & x > 0, \\ 0, & \text{otherwise} \end{cases}$$

$X_1, X_2, \dots, X_n$

$$\prod_{i=1}^n f(x_i; \lambda) = \begin{cases} \frac{1}{\lambda^n} \exp\left(-\frac{1}{\lambda} \sum_{i=1}^n x_i\right), & x_1, x_2, \dots, x_n > 0, \\ 0, & \text{otherwise} \end{cases}$$



1.5 ()

$$\Theta$$

$$f(x; \theta), \quad \theta \in \Theta,$$

$$\{f(x; \theta) : \theta \in \Theta\}$$

$$\left\{ \prod_{i=1}^n f(x_i; \theta) : \theta \in \Theta \right\}$$



$$N(\mu, \sigma^2) \quad \mu, \sigma^2 \quad E(\lambda) \quad \lambda \quad B(n, p) \quad p$$

$$f(x; \theta)$$

$$\theta \quad \text{“} \quad \text{”} \quad \text{“} \quad \theta \text{”}$$

1.6 ()



$$\Theta \quad \Theta \quad \text{“} \quad \text{”}$$

1.7 ()

$$X_1, X_2, \dots, X_n$$

$$T = T(X_1, X_2, \dots, X_n)$$

$$T$$

$$x_1, x_2, \dots, x_n$$

$$T(x_1, x_2, \dots, x_n)$$

$$(X_1, X_2, \dots, X_n)$$



1.8 ()

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

$$R = \max_{1 \leq i \leq n} X_i - \min_{1 \leq i \leq n} X_i.$$



2.2

1.9 ()

$X_1, X_2, \dots, X_n$ 独立同分布的随机变量

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)},$$

$(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ 的联合分布函数为

$$X_{(1)} = \min_{1 \leq i \leq n} \{X_i\}, \quad X_{(n)} = \max_{1 \leq i \leq n} \{X_i\}, \quad r = 1, 2, \dots, n.$$



1.3

“ ”

1. $X_1, X_2, \dots, X_n$

3 $n = 3$

- $X_1 = 1$
- $X_2 = 2$
- $X_3 = 3$

$F(x)$ “ 1000 ”

2. $X_{(1)}, X_{(2)}, \dots, X_{(n)}$

3 $X_1 = 1200, X_2 = 800, X_3 = 1050$

- $X_{(1)} = 800$
- $X_{(2)} = 1050$
- $X_{(3)} = 1200$

3. (2.2.2)

$$f_{(r)}(x) = r C_n^r F^{r-1}(x) [1 - F(x)]^{n-r} f(x) \quad \text{“ } x = 1000 \text{ ”}$$

- $1 - 1000 = f(x)$
- $r - 1 = 1 - 1000 = F(x)^{r-1}$
- $n - r = 1 - 1000 = [1 - F(x)]^{n-r}$
- $r C_n^r = \frac{n!}{(r-1)!(n-r)!}$

- $X_i = 10$

- $X_{(r)} = 10, 3$

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

1.1 ()

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim F(x)$ $(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ $1 \leq r \leq n$

$$P\{X_{(r)} \leq x\} = r C_n^r \int_0^{F(x)} t^{r-1} (1-t)^{n-r} dt. \quad (2.2.1)$$

$F(x)$ $f(x)$ $X_{(r)}$

$$f_{(r)}(x) = r C_n^r F^{r-1}(x) [1 - F(x)]^{n-r} f(x). \quad (2.2.2)$$



r $X_{(r)} \leq x$ “ r x ”

$$P\{X_{(r)} \leq x\} = \sum_{k=r}^n C_n^k [F(x)]^k [1 - F(x)]^{n-k}$$

2.2.3

$$\sum_{k=r}^n C_n^k p^k (1-p)^{n-k} = \frac{n!}{(r-1)!(n-r)!} \int_0^p t^{r-1} (1-t)^{n-r} dt$$

$$p = F(x)$$

“ ”

tip 1.4 ()

$$F(X) = P(X \leq x)$$

$$dt = F(r-1) - F(r-1) = F(r) - F(r-1) = P(r-1 < X \leq r) = P(X = r) = \frac{n!}{(r-1)!(n-r)!} p^{r-1} (1-p)^{n-r+1}$$

tip 1.5

$$c = P(F(x) = t) = I.$$

$$rC_n^r = \text{Multinomial Coefficient}$$

$$rC_n^r = r \cdot \frac{n!}{r!(n-r)!} = r \cdot \frac{n!}{r \cdot (r-1)!(n-r)!} = \frac{n!}{(r-1)!(n-r)!}$$

- $r-1$ x
- 1 x r
- $n-r$ x

1.4 “ ” n $X_1, X_2, \dots, X_n$ **1. (Sample Median) — “ ”**

- **A** ($n=5$) 60, 70, **75**, 85, 90 $n=2k+1=5$ $k=2$ $M = X_{(k+1)} = X_{(3)} = 75$
- **B** ($n=6$) 60, 70, **75, 80**, 85, 90 $n=2k=6$ $k=3$ $M = \frac{1}{2}(X_{(3)} + X_{(4)}) = \frac{75+80}{2} = 77.5$

2. α — “ 10% ”“ α ” α

- $n=10$ $\alpha=0.2$ 20%
- $[n(1-\alpha)+0.5] = [10(1-0.2)+0.5] = [8.5] = 8$
- 8 $X_{(8)}$ 0.2
- 10 8 9 10 30% 8 “ 20% ”

3. (x_α) — “ ”

- 0 100 $U[0, 100]$
- $F(x) = \frac{x}{100}$ $0 \leq x \leq 100$
- $\alpha=0.1$ 10% x_α $F(x_\alpha) \geq 1-0.1=0.9$ $\frac{x}{100}=0.9 \Rightarrow x=90$
- 90 90% 10%

4. $\inf ()$

“ ”

- “ 0.5 ” $x=3$ $F(3)=0.5$ x
- $\inf$ “ ” $\geq 1-\alpha$

- $M ()$ “ ”

- $x_\alpha ()$ “ ”

- “ ”

1.10 ()

$$F(x) \quad \alpha \in (0, 1) \quad x_\alpha$$

$$x_\alpha = \inf\{x : F(x) \geq 1 - \alpha\},$$

$$\alpha \quad 1 - \alpha \quad F$$

$$M_F = \frac{1}{2} (\sup\{x : F(x) \leq 0.5\} + \inf\{x : F(x) \geq 0.5\}).$$

$$X_1, X_2, \dots, X_n \quad F(x) \quad \alpha \quad X_{([n(1-\alpha)+0.5])}$$

$$M = \begin{cases} X_{(k+1)}, & n = 2k + 1; \\ \frac{1}{2}(X_{(k)} + X_{(k+1)}), & n = 2k, \end{cases}$$

k

$$F(x_\alpha) \geq 1 - \alpha \quad F(x_\alpha - 0) \leq 1 - \alpha \quad F(x_\alpha) = 1 - \alpha \quad X_{([n(1-\alpha)+0.5])} \quad M \quad x_\alpha \quad M_F$$



tip 1.6

$$\inf \quad \sup \quad x \in (0, 1)$$



I *Indicator Function* “ ” “ ” “ ” “ ” I 0 ”

$$I_{[X_i, \infty)}(x)$$

- x “ ” $x = 2$
- $[X_i, \infty)$ “ ” X_i
- I

- $x \in [X_i, \infty)$ $x \geq X_i$ $I = 1$
- $x \in [X_i, \infty)$ $x < X_i$ $I = 0$

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I_{[X_i, \infty)}(x) \quad I \quad X_i \quad “ x ” \quad I$$

1.5

$$n = 5$$

$$X_{(1)} = 5, \quad X_{(2)} = 10, \quad X_{(3)} = 12, \quad X_{(4)} = 15, \quad X_{(5)} = 25$$

1.

$$F_5(13) = 13$$

$$F_5(13) = \frac{1}{5} \sum_{i=1}^5 I_{[X_i, \infty)}(13)$$

- $X_1 = 5 \leq 13$ 1
- $X_2 = 10 \leq 13$ 1
- $X_3 = 12 \leq 13$ 1
- $X_4 = 15 > 13$ 0
- $X_5 = 25 > 13$ 0

$$3 \quad 3/5 = 0.6 \quad “ ”$$

2.

- 0.6
- n

3.

$$n \rightarrow \infty \quad “ ” \quad 1$$

1.11 ()

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim F(x)$

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I_{[X_i, \infty)}(x) = \frac{1}{n} \{X_i : X_i \leq x, i = 1, 2, \dots, n\}^\# \quad (2.2.6)$$

$$F_n(x) \quad X_1, X_2, \dots, X_n \quad I_A(x) \quad A \quad x \in A \quad I_A(x) = 1 \quad I_A(x) = 0$$

$$F_n(x) = \begin{cases} 0, & x < X_{(1)}; \\ k/n, & X_{(k)} \leq x < X_{(k+1)}, k = 1, 2, \dots, n-1; \\ 1, & x \geq X_{(n)}. \end{cases} \quad (2.2.7)$$



1.2 ()

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim F(x)$ $F_n(x)$

$$1. P\{\lim_{n \rightarrow \infty} F_n(x) = F(x)\} = 1.$$

$$2. \lim_{n \rightarrow \infty} P\left\{\frac{\sqrt{n}(F_n(x) - F(x))}{\sqrt{F(x)[1-F(x)]}} \leq y\right\} = \int_{-\infty}^y \phi(u) du \quad \phi(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \quad N(0, 1)$$



(2.2.2) $F_n(x)$ **Standardization**

1.

$$nF_n(x) \quad B(n, p) \quad p = F(x)$$

- $E[F_n(x)] = p = F(x)$
- $Var(F_n(x)) = \frac{p(1-p)}{n} = \frac{F(x)[1-F(x)]}{n}$

2.

CLT

$$\frac{F_n(x) - E[F_n(x)]}{\sqrt{Var(F_n(x))}} = \frac{F_n(x) - F(x)}{\sqrt{F(x)[1-F(x)]/n}} = \frac{\sqrt{n}(F_n(x) - F(x))}{\sqrt{F(x)[1-F(x)]}}$$

3.

$$\int_{-\infty}^y \phi(u) du \quad \Phi(y) \quad \frac{1}{\sqrt{n}}$$

1.3 (Glivenko)

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim F(x)$ $F_n(x)$ $D_n = \sup_{-\infty < x < \infty} |F_n(x) - F(x)|$

$$P\left\{\lim_{n \rightarrow \infty} D_n = 0\right\} = 1.$$



$$D_n = \sup |F_n(x) - F(x)|$$

$$F_n(x) \quad F(x) \quad \sup \quad x$$

$$P\{\lim_{n \rightarrow \infty} D_n = 0\} = 1$$

$$“ \quad ” \quad F(x)$$

2.3

1.12 (x)

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(0, 1)$ $\xi = \sum_{i=1}^n X_i^2$ $\xi \quad n \quad \chi^2 \quad \xi \sim \chi^2(n)$



tip 1.7 ()

$$E(Y) = k, \quad \text{Var}(Y) = 2k.$$



$$Y = Z_1^2 + \cdots + Z_n^2, \quad Z_1, \dots, Z_n \stackrel{i.i.d.}{\sim} N(0, 1),$$

$$Y \sim \chi^2(n) \quad n \quad n$$

$$\sum_{i=1}^n (X_i - \bar{X})^2$$

n

$$\sum_{i=1}^n (X_i - \bar{X}) = 0,$$

$$n-1 \quad n-1$$

tip 1.8

$$\begin{array}{ccccccc} xi & 1 & 10 & & 100 & & 5 \\ 10 & & 100000 & & & & 1 \end{array}$$



1.

$$n \quad X_i \sim N(0, 1) \quad \xi = \sum_{i=1}^n X_i^2 \quad n \quad \xi \sim \chi^2(n)$$

2.

- $\chi^2 = \sum X_i^2$
- $\chi^2 = \sum \frac{(O-E)^2}{E} \quad E$

3. “ ”

s^2

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\text{“ ” } \chi^2 \quad 0 \quad 1$$

1.4

$$\xi \sim \chi^2(n) \quad \xi$$

$$f(x) = \begin{cases} \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} e^{-\frac{x}{2}} x^{\frac{n}{2}-1}, & x > 0; \\ 0, & x \leq 0. \end{cases} \quad (1.1)$$



(Gamma Function)

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (\text{Re}(z) > 0) \quad (1.2)$$

tip 1.9

$$\begin{array}{ccccccc} 2 & n & x & x/2 & & n/2-1+1= & n/2 \end{array}$$



1.

“ ” “ ”

$$f(x) = \underbrace{\frac{1}{2^{n/2}\Gamma(n/2)}}_{\text{normalization}} \cdot \underbrace{e^{-x/2}}_{\text{exponential}} \cdot \underbrace{x^{n/2-1}}_{\text{power}} \quad (1.3)$$

- $(x^{n/2-1}) \quad x \rightarrow 0 \quad n = 2 \quad 1 \quad n > 2$
- $(e^{-x/2}) \quad \text{“ ”} \quad x \quad \text{“ ”}$
- 1

1.5 ($n \quad m \quad m+n$)

$$\xi \sim \chi^2(m) \quad \eta \sim \chi^2(n) \quad \xi \quad \eta$$

$$\xi + \eta \sim \chi^2(m+n).$$



1.1

$$X_1, X_2, \dots, X_n$$

$$X_i \sim N(\mu_i, \sigma^2), \quad i = 1, 2, \dots, n.$$

$$A = (a_{ij})_{n \times n} \quad n \times n$$

$$X = (X_1, X_2, \dots, X_n)^T, \quad Y = AX.$$

$$Y = Y_1, Y_2, \dots, Y_n$$

$$Y_i \sim N\left(\sum_{k=1}^n a_{ik}\mu_k, \sigma^2\right), \quad i = 1, 2, \dots, n.$$



1. $X_1, \dots, X_n \quad A \quad Y_1, \dots, Y_n$
2. $Y_i \quad \sigma^2 \quad \sum_{k=1}^n a_{ik}\mu_k$

1.6 (Cochran)

$$X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim N(0, 1) \quad A_1, A_2, \dots, A_k \quad n \times n$$

$$X = (X_1, X_2, \dots, X_n)^T,$$

$$Q_i = X^T A_i X, \quad i = 1, 2, \dots, k.$$

$$Q_1 + Q_2 + \dots + Q_k = \sum_{i=1}^n X_i^2,$$

$$\text{rank}(A_i) = n_i, \quad i = 1, 2, \dots, k,$$

$$Q_1, Q_2, \dots, Q_k$$

$$Q_i \sim \chi^2(n_i)$$

$$n = \sum_{i=1}^k n_i.$$



[Cochran “ ”]

- 1.6
- 1. $n \quad N(0, 1) \quad X_1, X_2, \dots, X_n \quad \sum_{i=1}^n X_i^2 \quad n \quad \chi^2(n)$
- 2. $k \quad Q_1 + Q_2 + \dots + Q_k = \sum_{i=1}^n X_i^2 \quad Q_i = X^T A_i X \quad \text{rank}(A_i) = n_i$
- $n = \sum_{i=1}^k n_i \quad k \quad Q_i$



$$Q_i \quad n_i \quad Q_i \sim \chi^2(n_i)$$

1.13 (t - student)

$$X, Y \quad X \sim N(0, 1) \quad Y \sim \chi^2(n)$$

$$\xi = \frac{X}{\sqrt{Y/n}},$$

$$\xi \quad n \quad t \quad \xi \sim t(n)$$



1. “ ”

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$\begin{aligned} & \sigma \quad \sigma \\ & \bullet \quad S \quad \sigma \\ & \bullet \quad \sigma \quad S \\ & \bullet \quad \xi = \frac{X}{\sqrt{Y/n}} \\ & \bullet \quad X \quad N(0, 1) \\ & \bullet \quad Y/n \\ & \bullet \end{aligned}$$

1.7

$$\xi \sim t(n) \quad \xi$$

$$f(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi} \Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}, \quad -\infty < x < \infty.$$



tip 1.10 (t)

$$\begin{aligned} & \bullet \quad \text{Kernel} \quad \text{Coefficient} \quad “ ” \\ & \bullet \quad 1. \quad “ ” \quad \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}} \\ & \bullet \quad \left(1 + \frac{x^2}{n}\right) \quad e^{-x^2/2} \quad “ ” \quad n \rightarrow \infty \quad (1 + 1/n)^n \rightarrow e \\ & \bullet \quad -\frac{n+1}{2} \quad n \quad 1 \quad 2 \quad “ ” \\ & \bullet \quad 2. \quad “ ” \quad \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi} \Gamma\left(\frac{n}{2}\right)} \\ & \bullet \quad 1 \\ & \bullet \quad \text{Gamma} \quad “ ” \quad \frac{n+1}{2} \quad “ ” \quad \frac{n}{2} \\ & \bullet \quad \sqrt{n\pi} \sqrt{\pi} \quad \sqrt{n} \quad x^2/n \end{aligned}$$

1.14 (- Fisher-Snedecor distribution)

$$X, Y \quad X \sim \chi^2(m) \quad Y \sim \chi^2(n)$$

$$\xi = \frac{X/m}{Y/n},$$

$$\xi \quad m \quad n \quad F \quad \xi \sim F(m, n)$$



1.

“ ”

$X \sim Y$ $m \sim n$ “ ”

$$\xi = \frac{X/m}{Y/n}$$

$(m, n) \sim F$

2.

F

- F
- **ANOVA** “ ” “ ”
-

1.8 (F)

$\xi \sim F(m, n)$ ξ

$$f(x) = \begin{cases} \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} m^{\frac{m}{2}} n^{\frac{n}{2}} x^{\frac{m}{2}-1} (n+mx)^{-\frac{m+n}{2}}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$



1.9 ()

$X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim N(\mu, \sigma^2)$ $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

1. $\bar{X} \sim N(\mu, \sigma^2/n)$

2. $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$

3. $\bar{X} \sim s^2$



1.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\sigma^2/n$$

2.

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

3.

$$\bar{X} \sim s^2$$

2

- 2.1** $X \sim N(\mu, 4)$ $\mu = \theta = \mu$ $\Theta = \mathbb{R}$ 4 $X = (X_1, X_2, X_3, X_4)^T$ $x = (8, 9, 11, 10)^T$
- $\hat{\mu}(X) = \bar{X} = \frac{1}{4}(X_1 + X_2 + X_3 + X_4)$
 - $\hat{\mu}(x) = \bar{x} = \frac{8+9+11+10}{4} = 9.5$
 - $\bar{X} \sim N\left(\mu, \frac{4}{4}\right) = N(\mu, 1)$ 95% $z_{0.975} = 1.96$ μ 95% $[\bar{X} - 1.96\frac{2}{\sqrt{4}}, \bar{X} + 1.96\frac{2}{\sqrt{4}}] = [\bar{X} - 1.96, \bar{X} + 1.96]$ $\bar{x} = 9.5$ $[9.5 - 1.96, 9.5 + 1.96] = [7.54, 11.46]$
 $\hat{\mu}(X) = \bar{X}$ $\hat{\mu}(x) = 9.5$ 95% $[7.54, 11.46]$

2.1 ()

$X_1, X_2, \dots, X_n$ i.i.d. X $\hat{\alpha}_k$ X k $\hat{\beta}_k$ X k



- $\hat{\beta}_2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$
- $\hat{\beta}_3 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3$ 0 0 0
- $\hat{\beta}_4 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4$

$X \sim F_\theta(x)$ X θ X $F_\theta(x)$ θ F x

2.2 μ μ $\theta = \mu^2$ $\alpha = \mu$ θ $\theta = \theta(\alpha) = \alpha^2$ μ $\alpha(X) = \bar{X}$ $\theta(\alpha) = \alpha^2$ α
 $\alpha(X) = \bar{X}$ θ

$$\theta(\alpha(X)) = (\bar{X})^2.$$

$$\mu^2 \quad \bar{X}^2$$

tip 2.1

k $\theta = (\theta_1, \theta_2, \dots, \theta_k)^T$ k $\alpha_1, \alpha_2, \dots, \alpha_k$ $\alpha_l = \alpha_l(\theta_1, \theta_2, \dots, \theta_k), l = 1, \dots, k$
 $\frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{i=1}^n X_i^2, \dots, \frac{1}{n} \sum_{i=1}^n X_i^k$ $\theta_1, \theta_2, \dots, \theta_k$

$$\begin{cases} \alpha_1(\theta_1, \theta_2, \dots, \theta_k) = \frac{1}{n} \sum_{i=1}^n X_i, \\ \alpha_2(\theta_1, \theta_2, \dots, \theta_k) = \frac{1}{n} \sum_{i=1}^n X_i^2, \\ \vdots \\ \alpha_k(\theta_1, \theta_2, \dots, \theta_k) = \frac{1}{n} \sum_{i=1}^n X_i^k. \end{cases}$$

$$\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k)^T \quad \theta$$

2.3 $X \sim N(\mu, \sigma^2)$ μ, σ^2 μ σ^2
 $E(X) = \mu$ $E(X^2) = \mu^2 + \sigma^2$ $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ $\frac{1}{n} \sum_{i=1}^n X_i^2$

$$\mu = \bar{X}, \quad \mu^2 + \sigma^2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2.$$

$$\mu \quad \sigma^2 \quad \bar{X} \quad \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$$

2.2 (likelihood function)

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x; \theta) \theta \in \Theta$

$$L(\theta; x) = L(\theta; x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$$

$$X = (X_1, \dots, X_n)^T \quad \theta \quad X \quad L(\theta; X) \quad X$$



tip 2.2

2.3 (MLE)

$$X = (X_1, X_2, \dots, X_n)^T \quad L(\theta; x) \quad \hat{\theta}(x) = \hat{\theta}(x_1, x_2, \dots, x_n)$$

$$L(\hat{\theta}(x); x) = \max_{\theta \in \Theta} L(\theta; x),$$

$$\hat{\theta}(X) \quad \theta \quad \text{MLE Maximum likelihood estimator}$$



2.1 ()

$$L(\theta; x)$$

$$\bullet \quad L(\theta; x) \quad \theta \quad \Theta$$

$$\frac{\partial L(\theta; x)}{\partial \theta_j} = 0, \quad j = 1, 2, \dots, k,$$

$$\frac{\partial \ln L(\theta; x)}{\partial \theta_j} = 0, \quad j = 1, 2, \dots, k.$$

$$\bullet \quad L(\theta; x) \quad \theta$$

$$\bullet \quad \theta \quad L(\theta; x)$$



2.2 (“ ”)

$$X \sim f(x; \theta), \theta \in \Theta, X \quad \hat{\theta}(X) \quad \theta \quad \eta = g(\theta) \quad \Theta \quad \hat{\Theta} \quad g(\hat{\theta}(X)) \quad \eta$$



$$\theta$$

2.4 (M)

$$X_1, \dots, X_n \quad \theta \quad \hat{\theta}$$

$$\sum_{i=1}^n \rho(X_i, \theta)$$

$$\rho \quad \hat{\theta} \quad \theta \quad M$$

$$\sum_{i=1}^n \psi(X_i, \theta) = 0,$$

$$\psi \quad \rho \quad M$$



tip 2.3

$$M$$

2.4

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 10,$$

M θ

1.

$$\rho(u) = u^2,$$

M

$$\sum_{i=1}^3 \rho(x_i - \theta) = (1 - \theta)^2 + (2 - \theta)^2 + (10 - \theta)^2.$$

$$\hat{\theta} = \frac{1 + 2 + 10}{3} = \frac{13}{3}.$$

2.

$$\rho(u) = |u|,$$

M

$$\sum_{i=1}^3 \rho(x_i - \theta) = |1 - \theta| + |2 - \theta| + |10 - \theta|.$$

$$\hat{\theta} = 2.$$

M ρ ρ

2.5 (Bayes)

$$X = (X_1, X_2, \dots, X_n)^T \quad f(x | \theta) \quad \theta \quad \pi(\theta) \quad X \quad \theta$$

$$p(\theta | X) = \frac{f(X | \theta) \pi(\theta)}{\int_{\Theta} f(X | \theta) \pi(\theta) d\theta}.$$

$g(\theta)$

$$\hat{g}(X) = \int_{\Theta} g(\theta) p(\theta | X) d\theta = \frac{\int_{\Theta} g(\theta) f(X | \theta) \pi(\theta) d\theta}{\int_{\Theta} f(X | \theta) \pi(\theta) d\theta}$$

$\hat{g}(X) \quad g(\theta) \quad \text{Bayes}$



tip 2.4 ($\propto$)

proportional to $A \propto B \quad A = c \cdot B \quad c$

$$p(\mu | \mathbf{X}) = \frac{f(\mathbf{X} | \mu) \cdot \pi(\mu)}{f(\mathbf{X})}$$

$f(\mathbf{X}) \quad \mu$

$$p(\mu | \mathbf{X}) \propto f(\mathbf{X} | \mu) \cdot \pi(\mu)$$



1. $\pi(\theta) \quad \theta \quad 0.5$

2. $f(X | \theta) \quad \theta \quad X$

3. $p(\theta | X) \quad X \quad \theta$

$$p(\theta | X) = \frac{f(X | \theta) \pi(\theta)}{\int_{\Theta} f(X | \theta) \pi(\theta) d\theta}$$

$$\propto \times 1$$

tip 2.5

2.5 Bayes $\theta \quad \pi(\theta) \sim \text{Beta}(2, 2) \quad 10 \quad 7 \quad \text{Beta-Binomial}$

$$\theta \mid X \sim \text{Beta}(2 + 7, 2 + 3) = \text{Beta}(9, 5),$$

$$\text{Bayes} \quad \hat{\theta} = \frac{9}{14} \approx 0.643 \quad \text{MLE } \hat{\theta}_{\text{MLE}} = 0.7 \quad 0.5$$

2.1 3.3

2.6 ($g(\theta)$)

$$\hat{g}(\mathbf{X}) \quad g(\theta) \quad \theta \in \Theta$$

$$E_{\theta}(\hat{g}(\mathbf{X})) = g(\theta),$$

$$\hat{g}(\mathbf{X}) \quad g(\theta)$$

2.6 $\mu \quad X_1, X_2, X_3 = 4, 5, 9$

$$\bar{X} = \frac{4 + 5 + 9}{3} = 6.$$

$$\mu = 6 \quad E(\bar{X}) = \mu = 6 \quad \bar{X} \quad \mu$$



-
-

2.7 ()

$$\hat{g}(\mathbf{X}) \quad \tilde{g}(\mathbf{X}) \quad g(\theta)$$

$$\text{Var}_{\theta}(\tilde{g}(\mathbf{X})) \geq \text{Var}_{\theta}(\hat{g}(\mathbf{X})), \quad \forall \theta \in \Theta,$$

$$\theta \in \Theta \quad \hat{g}(\mathbf{X}) \quad \tilde{g}(\mathbf{X})$$

$$e = \frac{\text{Var}_{\theta}(\hat{g}(\mathbf{X}))}{\text{Var}_{\theta}(\tilde{g}(\mathbf{X}))}$$

$$\tilde{g}(\mathbf{X}) \quad \hat{g}(\mathbf{X})$$



tip 2.6

$$g(\mathbf{X}) \quad \mathbf{X} \quad \theta \quad \text{Var}$$

2.7 $X \sim N(\mu, 1) \quad n = 5 \quad \bar{X} \quad m \quad \mu$

$$\text{Var}(\bar{X}) = \frac{1}{5} = 0.2, \quad \text{Var}(m) \approx \frac{\pi}{2 \times 5} \approx 0.314.$$

$$e = \frac{0.2}{0.314} \approx 0.637 \quad \bar{X} \quad m$$

2.8 ()

$$\hat{g}(\mathbf{X}) \quad g(\theta) \quad \text{Var}_{\theta}(\hat{g}(\mathbf{X})) < \infty \quad g(\theta) \quad g^*(\mathbf{X})$$

$$\text{Var}_{\theta}(\hat{g}(\mathbf{X})) \leq \text{Var}_{\theta}(g^*(\mathbf{X})), \quad \forall \theta \in \Theta,$$

$$\hat{g}(\mathbf{X}) = g(\theta)$$

UMVUE (Uniformly minimum variance unbiased estimator)



UMVUE

2.3 (UMVUE)

$$\mathcal{U} \quad \delta(\mathbf{X}) \in \mathcal{U} \quad g(\theta) \quad \text{UMVUE}$$

$$\delta_0(\mathbf{X}) \in \mathcal{U}_0$$

$$E_\theta(\delta(\mathbf{X})\delta_0(\mathbf{X})) = 0, \quad \forall \theta \in \Theta.$$



tip 2.7

$$U \quad U_0 \quad 0$$



$$\bullet \mathcal{U} \quad g(\theta)$$

$$\bullet \mathcal{U}_0 \quad E_\theta(\delta_0(\mathbf{X})) = 0$$

$$\bullet \delta(\mathbf{X}) \quad \text{UMVUE}$$

$$\delta(\mathbf{X}) \quad \text{UMVUE} \iff$$

$$E_\theta(\delta(\mathbf{X})\delta_0(\mathbf{X})) = 0, \quad \forall \theta \in \Theta, \forall \delta_0 \in \mathcal{U}_0.$$

$$\delta_0 \quad \delta$$

$$\text{UMVUE} \quad 1$$

2.4 (UMVUE)

$$\delta(\mathbf{X}) \quad \delta'(\mathbf{X}) \quad g(\theta) \quad \text{UMVUE}$$

$$P_\theta(\delta(\mathbf{X}) = \delta'(\mathbf{X})) = 1, \quad \forall \theta \in \Theta.$$



2.9 ()

$$\{f(x; \theta) : \theta \in \Theta\} \quad \theta$$

$$1. \quad \Theta$$

$$2. \quad A = \{x : f(x; \theta) > 0\} \quad \theta$$

$$3. \quad \forall x \in A \quad \frac{\partial f(x; \theta)}{\partial \theta} \quad \Theta \quad E_\theta\left(\frac{\partial \ln f(x; \theta)}{\partial \theta}\right) = 0 \quad \forall \theta \in \Theta$$

$$4. \quad \text{Fisher}$$

$$I(\theta) = E_\theta \left[\left(\frac{\partial \ln f(x; \theta)}{\partial \theta} \right)^2 \right] = \int_A \left(\frac{\partial \ln f(x; \theta)}{\partial \theta} \right)^2 f(x; \theta) dx > 0, \quad \forall \theta \in \Theta.$$



C-R

$$\bullet \quad 1 \quad \Theta \quad \theta$$

$$\bullet \quad 2 \quad \theta \quad \theta \quad U(0, \theta)$$

$$\bullet \quad 3 \quad E_\theta \left(\frac{\partial \ln f(x; \theta)}{\partial \theta} \right) = 0 \quad \text{C-R}$$

$$\bullet \quad 4 \quad I(\theta) > 0 \quad \text{Fisher} \quad \text{C-R} \quad \frac{1}{I(\theta)}$$

θ C-R



$$[Fisher] \quad Fisher \quad I(\theta) \quad X \quad \theta$$

$$I(\theta) = \text{Var} \left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right).$$

$$\frac{\partial}{\partial \theta} \ln f = \frac{1}{f} \cdot \frac{\partial f}{\partial \theta},$$

$$f$$

$$S(X; \theta) = \frac{\partial}{\partial \theta} \ln f(X; \theta) \quad E[S] = 0$$

$$I(\theta) = \text{Var}(S) = E(S^2),$$

$$I(\theta) \quad \theta$$

2.5 (Cramér-Rao)

$$\{f(x; \theta) : \theta \in \Theta\} \quad I \rightarrow g(\theta) \quad g'(\theta) \mathbf{X} = (X_1, X_2, \dots, X_n)^T \quad \hat{g}(\mathbf{X}) = \hat{g}(X_1, X_2, \dots, X_n)$$

$$g(\theta)$$

$$\frac{\partial}{\partial \theta} \int_{A^n} \hat{g}(x_1, \dots, x_n) \prod_{i=1}^n f(x_i; \theta) dx_1 \cdots dx_n = \int_{A^n} \hat{g}(x_1, \dots, x_n) \frac{\partial}{\partial \theta} \left(\prod_{i=1}^n f(x_i; \theta) \right) dx_1 \cdots dx_n,$$

$$I \quad I$$

$$\text{Var}_\theta(\hat{g}(\mathbf{X})) \geq \frac{(g'(\theta))^2}{nI(\theta)}, \quad \forall \theta \in \Theta. \quad (2.1)$$



$$\frac{(g'(\theta))^2}{nI(\theta)} \quad \mathbf{C-R}$$

1. $(g'(\theta))^2 \quad g(\theta)$
2. $I(\theta) \text{ Fisher} \quad I(\theta) \quad I$
3. $n \quad nI(\theta) \quad \mathbf{Fisher} \quad (X_1, X_2, \dots, X_n)^T$

$$\mathbf{C-R} \quad \mathbf{UMVUE}$$

2.2

2.10 ()

$$\hat{g}_n = \hat{g}_n(X_1, X_2, \dots, X_n) \quad g(\theta)$$

$$\theta \in \Theta \quad \varepsilon > 0$$

$$\lim_{n \rightarrow \infty} P_\theta\{|\hat{g}_n - g(\theta)| > \varepsilon\} = 0,$$

$$\hat{g}_n \quad g(\theta) \quad \hat{g}_n \xrightarrow{P} g(\theta)$$

$$\theta \in \Theta$$

$$P_\theta \left\{ \lim_{n \rightarrow \infty} \hat{g}_n = g(\theta) \right\} = 1,$$

$$\hat{g}_n \quad g(\theta) \quad \hat{g}_n \xrightarrow{a.s.} g(\theta)$$

- $\Leftrightarrow \xrightarrow{P}$
- $\Leftrightarrow \xrightarrow{a.s.} \text{almost surely}$

2.6 ()

$$G(\theta_1, \theta_2, \dots, \theta_k) \quad \Theta \subset \mathbf{R}^k \quad \hat{T}_{in} \quad \theta_i \quad i = 1, 2, \dots, k \quad G(\hat{T}_{1n}, \hat{T}_{2n}, \dots, \hat{T}_{kn}) \quad G(\theta_1, \theta_2, \dots, \theta_k)$$

2.7 ()

$$k \quad k \geq 2 \quad \alpha_i, i = 1, \dots, k \quad \beta_j, j = 2, \dots, k \quad g(\theta)$$

$$g(\theta) = G(\alpha_1, \alpha_2, \dots, \alpha_k, \beta_2, \dots, \beta_k)$$

$$G \quad \hat{\alpha}_i \quad \hat{\beta}_j$$

$$G(\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_k, \hat{\beta}_2, \dots, \hat{\beta}_k)$$

$$g(\theta)$$



$$g(\theta)$$

2.8 ()

$$\{f(x; \theta) : \theta \in \Theta\}$$

$$1. \Theta$$

$$2. \theta \theta'$$

$$3. \{f(x; \theta) : \theta \in \Theta\} \quad A = \{x : f(x; \theta) > 0\} \quad \theta$$

$$(X_1, X_2, \dots, X_n)^T \theta \quad \hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n) \quad \hat{\theta}_n \theta$$



$$\Theta \quad \theta$$

$$” \quad ”$$

$$\theta$$

$$\theta$$

$$\theta$$

$$\theta$$

2.9 (MLE Consistency)

$$\{f(x; \theta) : \theta \in \Theta\}$$

$$(1) \Theta \quad \mathbf{R} \quad \theta$$

$$(2) \quad \theta \quad \theta' \quad \theta$$

$$(3) \{f(x; \theta) : \theta \in \Theta\} \quad A$$

$$(4) f(x; \theta) \quad \theta \quad \frac{\partial f(x; \theta)}{\partial \theta} \quad \Theta \quad (X_1, X_2, \dots, X_n)^T \in A^n$$

$$\frac{\partial}{\partial \theta} \prod_{i=1}^n f(X_i; \theta) = 0$$

$$\hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n) \quad \text{MLE} \quad \text{---}$$

$$\hat{\theta}_n \xrightarrow{P} \theta \quad (n \rightarrow \infty) \quad \hat{\theta}_n \quad \theta$$



$$\text{MLE} \quad \text{---} \quad n \rightarrow \infty$$

$$\text{Consistency}$$

$$\hat{\theta}_n \xrightarrow{P} \theta \quad (n \rightarrow \infty).$$

$$0 \quad (1)(3)(4) \quad \text{MLE} \quad (2)$$

2.11 ()

$$g(\theta) \quad \hat{g}_n = \hat{g}_n(X_1, X_2, \dots, X_n)$$

$$\sqrt{n}(\hat{g}_n - g(\theta)) \xrightarrow{L} N(0, V(\theta)) \quad (n \rightarrow \infty),$$

$$\hat{g}_n \quad n$$

$$\hat{g}_n \sim N\left(g(\theta), \frac{V(\theta)}{n}\right),$$

$$\hat{g}_n \quad V(\theta)/n \quad \hat{g}_n$$



$$\hat{g}_n \quad \sqrt{n}$$

$$\hat{g}_n - g(\theta) \quad n \quad 0 \quad \sqrt{n} \quad ” \quad ” \quad \xrightarrow{L}$$

$$\hat{g}_n \sim N(g(\theta), V(\theta)/n) \quad \hat{g}_n$$

L “Loi” ” / ”

2.10 (MLE)

$$X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim f(x, \theta) \quad \theta \in \Theta \quad \{f(x; \theta) : \theta \in \Theta\}$$

$$(1) \Theta \quad \theta$$

$$\begin{aligned}
 (2) \quad & A = \{x : f(x; \theta) > 0\} \quad \theta \\
 (3) \quad & \forall x \in A \quad f(x; \theta) \quad \theta \quad \text{Taylor} \\
 (4) \quad & \int_A f(x; \theta) dx \quad \theta \quad \text{Fisher} \\
 (5) \quad & \text{Fisher} \quad I(\theta) \quad 0 < I(\theta) < \infty \quad 1/I(\theta) \\
 (6) \quad & \forall \theta_0 \in \Theta \quad a \quad M(x) \\
 & \left| \frac{\partial^3 \ln f(x; \theta)}{\partial \theta^3} \right| \leq M(x), \quad x \in A, \quad \theta_0 - a < \theta < \theta_0 + a \\
 & E_{\theta_0}(|M(X)|) < \infty \quad \text{Taylor} \\
 & \hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n) \quad \hat{\theta}_n \xrightarrow{P} \theta_0 \\
 & \sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{L} N\left(0, \frac{1}{I(\theta_0)}\right).
 \end{aligned}$$



2.11 (Delta - y=x)

$$\begin{aligned}
 & \hat{\theta}_n \quad \theta \\
 & \sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{L} N(0, V(\theta)), \\
 & h(\theta) \quad \Theta \quad h'(\theta) \neq 0 \\
 & \sqrt{n}(h(\hat{\theta}_n) - h(\theta)) \xrightarrow{L} N(0, (h'(\theta))^2 V(\theta)) \quad (n \rightarrow \infty).
 \end{aligned}$$



2.1 (MLE C-R)

3.4.5 3.4.6

$$\sqrt{n}(h(\hat{\theta}) - h(\theta)) \xrightarrow{L} N\left(0, \frac{[h'(\theta)]^2}{I(\theta)}\right).$$

$$h(\theta) \quad h(\hat{\theta}) \quad C-R \quad h(\hat{\theta}) \quad C-R$$



$$\begin{aligned}
 & I \quad J \quad F \\
 & \bullet \quad F \quad F \quad k \quad \text{Var}(T) \\
 & \bullet \quad F_c \quad c \quad \hat{c} \quad F_{\hat{c}} \quad B
 \end{aligned}$$

$$\widehat{\text{MSE}}_{pb}(T) = \frac{1}{B} \sum_{j=1}^B (T^{(j)} - \theta(F_{\hat{c}}))^2.$$

$$\bullet \quad F \quad F_n \quad F_{\hat{c}} \quad \{x_1, x_2, \dots, x_n\} \quad n \quad B \quad T^{(j)}$$

2.3



$$\begin{aligned}
 & (X_1, X_2, \dots, X_n)^T \quad F(x, \theta) \quad \hat{\theta}_{1n} \quad \hat{\theta}_{2n} \quad \theta \\
 & \sqrt{n}(\hat{\theta}_{1n} - \theta) \xrightarrow{L} N(0, \sigma_1^2), \quad \sqrt{n}(\hat{\theta}_{2n} - \theta) \xrightarrow{L} N(0, \sigma_2^2), \\
 & \sigma_1^2/\sigma_2^2 \quad \hat{\theta}_{2n} \quad \hat{\theta}_{1n}
 \end{aligned}$$

2.12 ()

$$\begin{aligned}
 & \hat{\theta}_n \quad \theta \quad (X_1, X_2, \dots, X_n)^T \quad \sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{L} N(0, \sigma_1^2) \quad n \quad m = m(n) \\
 & \lim_{n \rightarrow \infty} \frac{m(n)}{n} = \gamma \quad (\gamma > 0),
 \end{aligned}$$

$$\sqrt{n}(\hat{\theta}_{m(n)} - \theta) \xrightarrow{L} N\left(0, \frac{\sigma_1^2}{\gamma}\right).$$



tip 2.8

$$\frac{\hat{\theta}_{2n}}{\sigma_1^2} \quad \frac{\hat{\theta}_{1n}}{\sigma_1^2/\gamma} \quad e = \sigma_1^2/\sigma_2^2 \quad \frac{\hat{\theta}_{1n}}{n} \quad \frac{\hat{\theta}_{2n}}{en} \quad 3.4.7 \quad \text{---} \quad n \quad m(n) \approx \gamma n$$

$$1. \quad \mu \quad \bar{X} \quad \sigma^2 \quad \pi\sigma^2/2 \quad \bar{X} \quad 2/\pi \approx 0.637 \quad 57\%$$

$$2. \quad \hat{\theta}_{1n} \quad \text{C-R} \quad 1/I(\theta) \quad \text{MLE} \quad e = \frac{1/I(\theta)}{\sigma_2^2}.$$

$$3. \quad e = 1 \quad \hat{\theta}_{2n} \quad e \quad e \quad 0.8 \quad 25\%$$

2.3.1

2.12 ()

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x; \theta) \quad \theta \in \Theta \quad 0 < \alpha < 1 \quad \theta_1(\mathbf{X}) = \theta_1(X_1, X_2, \dots, X_n) \quad \theta_2(\mathbf{X}) = \theta_2(X_1, X_2, \dots, X_n) \quad \theta_1(\mathbf{X}) \leq \theta_2(\mathbf{X})$

$$P_\theta\{\theta_1(\mathbf{X}) \leq \theta \leq \theta_2(\mathbf{X})\} \geq 1 - \alpha, \quad \forall \theta \in \Theta,$$

$$[\theta_1(\mathbf{X}), \theta_2(\mathbf{X})] \quad \theta \quad 100(1 - \alpha)\%$$

$$P_\theta\{\theta \leq \theta_2(\mathbf{X})\} \geq 1 - \alpha, \quad \forall \theta \in \Theta,$$

$$\theta_2(\mathbf{X}) \quad \theta \quad 100(1 - \alpha)\%$$

$$P_\theta\{\theta \geq \theta_1(\mathbf{X})\} \geq 1 - \alpha, \quad \forall \theta \in \Theta,$$

$$\theta_1(\mathbf{X}) \quad \theta \quad 100(1 - \alpha)\%$$



tip 2.9

θ



$$\theta \quad [\theta_1(\mathbf{X}), \theta_2(\mathbf{X})] \quad 1 - \alpha \quad 100(1 - \alpha)\% \quad \theta$$

$$1 - \alpha \quad \theta \quad 1 - \alpha \quad \text{---} \theta \quad \theta$$

$$\theta \quad \theta$$

2.13 ()

$$1. \quad \theta \quad T = T(\mathbf{X})$$

$$2. \quad T \quad \theta \quad G(T, \theta) \quad G(T, \theta)$$

$$3. \quad G(T, \theta) \quad S \quad P(G(T, \theta) \in S) = 1 - \alpha \quad S$$

$$4. \quad G(T, \theta) \in S \quad \theta_1(\mathbf{X}) \leq \theta \leq \theta_2(\mathbf{X}) \quad [\theta_1(\mathbf{X}), \theta_2(\mathbf{X})] \quad 100(1 - \alpha)\%$$



$$\text{“ ”} \quad G(T, \theta) \quad \bar{X} \quad \mu \quad \mu \quad \mu$$

tip 2.10

$T \quad T \quad \theta$ θ $\theta \quad T$ T
 T $N(0,1) \quad \chi^2$ *Beta* *CLT* T

2.3.2



$$Z \sim \mathcal{N}(0, 1) \quad Q = Z^2 \quad Q \sim \chi^2(1)$$

$$q = z^2 \quad q$$

$$|z| > 1 \Rightarrow q > 1 \quad , \quad |z| < 1 \Rightarrow q < 1$$

$$\pm z \quad \frac{Q}{q} \quad 2 \, dz \quad dq$$

$$f_Q(q) = 2 \cdot f_Z(\sqrt{q}) \cdot \frac{1}{2\sqrt{q}} = \frac{f_Z(\sqrt{q})}{\sqrt{q}}, \quad q > 0$$

 Z

$$E[Q] = E[Z^2] = \int_{-\infty}^{+\infty} z^2 f_Z(z) \, dz = 1$$

3

3.1



- “ H_0 ” “ ” H_0 ”
- “ ” H_0 ”
- H_0 H_0

3.1 ()

1.

H_0 / H_1 / H_1 H_0 H_0

2.

S S^c

$$\phi(\mathbf{X}) = \begin{cases} 1, & \mathbf{X} \in S \\ 0, & \mathbf{X} \in S^c \end{cases}$$

3.

H_0 H_0



3.2 ()

$$\beta_\phi(\theta) = E_\theta(\phi(\mathbf{X})) = P_\theta(\mathbf{X} \in S)$$

- $\theta \in \Theta_0$ $\beta_\phi(\theta)$
- $\theta \in \Theta_1$ $\beta_\phi(\theta)$

$$\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha$$

α 0.01, 0.05, 0.1

α



3.1 p $n = 5$

$$H_0 : p \leq 0.04 \quad H_1 : p > 0.04$$

$$Y = \sum_{i=1}^5 X_i \quad 5 \quad Y \sim B(5, p) \quad k = 1 \quad S = \{Y > 1\}$$

$$\phi(\mathbf{X}) = \begin{cases} 1, & Y > 1 \\ 0, & Y \leq 1 \end{cases}$$

$p = 0.04$

$$P_{0.04}(Y > 1) = \sum_{j=2}^5 \binom{5}{j} (0.04)^j (0.96)^{5-j} \approx 0.0148 \leq \alpha = 0.05$$

$$\beta_\phi(p) = P_p(Y > 1) \quad p = 0.5 \quad \beta_\phi(0.5) \approx 0.81 \quad 0.5 \quad 81\% \quad H_0$$

y $y > 1$ H_0 H_0

$$Y \quad H_1 \quad Y \quad Y \quad H_0 \quad E(Y) = np \quad p \quad Y$$

$$\alpha \quad P_{p \leq 0.04}(Y > k) \leq \alpha \quad k \quad \alpha \quad k$$

$$\text{---} \quad p \quad p$$



$$H_0: \theta \in \Theta_0 \quad \theta \in \Theta_1$$

3.2

3.3 (—)

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2) \quad H_0 \leftrightarrow H_1$$

- σ^2 μ $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ Z
- σ^2 μ $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$ t
- μ σ^2 $\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$ χ^2
- σ_1^2, σ_2^2 $\mu_1 - \mu_2$ Z
- $\sigma_1^2 = \sigma_2^2$ $\mu_1 - \mu_2$ t
- μ_1, μ_2 σ_1^2/σ_2^2 $F = \frac{S_1^2}{S_2^2} \sim F(m-1, n-1)$ F



3.4 (—)

$$H_1: \theta \neq \theta_0 \quad c_{\alpha/2} \quad -c_{\alpha/2}$$

- $H_1: \theta \neq \theta_0 \quad |T| > c_{\alpha/2}$
- $H_1: \theta > \theta_0 \quad T > c_{\alpha/2}$
- $H_1: \theta < \theta_0 \quad T < -c_{\alpha/2}$



$$\begin{array}{l} \text{vs } H_1: \mu \neq \mu_0 \quad z_{\alpha/2} \quad H_1: \mu > \mu_0 \quad H_1: \mu < \mu_0 \quad z_{\alpha} \\ \sigma^2 \quad Z \quad t \quad \sigma^2 \quad S \quad t \quad n-1 \end{array}$$

3.1 ()

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2) \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$H_0: \sigma^2 = \sigma_0^2$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$$

$$n-1 \quad S^2 \quad \bar{X}$$

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1-1, n_2-1)$$



3.3

3.4

3.5 ()

$$\phi(\mathbf{X}) \leq \alpha \quad \alpha \quad \phi^*(\mathbf{X})$$

$$\beta_\phi(\theta) \geq \beta_{\phi^*}(\theta), \quad \forall \theta \in \Theta_1$$

$$\beta_\phi(\theta) \geq \beta_{\phi^*}(\theta) \quad \phi \quad \phi^* \quad \phi \quad \alpha \quad \text{Uniformly Most Powerful Test} \quad \text{UMP}$$



UMP

$$1. \quad \alpha \quad \beta_\phi(\theta) \leq \alpha, \quad \forall \theta \in \Theta_0$$

2.

$$3. \quad \phi \quad \theta \in \Theta_1$$

$$\beta_\phi(\theta) \geq \beta_{\phi^*}(\theta), \quad \forall \phi^*$$

 ϕ UMP
“ ϕ ”

UMP



$$\text{UMP} \quad \beta_\phi(\theta) \leq \alpha, \forall \theta \in \Theta_0 \quad \theta \in \Theta_1$$

“Uniformly” “ ” $\theta \quad \Theta_1$

tip 3.1

UMP

1.

2. Neyman-Pearson

$$\left\{ \frac{f(X; \theta_1)}{f(X; \theta_0)} > c \right\}$$

UMP

UMP N-P

Lambda $\Lambda \quad \lambda$

3.2 (Neyman-Pearson)

$$\alpha \in [0, 1] \quad H_0 : f = f_0 \text{ vs } H_1 : f = f_1$$

1.

$$\phi(\mathbf{X}) = \begin{cases} 1, & \Lambda(\mathbf{X}) > k \\ r, & \Lambda(\mathbf{X}) = k \\ 0, & \Lambda(\mathbf{X}) < k \end{cases}$$

$$\Lambda(\mathbf{X}) = \prod_{i=1}^n f_1(X_i) / \prod_{i=1}^n f_0(X_i) \quad k > 0 \quad r \in [0, 1] \quad E_0[\phi(\mathbf{X})] = \alpha$$

$$2. \quad \phi \leq \alpha \quad \text{UMP}$$

$$3. \quad \phi \leq \alpha \quad \text{UMP} \quad k \quad E_1(\phi(\mathbf{X})) < 1 \quad E_0(\phi(\mathbf{X})) = \alpha$$



$$\text{MLR} \quad \theta_1 \quad T(\mathbf{X}) \quad \theta_1 \quad \theta_1 \quad T(\mathbf{X}) > c \quad \text{UMP}$$

3.6 ()

$$\{f(x; \theta), \theta \in \Theta\}$$

1.

$$2. \quad T = T(\mathbf{X}) \quad \theta_1 < \theta_2$$

$$\frac{\prod_{i=1}^n f(X_i; \theta_2)}{\prod_{i=1}^n f(X_i; \theta_1)}$$

$$\mathbf{X} \quad T(\mathbf{X}) \quad T(\mathbf{X})$$



$$\text{MLR} \quad \theta_1 < \theta_2 \quad \frac{\prod f(X_i; \theta_2)}{\prod f(X_i; \theta_1)} \quad T(\mathbf{X}) \quad \text{MLR} \quad \text{UMP}$$

tip 3.2

$$\theta_1 \quad \theta_2 \quad X_i \quad \mathbf{X} \quad \theta_2 \quad \theta_1$$

$$3.2 \quad X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1) \quad \theta_0 < \theta_1$$

$$\Lambda_{\theta_1}(\mathbf{X}) = e^{n(\theta_1 - \theta_0)\bar{X} - \frac{n}{2}(\theta_1^2 - \theta_0^2)}$$

$$e^{-\frac{n}{2}(\theta_1^2 - \theta_0^2)} \quad \theta \quad \bar{X} \quad \theta_1 > \theta_0 \quad n(\theta_1 - \theta_0) > 0 \quad \Lambda_{\theta_1}(\mathbf{X}) \quad \bar{X} \quad T(\mathbf{X}) = \bar{X} \quad \theta_1 \quad \bar{X}$$

MLR

3.3 (UMP)

$$\{f(x; \theta), \theta \in \Theta\} \quad \theta_0 \in \Theta \quad \{\theta > \theta_0\} \cap \Theta$$

$$H_0 : \theta \leq \theta_0; \quad H_1 : \theta > \theta_0$$

$$\alpha \in (0, 1)$$

1.

$$\phi(\mathbf{X}) = \begin{cases} 1, & T(\mathbf{X}) > c \\ r, & T(\mathbf{X}) = c \\ 0, & T(\mathbf{X}) < c \end{cases}$$

$$\phi \quad E_{\theta_0}(\phi(\mathbf{X})) = \alpha \quad \alpha \quad \text{UMP}$$

$$2. \quad \beta_\phi(\theta) \quad \{\theta : 0 < \beta_\phi(\theta) < 1\}$$

$$3. \quad \theta < \theta_0$$

$$\beta_\phi(\theta) = \min\{\beta_{\phi^*}(\theta) : E_{\theta_0}(\phi^*(\mathbf{X})) = \alpha\}.$$



$$1. \quad P_{H_0}(X > c) = \alpha \quad c = 1.645$$

$$2. \quad c \quad x \quad x \quad f x$$

$$3. \quad c = \ln k + 0.5 \quad k \approx 3.14$$

k

3.4 ()

$$H_0 : \theta \geq \theta_0; \quad H_1 : \theta < \theta_0$$

UMP

$$\phi(\mathbf{X}) = \begin{cases} 1, & T(\mathbf{X}) < c \\ r, & T(\mathbf{X}) = c \\ 0, & T(\mathbf{X}) > c \end{cases}$$

$$\beta_\phi(\theta)$$



3.3 $f(x; \theta) = \frac{1}{\pi[1 + (x - \theta)^2]}$ MLR $\theta_1 = 0 \theta_2 = 1$

$$\Lambda(x) = \frac{1 + x^2}{1 + (x - 1)^2}$$

x	-2	0	2	10	100
$\Lambda(x)$	0.5	0.5	2.5	1.23	1.02

 $\Lambda(x)$

MLR

UMP

3.1 (4.4.3) X

$$f(x; \theta) = C(\theta)h(x) \exp(Q(\theta)t(x)), \quad \theta \in \Theta \subset \mathbf{R},$$

$$\Theta = \left\{ \theta : \int_{-\infty}^{+\infty} h(x) \exp(Q(\theta)t(x)) dx < +\infty \right\} \quad Q(\theta) \quad \theta$$

$$T(\mathbf{X}) = T(X_1, X_2, \dots, X_n) = \sum_{i=1}^n t(X_i),$$

(1) $Q(\theta)$ (4.4.12) α UMP (4.4.13) (4.4.14)

(2) $Q(\theta)$ (4.4.12) α UMP (4.4.20) (4.4.14)

**3.5 4.5****3.6 4.6****3.5 (4.6.1)**

(4.6.1) H_0

$$U_i = F_0(X_i), \quad i = 1, 2, \dots, n$$

$$U(0, 1)$$

**3.7 (PP QQ)**

$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ F_0

PP - $F_0(X_{(i)})$ $\frac{i}{n}$ H_0 45°

QQ - $F_0^{-1}\left(\frac{i}{n}\right)$ $X_{(i)}$ H_0 45°



3.8 (χ^2)

$$X = a_1, a_2, \dots, a_M \quad n_j \quad a_j \quad np_j \quad \chi^2$$

$$\chi^2 = \sum_{j=1}^M \frac{(n_j - np_j)^2}{np_j}$$



3.6 (4.6.2)

$$H_0 : \chi^2 \xrightarrow{L} \chi^2(M-1) \quad n \rightarrow \infty$$

3.9 (χ^2)

$$H_0 : P(X = a_j) = p_j, \quad j = 1, 2, \dots, M$$

$$p_j \quad \sum_{j=1}^M p_j = 1 \quad \alpha$$

$$\varphi = \begin{cases} 1, & \chi^2 > \chi_{\alpha}^2(M-1) \\ 0, & \chi^2 \leq \chi_{\alpha}^2(M-1) \end{cases}$$

$$np_j \geq 5$$



$$\chi^2$$

$$1. \quad p_j \quad \chi^2 \quad M-1$$

$$2. \quad 0 \quad M \quad \chi^2$$

$$3. \quad p_j \quad MLE$$

$$\chi^2 \xrightarrow{L} \chi^2(M-1-k)$$

$$k$$

4

4.1 (5.1.1)

x

$$E(Y | X = x) = a + bx, \quad (5.1.4)$$

$$E(Y | X = x) = \mu_2 + \rho \sqrt{\frac{\text{Var}(Y)}{\text{Var}(X)}} (x - \mu_1),$$

$$\mu_1 = E(X), \quad \mu_2 = E(Y), \quad \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$



4.1

4.1 ()

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

$$\varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2) \quad x_i \quad \beta_0, \beta_1, \sigma^2$$



4.2 (OLS)

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$



4.3 ()

x^*

$$\hat{Y}(x^*) = \hat{\beta}_0 + \hat{\beta}_1 x^*.$$



4.4 (OLS)

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 \quad \beta_0, \beta_1$$

$$\frac{\partial Q}{\partial \beta_0} = 0 \Rightarrow \sum (Y_i - \beta_0 - \beta_1 x_i) = 0, \quad \frac{\partial Q}{\partial \beta_1} = 0 \Rightarrow \sum x_i (Y_i - \beta_0 - \beta_1 x_i) = 0.$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

$$S_{xx} = \sum (x_i - \bar{x})^2 \quad S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$$



$|e_i|$

e_i^2

$OLS = MLE \text{ Gauss-Markov}$

$OLS \text{ BLUE}$

tip 4.1

$$= \div \quad \hat{\beta}_1 = S_{xy}/S_{xx} = \widehat{\text{Cov}}(x, Y)/\widehat{\text{Var}}(x) \quad " \quad (\bar{x}, \bar{Y})"$$

4.2

4.2 ()

$$P^2 = P$$

4.5 (χ^2)

$$\mathbf{X} \sim N(\mathbf{0}, \sigma^2 I_n) \quad P \text{ } n \times n \quad \text{rank}(P) = r$$

$$\frac{\mathbf{X}^T P \mathbf{X}}{\sigma^2} \sim \chi^2(r).$$



$$\mathbf{u} = \frac{1}{\sqrt{n}}(1, 1, \dots, 1)^T \quad \mathbf{u}^T \mathbf{u} = 1 \quad P = I_n - \mathbf{u} \mathbf{u}^T$$

$$P^2 = (I_n - \mathbf{u} \mathbf{u}^T)^2 = I_n - 2\mathbf{u} \mathbf{u}^T + \underbrace{\mathbf{u} (\mathbf{u}^T \mathbf{u}) \mathbf{u}^T}_{=1} = I_n - \mathbf{u} \mathbf{u}^T = P. \quad \checkmark$$

$$\text{rank}(\mathbf{u} \mathbf{u}^T) = 1 \quad \text{rank}(P) = n - 1$$

$$\frac{\mathbf{X}^T (I_n - \mathbf{u} \mathbf{u}^T) \mathbf{X}}{\sigma^2} \sim \chi^2(n - 1).$$

5

5.1

3	720												
	668	711	625	701	688	667	694	630	547	703	688	697	703
	656	677	700	702	688	691	664	688	679	708	699	667	703

- 1.
- 2.
- 3.

```
(1) R      stem()                                scale 0.5
```

```
> data <- c(668, 711, 625, 701, 688, 667, 694, 630, 547, 703, 688, 697, 703,
+           656, 677, 700, 702, 688, 691, 664, 688, 679, 708, 699, 667, 703)
> stem(data, scale = 0.5)
```

The decimal point **is** 1 digit(s) to the right of the /

54 | 7
56 |
58 |
60 |
62 | 50
64 | 6
66 | 4777879
68 | 88881479
70 | 01233381

(2) R *mean()* *median()*

- $\bar{x} \approx 678.6154$
- $M = 688$

(3) $(678.6154) < (688)$ 547 “ ”

4	(mg/L)	20										
			133	137	148	149	152	167	174	179	189	192
			201	209	210	211	218	238	245	248	253	257

(1) (2)

(1) R $mean()$ $median()$

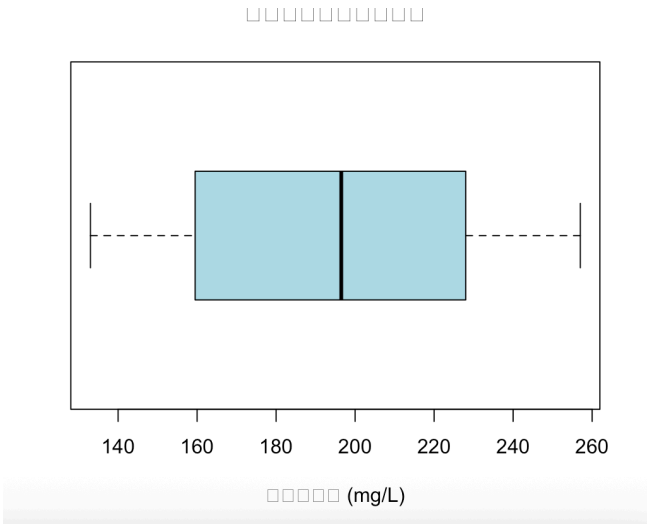
- $\bar{x} = 195.5$
- $M = 196.5$

$$(2) \quad R$$

- **(Range)** $R = \max(x) - \min(x) = 257 - 133 = 124$
- **(IQR)** $IQR() \quad IQR = Q_3 - Q_1 = 59.75$
- **(Sample Variance)** $var() \quad n - 1$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \approx 1605.842$$

5



- R
- Q_3 whisker
- $(\bar{x} = 195.5)$ $(M = 196.5)$
-

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3 $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim N(\mu, \sigma^2)$ μ σ^2

$$\sum_{i=1}^n X_i, \quad X_1 + 2\mu, \quad \max_{1 \leq i \leq n} \{X_i\}, \quad \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}, \quad \sum_{i=1}^n (X_i - \mu)^2.$$

tip 5.1
 $X_1, \dots, X_n$



$$x_1, x_2, \dots, x_n,$$

4 $N(\mu, \sigma^2)$ μ, σ^2 μ, σ^2 n n

$$x_1, x_2, \dots, x_n.$$

$$X_1, X_2, \dots, X_n$$

n

$$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2),$$

μ, σ^2

- $\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}.$

- $(X_1, X_2, \dots, X_n)$

$$f(x_1, x_2, \dots, x_n; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma}} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right], \quad (\mu, \sigma^2) \in \Theta.$$

$$\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$$

$$\left\{P_{\mu, \sigma^2} : (X_1, \dots, X_n) \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2), (\mu, \sigma^2) \in \Theta\right\}$$

tip 5.2

 6 $X_1, X_2, \dots, X_n$ i.i.d. X_1

$$f(x, \sigma) = \frac{1}{\pi\sigma} \frac{1}{1 + (x/\sigma)^2}, \quad -\infty < x < \infty,$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\varphi_X(t) = E(e^{itX}).$$

$$X_i \sim \text{Cauchy}(0, \sigma) \implies \varphi_{X_i}(t) = e^{-\sigma|t|}.$$

$$X_1, X_2, \dots, X_n$$

$$\varphi_{\bar{X}}(t) = E(e^{it\bar{X}}) = E\left(e^{it\frac{1}{n} \sum_{i=1}^n X_i}\right) = \prod_{i=1}^n E\left(e^{i(t/n)X_i}\right).$$

$$\varphi_{\bar{X}}(t) = \prod_{i=1}^n \varphi_{X_i}\left(\frac{t}{n}\right) = \left(e^{-\sigma|t|/n}\right)^n = e^{-\sigma|t|}.$$

tip 5.3

independent and identically distributed

n

e

X_i

X_i

ϕ_{X_i}

n

$1/n$

σ

$$\bar{X} \sim \text{Cauchy}(0, \sigma).$$

$\bar{X}$

$$f_{\bar{X}}(x) = \frac{1}{\pi\sigma} \frac{1}{1 + (x/\sigma)^2}, \quad -\infty < x < \infty.$$

tip 5.4

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$\bar{X}$

$\bar{X}$

$\bar{X}$

$\bar{X}$

$\bar{X}$

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 1 (2.2.3)

$$\sum_{k=r}^n C_n^k p^k (1-p)^{n-k} = r C_n^r \int_0^p t^{r-1} (1-t)^{n-r} dt.$$

$$S(p) = \sum_{k=r}^n C_n^k p^k (1-p)^{n-k} \quad p$$

$$k C_n^k = n C_{n-1}^{k-1}, \quad (n-k) C_n^k = n C_{n-1}^k$$

$$S'(p) = n \sum_{k=r}^n C_{n-1}^{k-1} p^{k-1} (1-p)^{n-k} - n \sum_{k=r}^n C_{n-1}^k p^k (1-p)^{n-k-1}$$

$$k = r$$

$$S'(p) = n C_{n-1}^{r-1} p^{r-1} (1-p)^{n-r} = r C_n^r p^{r-1} (1-p)^{n-r}$$

$$0 \leq p \leq 1 \quad S(0) = 0$$

$$S(p) = r C_n^r \int_0^p t^{r-1} (1-t)^{n-r} dt. \quad \blacksquare$$

tip 5.5

$$0 \leq p \leq 1 \quad r C_n^r \quad r \quad C_n^k \quad p \quad k \quad n$$



[]

$$C_n^k = \frac{n!}{k!(n-k)!}$$

tip 5.6

$$\begin{aligned} & (n-k)! \quad k! \\ & k C_n^k \quad k \quad k! \quad (k-1)! \quad n! \quad n \cdot (n-1)! \quad (k-1)!(n-k)! \quad n C_{n-1}^{k-1} \\ & (n-k) C_n^k \quad (n-k) \quad (n-k)! \quad (n-k-1)! \quad n! \quad n \cdot (n-1)! \quad k!(n-k-1)! \quad k!(n-1-k)! \\ & n C_{n-1}^k \\ & n \quad n-1 \quad k \quad n-k \end{aligned}$$

$$\bullet k C_n^k = n C_{n-1}^{k-1} \quad n \quad k \quad 1 \quad n-1 \quad k-1$$

$$\bullet (n-k) C_n^k = n C_{n-1}^k \quad n \quad k \quad n-k \quad 1 \quad n-1 \quad k$$



2 X

$$P\{X = k\} = p(1-p)^{k-1}, \quad k = 1, 2, \dots,$$

$$X_1, X_2, \dots, X_n \quad X \quad X_{(1)}, X_{(n)} \\ q = 1-p \quad P(X = k) = pq^{k-1}$$

$$P(X > m) = q^m, \quad P(X \leq m) = 1 - q^m, \quad m = 0, 1, 2, \dots$$

$$1. X_{(1)} \quad P(X_{(1)} > m) = (P(X > m))^n = (q^n)^m = 1 - q^n \quad X_{(1)} \sim \text{Geo}(1 - q^n)$$

$$P(X_{(1)} = m) = (1 - q^n)(q^n)^{m-1}, \quad m = 1, 2, \dots$$

$$2. X_{(n)} \quad P(X_{(n)} \leq m) = (P(X \leq m))^n = (1 - q^m)^n$$

$$P(X_{(n)} = m) = (1 - q^m)^n - (1 - q^{m-1})^n, \quad m = 1, 2, \dots$$

$$X_{(n)}$$

tip 5.7 (F)

$$P(X \leq x) = F(x) \longrightarrow \text{CDF (Cumulative Distribution Function)} \quad / \quad$$

$$X \quad x, \quad 5 \quad 5$$

tip 5.8

$$X_1 \quad m \quad m$$

$$X_1$$

 **3** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$ $X_{(1)} \quad X_{(n)}$
 $X \sim U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$

$$f(x) = 1, \quad F(x) = x - \theta + \frac{1}{2}, \quad \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}.$$

$$f_{(n)}(x) = n[F(x)]^{n-1}f(x) \quad f_{(1)}(x) = n[1 - F(x)]^{n-1}f(x)$$

$$f_{X_{(n)}}(x) = n \left(x - \theta + \frac{1}{2} \right)^{n-1}, \quad f_{X_{(1)}}(x) = n \left(\theta + \frac{1}{2} - x \right)^{n-1},$$

$$\theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2} \quad 0$$

tip 5.9

$$F_{X_{(1)}}(x) = P(X_{(1)} \leq x) \quad \{X_{(1)} \leq x\} = \{\exists i, X_i \leq x\} \quad \text{'' ''}$$

$$1 - P(X_{(1)} > x) \quad \{X_{(1)} > x\} = \{X_i > x\} \quad \text{'' ''}$$

$$X_i \quad P(X_i > x) = 1 - F(x) \quad \text{''} \leq x \text{''} \quad F(x) \quad \text{''} > x \text{''}$$

$$P(X_{(1)} > x) = [1 - F(x)]^n \implies F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n.$$

$$F'_{X_{(1)}}(x) \quad f_{X_{(1)}}(x)$$

5.2**5.2.1 45,3,4,6**

 **4** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim F(x), F(x) \quad f(x),$

1. $X_{(1)} \quad X_{(n)}$

2. $R = X_{(n)} - X_{(1)}$

$U = X_{(1)} \quad V = X_{(n)}$

$$F_{U,V}(x, y) = P(U \leq x, V \leq y) = P(X_{(1)} \leq x, X_{(n)} \leq y).$$

$$x \leq y$$

$$F_{U,V}(x, y) = P(V \leq y) - P(U > x, V \leq y) = [F(y)]^n - [F(y) - F(x)]^n$$

$$x > y \quad [F(y)]^n \quad \partial^2 / \partial x \partial y \quad x < y$$

$$f_{U,V}(x, y) = n(n-1)[F(y) - F(x)]^{n-2}f(x)f(y), \quad x < y.$$

$$R = V - U, \quad T = U \quad U = T, \quad V = T + R \quad J = 1 \quad T$$

$$f_R(r) = \int_{-\infty}^{\infty} f_{U,V}(t, t+r) dt = n(n-1) \int_{-\infty}^{\infty} [F(t+r) - F(t)]^{n-2} f(t) f(t+r) dt, \quad r > 0,$$

$r \leq 0 \quad f_R(r) = 0$

tip 5.10

$$\begin{array}{ccccccc} & & X_{(1)} \leq x & & X_{(n)} \leq y & & \\ X_{(n)} & y & & y & & y & y \\ X_{(1)} & x & & x & & & \\ X_{(n)} & y & X_{(n)} & y & X_{(1)} & x & x & y \end{array}$$

tip 5.11

 5 5

$-2.8, -1, 1.5, 2.1, 3.4.$

$x_{(1)} = -2.8, \quad x_{(2)} = -1, \quad x_{(3)} = 1.5, \quad x_{(4)} = 2.1, \quad x_{(5)} = 3.4.$

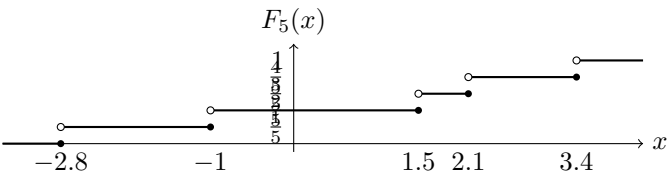
$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I\{x_i \leq x\}.$$

$n = 5$

$$F_5(x) = \begin{cases} 0, & x < -2.8, \\ \frac{1}{5}, & -2.8 \leq x < -1, \\ \frac{2}{5}, & -1 \leq x < 1.5, \\ \frac{3}{5}, & 1.5 \leq x < 2.1, \\ \frac{4}{5}, & 2.1 \leq x < 3.4, \\ 1, & x \geq 3.4. \end{cases}$$

$x = -2.8, -1, 1.5, 2.1, 3.4$

$\frac{1}{5}$
TikZ



tip 5.12

$x \qquad F5 \ x$

 6 $N(\mu, \sigma^2) \quad \alpha \ (0 < \alpha < 1) \quad z_\alpha \qquad \alpha \quad u_\alpha$

$$z_\alpha = \mu + \sigma u_\alpha$$

$$Y = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$P(X > z_\alpha) = P\left(Y > \frac{z_\alpha - \mu}{\sigma}\right).$$

$$P(X > z_\alpha) = \alpha = P(Y > u_\alpha)$$

$$\frac{z_\alpha - \mu}{\sigma} = u_\alpha \implies z_\alpha = \mu + \sigma u_\alpha.$$

tip 5.13 α z_α $\mu\alpha$ **5.2.2 46,1,2,3**

$$1 \quad X \sim \chi^2(n)$$

$$E(X^k) = 2^k \left(\frac{n}{2} + k - 1\right) \left(\frac{n}{2} + k - 2\right) \cdots \frac{n}{2}, \quad k = 1, 2, \dots$$

$$E(X) = n, \text{Var}(X) = 2n$$

$$X \sim \chi^2(n)$$

$$f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-x/2}, \quad x > 0.$$

$$E(X^k) = \int_0^\infty x^k f(x) dx = \frac{1}{2^{n/2}\Gamma(n/2)} \int_0^\infty x^{\frac{n}{2}+k-1} e^{-x/2} dx.$$

$$\int_0^\infty x^{a-1} e^{-x/2} dx = 2^a \Gamma(a) \quad a = \frac{n}{2} + k$$

$$E(X^k) = \frac{2^{\frac{n}{2}+k} \Gamma(\frac{n}{2} + k)}{2^{n/2}\Gamma(n/2)} = 2^k \frac{\Gamma(\frac{n}{2} + k)}{\Gamma(n/2)}.$$

$$\Gamma(z+1) = z\Gamma(z)$$

$$E(X^k) = 2^k \frac{\Gamma(\frac{n}{2} + k)}{\Gamma(n/2)} = 2^k \left(\frac{n}{2} + k - 1\right) \left(\frac{n}{2} + k - 2\right) \cdots \frac{n}{2}.$$

$$k = 1$$

$$E(X) = 2 \cdot \frac{n}{2} = n.$$

$$k = 2$$

$$E(X^2) = 2^2 \left(\frac{n}{2} + 1\right) \frac{n}{2} = 4 \cdot \frac{n+2}{2} \cdot \frac{n}{2} = n(n+2) = n^2 + 2n.$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = (n^2 + 2n) - n^2 = 2n.$$

tip 5.14

$$\frac{\Gamma(n/2) \cdot 2^{n/2}}{\frac{n}{2} + k - 1} \cdot \frac{x^{\frac{n}{2}-1}}{k} \cdot \frac{\Gamma(\frac{n}{2})}{2^{n/2}} = \frac{x^k}{\Gamma(\frac{n}{2})} \cdot \frac{\Gamma(\frac{n}{2} + k)}{2^k}$$



$$2 \quad X \sim t(n) \quad k < n \quad E(X^k)$$

$$E(X^k) = \begin{cases} 0, & k < 0 \\ n^{\frac{k}{2}} \frac{\Gamma(\frac{k+1}{2}) \Gamma(\frac{n-k}{2})}{\sqrt{\pi} \Gamma(\frac{n}{2})}, & 0 < k < n \end{cases}$$

$$k \geq n \quad E(X^k)$$

tip 5.15

$$X \sim t(n) \quad f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}} \quad |x| \rightarrow \infty$$

$$|x|^k f(x) \sim C|x|^{k-n-1} \quad (C > 0).$$

$$\int_{-\infty}^{\infty} x^{k-n-1} dx \iff k-n-1 < -1 \iff k < n$$

$$E(|X|^k) < \infty \iff k < n \quad t(n) \quad k < n$$

tip 5.16

x

3 $X_1 \sim \chi^2(m), X_2 \sim \chi^2(n)$ X_1, X_2 $X_1 + X_2$ X_1/X_2

$$U = \frac{X_1}{X_1 + X_2}, W = \frac{X_1}{X_1 + X_2} \quad u > 0, 0 < w < 1 \quad X_1 = Uw, X_2 = U(1-w) \quad \left| \frac{\partial(x_1, x_2)}{\partial(u, w)} \right| = |1-w \quad -u| = u$$

$$X_1 \sim \chi^2(m), X_2 \sim \chi^2(n)$$

$$f_{U,W}(u, w) = f_{X_1, X_2}(uw, u(1-w)) \cdot u = \underbrace{\frac{1}{2^{\frac{m+n}{2}} \Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} u^{\frac{m+n}{2}-1} e^{-u/2}}_{g(u)} \cdot \underbrace{w^{\frac{m}{2}-1} (1-w)^{\frac{n}{2}-1}}_{h(w)}, \quad u > 0, 0 < w < 1.$$

$$f_{U,W} = g(u)h(w) \quad U \perp W \quad \frac{X_1}{X_2} = \frac{w}{1-w} \quad W \quad U \perp \frac{X_1}{X_2} \quad X_1 + X_2 \quad \frac{X_1}{X_2}$$

tip 5.17

u

u

$$X_1 + X_2 \sim \chi^2(m+n).$$

$$g(u) = f_U(u) = \frac{1}{2^{\frac{m+n}{2}} \Gamma(\frac{m+n}{2})} u^{\frac{m+n}{2}-1} e^{-u/2}, \quad u > 0.$$



$$V = \frac{X_1}{X_2} \quad W = \frac{X_1}{X_1 + X_2}$$

$$V = \frac{W}{1-W} \quad (V > W \quad W = \frac{V}{1+V}).$$

$$U > W \quad U < V$$



$$F \sim F(m, n)$$

$$F_{1-\alpha}(m, n) F_{\alpha}(n, m) = 1, \quad \forall \alpha \in (0, 1).$$

$$X \sim F(m, n) \quad F$$

$$\frac{1}{X} \sim F(n, m).$$

$$c = F_{1-\alpha}(m, n),$$

$$P(X \leq c) = 1 - \alpha, \quad P(X > c) = \alpha.$$

$$P\left(\frac{1}{X} < \frac{1}{c}\right) = P(X > c) = \alpha.$$

$\frac{1}{X} \sim F(n, m) \quad F(n, m) \quad \alpha$

$F_{\alpha}(n, m) = \frac{1}{c} = \frac{1}{F_{1-\alpha}(m, n)}.$

$F_{1-\alpha}(m, n) F_{\alpha}(n, m) = 1.$

tip 5.18

$nm \quad mn \quad l/x \quad X$

$X_1, X_2, \cdots, X_n \quad A_{2 \times n}$

$Y = AX,$

$Y \quad X$

$X = (X_1, X_2, \cdots, X_n)^T, \quad Y = AX,$

$A \quad 2 \times n$

$Y \quad X$

$Y \quad X_1, \ldots, X_n \quad X_i \quad A$
 $X_1, \ldots, X_n \quad Y = AX \quad Y \quad X$

$\chi^2 \quad X_1, X_2, \ldots, X_n \text{ i.i.d. } X_1 \sim N(0, 1)$

$\mathbf{X} = (X_1, \cdots, X_n)^T,$

$A \quad n$

$\mathbf{X}^T A \mathbf{X} \sim \chi^2(r)$

A

$A^2 = A,$

$\text{rank}(A) = r.$

$A \quad Q$

$Q^T A Q = \Lambda = \text{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n).$

$Y = Q^T X \quad X \sim N_n(0, I_n) \quad Q$

$Y \sim N_n(0, I_n),$

$Y_1, \cdots, Y_n \quad N(0, 1)$

$X^T A X = X^T Q \Lambda Q^T X = Y^T \Lambda Y = \sum_{i=1}^n \lambda_i Y_i^2.$

$A^2 = A$

$\Lambda^2 = \Lambda,$

$\lambda_i^2 = \lambda_i \quad \lambda_i = 0 \quad 1$

$\text{rank}(A) = \text{rank}(\Lambda) = r,$

$$r \quad \lambda_i = 1 \quad 0$$

$$X^T A X = \sum_{i=1}^r Y_i^2 \sim \chi^2(r).$$

$$X^T A X \sim \chi^2(r)$$

$$M_{X^T A X}(t) = \prod_{i=1}^n (1 - 2\lambda_i t)^{-1/2} = (1 - 2t)^{-r/2}.$$

$$r \quad \lambda_i = 1 \quad \lambda_i = 0$$

$$\Lambda^2 = \Lambda, \quad \text{rank}(\Lambda) = r.$$

$$A = Q \Lambda Q^T$$

$$A^2 = Q \Lambda^2 Q^T = Q \Lambda Q^T = A, \quad \text{rank}(A) = r.$$

$$X^T A X \sim \chi^2(r)$$

$$A \quad \text{rank}(A) = r$$

5.3



$$X_1, \dots, X_n \text{ i.i.d.}, X_i \sim N(\mu, \sigma^2) \quad X_{n+1} \sim N(\mu, \sigma^2) \quad X_1, \dots, X_n$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$T = \sqrt{\frac{n}{n+1}} \frac{\bar{X} - X_{n+1}}{s} \sim t(n-1).$$

$$\bar{X} \sim N(\mu, \sigma^2/n) \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1) \quad X_{n+1} \sim N(\mu, \sigma^2) \quad X_1, \dots, X_n \quad \bar{X}, s^2$$

$$\bar{X} - X_{n+1} \sim N\left(0, \frac{n+1}{n} \sigma^2\right) \quad s^2.$$

$$Z = \frac{\bar{X} - X_{n+1}}{\sqrt{\frac{n+1}{n}} \sigma} \sim N(0, 1), \quad U = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1),$$

$$Z \perp U \quad t$$

$$\frac{Z}{\sqrt{U/(n-1)}} = \sqrt{\frac{n}{n+1}} \cdot \frac{\bar{X} - X_{n+1}}{s} = T \sim t(n-1).$$

tip 5.19

$$t \quad \bar{X} - X_{n+1} \quad 0 \quad \sqrt{\frac{n+1}{n}} \quad n-1 \quad \sigma^2 \quad \text{---} \quad s \quad s$$



$$\mathbf{2.4 \ 1} \quad X_1, \dots, X_n \text{ i.i.d. } X_1 \sim N(0, 1)$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

$$\text{Var}(s^2)$$

$$X_1, \dots, X_n \sim N(0, 1) \quad \sigma^2 = 1$$

$$(n-1)s^2 \sim \chi^2(n-1) \implies D((n-1)s^2) = 2(n-1).$$

$$D((n-1)s^2) = (n-1)^2 D(s^2)$$

$$(n-1)^2 D(s^2) = 2(n-1) \implies D(s^2) = \frac{2}{n-1}.$$

tip 5.20

$$\chi^2(n-1) \quad 2(n-1) \quad "2" \quad (n-1)s^2 \quad s^2 \quad (n-1) \quad (n-1)^2 \quad D((n-1)s^2) = (n-1)^2 D(s^2)$$

$$D(s^2) = \frac{2}{n-1}$$

 **2.4 2** $X_1, \dots, X_m$ i.i.d. $X_1 \sim N(\mu_1, \sigma^2)$ $Y_1, \dots, Y_n$ i.i.d. $Y_1 \sim N(\mu_2, \sigma^2)$

$$s_1^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - \bar{X})^2, \quad s_2^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

1.

$$\frac{(m-1)s_1^2 + (n-1)s_2^2}{\sigma^2}$$

2. s_1^2

$$\frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2}$$

1. $\frac{(m-1)s_1^2}{\sigma^2} \sim \chi^2(m-1) \quad \frac{(n-1)s_2^2}{\sigma^2} \sim \chi^2(n-1)$

$$\frac{(m-1)s_1^2 + (n-1)s_2^2}{\sigma^2} \sim \chi^2(m+n-2).$$

2. $W = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} \quad \chi^2(k) \quad 2k \quad D(aX) = a^2 D(X)$

$$\frac{(m-1)s_1^2}{\sigma^2} \sim \chi^2(m-1) \implies D(s_1^2) = \frac{2\sigma^4}{m-1},$$

$$\frac{(m+n-2)W}{\sigma^2} \sim \chi^2(m+n-2) \implies D(W) = \frac{2\sigma^4}{m+n-2}.$$

$$m+n-2 > m-1 \quad D(W) < D(s_1^2) \quad W$$

tip 5.21

 **2.4 3** $X_1, \dots, X_n$ i.i.d. $X_1 \sim N(0, 1)$

$$Z = \frac{\sum_{i=1}^{n-1} X_i}{X_n}$$

$$Y = \sum_{i=1}^{n-1} X_i \sim N(0, n-1) \quad X_n \sim N(0, 1) \quad Y \quad X_1, \dots, X_{n-1} \quad \frac{Y}{\sqrt{n-1}} \sim N(0, 1) \quad X_n$$

$$\frac{Y/\sqrt{n-1}}{X_n} \sim C(0, 1).$$

$$Z = \frac{Y}{X_n} = \sqrt{n-1} \cdot \frac{Y/\sqrt{n-1}}{X_n}$$

$$aC(0, 1) = C(0, a)$$

$$Z \sim C(0, \sqrt{n-1}).$$

tip 5.22

I

 **1** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim P(\lambda)$ λ

$$E(X) = \lambda, \quad E(X^2) = \lambda^2 + \lambda$$

1. $\bar{X} = \lambda \quad \hat{\lambda}_1 = \bar{X}$

2. $\frac{1}{n} \sum_{i=1}^n X_i^2 = \lambda^2 + \lambda$

$$\hat{\lambda}_2 = \frac{-1 + \sqrt{1 + \frac{4}{n} \sum_{i=1}^n X_i^2}}{2}.$$

tip 5.23

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

 **2** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim B(k, p)$ p

$$X_1 \sim B(k, p)$$

$$E(X) = kp.$$

$$\bar{X} = kp.$$

p

$$\hat{p} = \frac{\bar{X}}{k}.$$

tip 5.24

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

 **3** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim U(\theta_1, \theta_2)$ θ_1, θ_2

$$E(X) = \frac{\theta_1 + \theta_2}{2} \quad E(X^2) = \frac{\theta_1^2 + \theta_1\theta_2 + \theta_2^2}{3} \quad m_2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

$$\theta_1 + \theta_2 = 2\bar{X}, \quad \theta_1^2 + \theta_1\theta_2 + \theta_2^2 = 3m_2.$$

$$(\theta_1 + \theta_2)^2 = \theta_1^2 + 2\theta_1\theta_2 + \theta_2^2$$

$$\theta_1\theta_2 = (2\bar{X})^2 - 3m_2 = 4\bar{X}^2 - 3m_2.$$

$$\theta_1, \theta_2 \quad t^2 - 2\bar{X}t + (4\bar{X}^2 - 3m_2) = 0$$

$$\hat{\theta}_{1,2} = \bar{X} \mp \sqrt{3(m_2 - \bar{X}^2)}.$$

 $[U(\theta, \theta)]$

$$E(X^k) = \int_{\theta_1}^{\theta_2} x^k \cdot \frac{1}{\theta_2 - \theta_1} dx = \frac{\theta_2^{k+1} - \theta_1^{k+1}}{(k+1)(\theta_2 - \theta_1)}.$$

$$(\theta_2 - \theta_1) a^2 - b^2 = (a - b)(a + b) \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$E(X) = \frac{\theta_1 + \theta_2}{2}, \quad E(X^2) = \frac{\theta_1^2 + \theta_1\theta_2 + \theta_2^2}{3}.$$



$$t^2 - St + P = 0$$

r_1, r_2

$$r_1 + r_2 = S, \quad r_1 r_2 = P.$$

5 $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim f(x)$

$$f(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right),$$

σ

$$f(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right)$$

0

$$E(X) = 0, \quad E(X^2) = 2\sigma^2.$$

$$E(X) = \sigma$$

$$\frac{1}{n} \sum_{i=1}^n X_i^2 = E(X^2) = 2\sigma^2,$$

$$\hat{\sigma} = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}.$$

σ

$$\hat{\sigma} = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}.$$

tip 5.25

σ



$\hat{\theta}$ is a function of θ and $X_1, \dots, X_n$ — $\hat{\theta}$ is a function of θ

5.4

7 $X_1, X_2, \dots, X_n$ i.i.d. θ

$$2. \quad f(x; \theta) = \frac{x}{\theta^2} e^{-x^2/\theta^2}, \quad x > 0, \theta > 0.$$

tip 5.26

$$L(\theta) = \prod_{i=1}^n \frac{x_i}{\theta^2} e^{-x_i^2/\theta^2} = \frac{\prod_{i=1}^n x_i}{\theta^{2n}} \exp\left(-\frac{\sum_{i=1}^n x_i^2}{\theta^2}\right)$$

$$\ln L(\theta) = \sum_{i=1}^n \ln x_i - 2n \ln \theta - \frac{\sum_{i=1}^n x_i^2}{\theta^2} \quad \frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{2n}{\theta} + \frac{2 \sum_{i=1}^n x_i^2}{\theta^3} = 0 \quad \sum_{i=1}^n x_i^2 = n\theta^2$$

$$\hat{\theta} = \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2}$$



8 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right)$

$$T \in \left[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}\right]$$

θ

$$f(x; \theta) = \begin{cases} 1 & \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) \quad x_i \in [\theta - \frac{1}{2}, \theta + \frac{1}{2}] \quad L(\theta) = 1 \quad L(\theta) = 0$$

$$\theta - \frac{1}{2} \leq x_{(1)} \quad x_{(n)} \leq \theta + \frac{1}{2}$$

$$x_{(n)} - \frac{1}{2} \leq \theta \leq x_{(1)} + \frac{1}{2} \quad T \in [X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}] \quad L(T) = 1 = \max_{\theta} L(\theta) \quad T = \theta$$

tip 5.27

$$x_i \in [\theta - 1/2, \theta + 1/2]$$

 **10** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x; \alpha, \beta)$

$$f(x; \alpha, \beta) = \begin{cases} \beta \exp\{-\beta(x - \alpha)\}, & x \geq \alpha, \\ 0, & \end{cases}$$

$$\beta > 0 \quad \alpha, \beta$$

$$L(\alpha, \beta) = \prod_{i=1}^n \beta e^{-\beta(x_i - \alpha)} \cdot \mathbf{1}(\alpha \leq x_{(1)}) = \beta^n e^{-\beta \sum_{i=1}^n (x_i - \alpha)} \cdot \mathbf{1}(\alpha \leq x_{(1)})$$

$$\beta \quad L(\alpha, \beta) \propto e^{n\beta\alpha} \quad \alpha \leq x_{(1)} \quad \hat{\alpha} = X_{(1)} \quad \hat{\alpha}$$

$$\ln L(\beta) = n \ln \beta - \beta \sum_{i=1}^n (x_i - x_{(1)})$$

$$\beta \quad \frac{d \ln L}{d \beta} = \frac{n}{\beta} - \sum_{i=1}^n (x_i - x_{(1)}) = 0$$

$$\hat{\beta} = \frac{n}{\sum_{i=1}^n (X_i - X_{(1)})}$$

tip 5.28

$$\alpha \quad .$$

 $[r, r+1] \quad X = r+1 \quad " \quad " \quad r \quad T > r \quad r+1 \quad " \quad r+1 \quad "$

$$P(X = r+1) = P(T > r) = \theta^r, \quad \sum_{k=1}^r \theta^{k-1}(1 - \theta) + \theta^r = 1 \quad \checkmark$$

tip 5.29

$$P(X = k) \quad k \quad 1 - \theta \quad k - 1 \quad \theta$$

 **11** $X_1, X_2, \dots, X_n$ i.i.d. X

$$P\{X = k\} = \theta^{k-1}(1 - \theta), \quad k = 1, 2, \dots, r;$$

$$P\{X = r+1\} = \theta^r.$$

$$M = \#\{i : X_i = r+1, i = 1, 2, \dots, n\},$$

θ MLE

$$\hat{\theta} = \frac{\sum_{i=1}^n X_i - n}{\sum_{i=1}^n X_i - M}.$$

k PMF

$$x_i \quad M = \#\{i : x_i = r+1\} \quad S = \sum_{i=1}^n x_i$$

$$L(\theta) = \prod_{x_i \leq r} \theta^{x_i-1}(1 - \theta) \cdot \prod_{x_i = r+1} \theta^r = \theta^{\sum_{x_i \leq r} (x_i-1) + rM} (1 - \theta)^{n-M}$$

$$x_i = r + 1 \quad \sum_{x_i=r+1} x_i = (r+1)M$$

$$\sum_{x_i \leq r} (x_i - 1) + rM = \underbrace{\sum_{x_i \leq r} x_i - (n - M) + rM}_{=\sum_{x_i \leq r} (x_i - 1)} = \underbrace{\sum_{x_i \leq r} x_i + (r+1)M - n}_{=S} = S - n$$

$$\ln L(\theta) = (S - n) \ln \theta + (n - M) \ln(1 - \theta)$$

$$\frac{d \ln L}{d\theta} = \frac{S - n}{\theta} - \frac{n - M}{1 - \theta} = 0 \quad (S - n)(1 - \theta) = (n - M)\theta$$

$$\hat{\theta} = \frac{S - n}{S - M} = \frac{\sum_{i=1}^n X_i - n}{\sum_{i=1}^n X_i - M}$$

tip 5.30

$$P\{X = k\} = \binom{x_i}{k} \theta^k (1 - \theta)^{x_i - k}$$

$$\bullet \quad x_i \leq r \quad \theta^{x_i - 1} (1 - \theta) \quad x_i = 1 \quad (1 - \theta)$$

$$\bullet \quad x_i = r + 1 \quad \theta^r$$

$$M = \#\{i : x_i = r + 1\} \geq r + 1 \quad 5, 6, 7, 8, \dots \quad r + 1 \quad M$$

$$\theta^r \quad M \quad rM$$

$$\prod_{x_i=r+1} \theta^r = \theta^{rM}$$

$$M \quad x_i$$

 **12** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x, \theta_1, \theta_2)$

$$f(x, \theta_1, \theta_2) = \begin{cases} \frac{\theta_1}{\theta_2} x^{\theta_1 - 1} \exp\left(-\frac{x^{\theta_1}}{\theta_2}\right), & x \geq 0, \\ 0, & \end{cases}$$

$$\theta_1 \quad \theta_2$$

$$L(\theta_2) = \prod_{i=1}^n \frac{\theta_1}{\theta_2} x_i^{\theta_1 - 1} \exp\left(-\frac{x_i^{\theta_1}}{\theta_2}\right)$$

$$\ln L(\theta_2) = n \ln \theta_1 - n \ln \theta_2 + (\theta_1 - 1) \sum_{i=1}^n \ln x_i - \frac{1}{\theta_2} \sum_{i=1}^n x_i^{\theta_1}$$

$$\theta_2$$

$$\frac{d \ln L}{d\theta_2} = -\frac{n}{\theta_2} + \frac{\sum_{i=1}^n x_i^{\theta_1}}{\theta_2^2} = 0$$

$$\theta_2$$

$$\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n X_i^{\theta_1}$$

tip 5.31

 **18** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, \sigma^2) \quad \sigma^2 \quad \mu \quad N(\nu, \tau^2)$

1. μ

2. μ

$$f(\mathbf{X}|\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \propto \exp\left(-\frac{n(\bar{x} - \mu)^2}{2\sigma^2}\right)$$

$$\pi(\mu) \propto \exp\left(-\frac{(\mu - \nu)^2}{2\tau^2}\right)$$

$$p(\mu|\mathbf{X}) \propto f(\mathbf{X}|\mu) \cdot \pi(\mu)$$

$$-\frac{n(\bar{x} - \mu)^2}{2\sigma^2} - \frac{(\mu - \nu)^2}{2\tau^2} = -\frac{1}{2} \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2} \right) \left(\mu - \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\nu}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}} \right)^2 +$$

$$\frac{1}{\tau_1^2} = \frac{n}{\sigma^2} + \frac{1}{\tau^2} \quad \mu_1 = \tau_1^2 \left(\frac{n\bar{x}}{\sigma^2} + \frac{\nu}{\tau^2} \right)$$

1.

$$\mu|\mathbf{X} \sim N(\mu_1, \tau_1^2)$$

2. μ Bayes

$$\hat{\mu} = \mu_1 = \frac{\frac{n}{\sigma^2}\bar{X} + \frac{1}{\tau^2}\nu}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

tip 5.32

$$\mu \quad \sigma \quad \mu$$

tip 5.33

$$l/\exp \quad \mu \quad \mu l$$



$$\sum_{i=1}^n (x_i - \mu)^2 = n(\bar{x} - \mu)^2 + \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\mu \propto \frac{1}{\sqrt{2\pi}\sigma}$$

$$f(\mathbf{X}|\mu) \propto \exp\left(-\frac{n(\bar{x} - \mu)^2}{2\sigma^2}\right)$$

20 Pareto $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(0, \theta)$ θ Pareto(a, b)

$$\pi(\theta) = \frac{ba^b}{\theta^{b+1}}, \quad a < \theta < \infty, \quad a > 0, \quad b > 0.$$

1. θ 2. θ

$$f(\mathbf{X}|\theta) = \prod_{i=1}^n \frac{1}{\theta} \cdot \mathbf{1}(\theta \geq x_{(n)}) = \theta^{-n} \cdot \mathbf{1}(\theta \geq x_{(n)})$$

$$\pi(\theta) \propto \theta^{-(b+1)} \cdot \mathbf{1}(\theta > a)$$

$$p(\theta|\mathbf{X}) \propto \theta^{-n} \cdot \theta^{-(b+1)} \cdot \mathbf{1}(\theta \geq \max(x_{(n)}, a)) = \theta^{-(n+b+1)} \cdot \mathbf{1}(\theta \geq \max(x_{(n)}, a))$$

$$a' = \max(x_{(n)}, a) \quad b' = n + b$$

1.

$$\theta|\mathbf{X} \sim \text{Pareto}(\max(x_{(n)}, a), n + b)$$

2. θ Bayes

$$\hat{\theta} = \frac{(n + b) \max(X_{(n)}, a)}{n + b - 1}$$

tip 5.34

$$a \quad b$$



$$p(\theta | \mathbf{X}) \propto \theta^{-n} \cdot \theta^{-(b+1)} \cdot \mathbf{1}(\theta \geq \max(x_{(n)}, a)).$$

$$p(\theta | \mathbf{X}) \propto \theta^{-(n+b+1)} \cdot \mathbf{1}(\theta \geq \max(x_{(n)}, a)).$$

$$\text{Pareto}(a', b') \propto \theta^{-(b'+1)} \cdot \mathbf{1}(\theta > a') \quad b' = n + b \quad a' = \max(x_{(n)}, a)$$

$$\theta | \mathbf{X} \sim \text{Pareto}(\max(x_{(n)}, a), n + b).$$

$$E[\theta] = \frac{b'a'}{b'-1} \quad \text{Bayes}$$

$$\hat{\theta} = \frac{(n+b) \max(x_{(n)}, a)}{n+b-1}.$$

5.5



$$1 \quad X \sim P(\lambda) \quad 1 \quad g(\lambda) = e^{-2\lambda}$$

tip 5.35

$$h(X) = X \quad e^{-2\lambda}$$

$$\hat{g}(X) = g(\lambda) = e^{-2\lambda}$$

$$E[\hat{g}(X)] = \sum_{k=0}^{\infty} \hat{g}(k) \cdot \frac{\lambda^k e^{-\lambda}}{k!} = e^{-2\lambda}$$

$$X = g(X)$$

$$E[h(X)] = \sum_{k=0}^{\infty} h(k) \cdot P(X = k) = \sum_{k=0}^{\infty} h(k) \cdot \frac{\lambda^k e^{-\lambda}}{k!}$$

$$e^{-2\lambda} = e^{-\lambda} \cdot e^{-\lambda} \quad k=0 \quad P(X=0) = e^{-\lambda}$$

$$\hat{g}(k) = (-1)^k$$

$$E[(-1)^X] = \sum_{k=0}^{\infty} (-1)^k \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} = e^{-\lambda} \cdot e^{-\lambda} = e^{-2\lambda} \checkmark$$

$$\hat{g}(X) = (-1)^X \quad g(\lambda) = e^{-2\lambda}$$

$$g(\lambda) = e^{-2\lambda} \in (0, 1) \quad (-1)^X = \pm 1 \quad X \quad -1 < 0 \quad g(\lambda)$$



$$P(X=0)$$

tip 5.36 (e)

$$k \quad k \quad l \quad e \quad .e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$



$$2 \quad X \sim B(n, p) \quad g(p) = \frac{1}{1+p^2}$$

$$g(p) = \frac{1}{1+p^2} \quad \hat{g}(X)$$

$$E[\hat{g}(X)] = \sum_{k=0}^n \hat{g}(k) \binom{n}{k} p^k (1-p)^{n-k} = \frac{1}{1+p^2}$$

$$p \in (0, 1) \quad p \quad n \quad \frac{1}{1+p^2} \quad g(p) = \frac{1}{1+p^2}$$

tip 5.37



$$E[\hat{g}(\mathbf{X})] = g(\theta), \quad \forall \theta$$

$$E[\hat{g}(\mathbf{X})] = g(\theta), \quad \forall \theta$$

θ

- $g(\theta)$
- $\theta \rightarrow g(\theta)$

$g(\theta)$



3 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, 1)$ $g(\mu) = |\mu|$

$$g(\mu) = |\mu| \quad \hat{g}(\mathbf{X})$$

$$E[\hat{g}(\mathbf{X})] = |\mu|, \quad \forall \mu \in \mathbb{R}$$

$$E[\hat{g}(\mathbf{X})] = \int_{-\infty}^{\infty} \hat{g}(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}} dx$$

$$g(\mu) = |\mu| \quad \mu = 0$$



4 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(\theta_1, \theta_2)$ θ_1, θ_2

$$U(\theta_1, \theta_2)$$

$$E[X_{(1)}] = \frac{n\theta_1 + \theta_2}{n+1}, \quad E[X_{(n)}] = \frac{\theta_1 + n\theta_2}{n+1}$$

$$\theta_1 = \frac{nE[X_{(1)}] - E[X_{(n)}]}{n-1}, \quad \theta_2 = \frac{nE[X_{(n)}] - E[X_{(1)}]}{n-1}$$

θ_1, θ_2

$$\hat{\theta}_1 = \frac{nX_{(1)} - X_{(n)}}{n-1}, \quad \hat{\theta}_2 = \frac{nX_{(n)} - X_{(1)}}{n-1}$$

tip 5.38

$$E(X_1) = X_1, \quad \theta_1 = X_1, \quad X_1$$



$$\hat{\theta}_1 = \frac{nX_{(1)} - X_{(n)}}{n-1}$$

$$E[\hat{\theta}_1] = \frac{nE[X_{(1)}] - E[X_{(n)}]}{n-1} = \theta_1$$

θ_1

$$\hat{\theta}_1 = X_{(1)}$$

tip 5.39 ()

$$f(x) = \frac{1}{\theta_2 - \theta_1}, \quad F(x) = \frac{x - \theta_1}{\theta_2 - \theta_1}$$

$$E[X_{(1)}] = \frac{n\theta_1 + \theta_2}{n+1}, \quad E[X_{(n)}] = \frac{\theta_1 + n\theta_2}{n+1}$$



$$E[X_{(1)}] = E[X_{(n)}] \quad X_{(1)} = X_{(n)}$$



$$\hat{\theta} = \theta^* \quad \theta \quad \tilde{\theta} = a\hat{\theta} + b\theta^* \quad a, b \quad \tilde{\theta} = \theta$$

$$E[\tilde{\theta}] = E[a\hat{\theta} + b\theta^*] = aE[\hat{\theta}] + bE[\theta^*] = a\theta + b\theta = (a+b)\theta$$

$$\tilde{\theta} = \theta \quad E[\tilde{\theta}] = \theta$$

$$a + b = 1$$

tip 5.40

$$\tilde{\theta} = \theta$$

6. $X \sim U(0, \theta)$ $X_1, X_2, \dots, X_n$ i.i.d. $X \sim U(0, \theta)$ θ^2, θ^{-1}

$$E[X_{(n)}] = \frac{n\theta}{n+1}$$

$$E\left[\frac{n+1}{n}X_{(n)}\right] = \theta$$

$$\hat{\theta} = \frac{n+1}{n}X_{(n)}$$

$$f_{(n)}(x) = \frac{nx^{n-1}}{\theta^n}$$

$$E[X_{(n)}^2] = \int_0^\theta x^2 \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n\theta^2}{n+2}$$

$$\theta^2$$

$$\hat{\theta}^2 = \frac{n+2}{n}X_{(n)}^2$$

$$\theta^{-1}$$

$$E[X_{(n)}^{-1}] = \int_0^\theta x^{-1} \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{\theta(n-1)}$$

$$\theta^{-1}$$

$$\widehat{\theta^{-1}} = \frac{n-1}{n}X_{(n)}^{-1} = \frac{n-1}{nX_{(n)}}$$

tip 5.41

$$\bullet \quad X_i \sim U(0, \theta) \quad \theta^2, \theta^{-1}$$

$$\bullet \quad \theta \quad X_{(n)}$$

$$\bullet \quad E[X_{(n)}^2] = \frac{2}{3}\theta^2 \quad E[X_{(n)}^{-1}] = \frac{1}{\theta}$$

$$1. \quad E[X_{(n)}^k] = \frac{k}{k+1}\theta^k$$

$$2. \quad \theta^2, \theta^{-1}$$

5.6

1

$$I(\theta) = \text{Var}\left(\frac{\partial}{\partial\theta} \ln f(X; \theta)\right) = -E\left(\frac{\partial^2}{\partial\theta^2} \ln f(X; \theta)\right)$$

tip 5.42

$$S(X) = \frac{\partial}{\partial\theta} \ln f(X; \theta) \quad S(X) = \frac{\partial}{\partial\theta} \ln f(X; \theta) \quad S(X) = \frac{\partial}{\partial\theta} \ln f(X; \theta) \quad S(X) = \frac{\partial}{\partial\theta} \ln f(X; \theta)$$

$$S(X; \theta) = \frac{\partial}{\partial\theta} \ln f(X; \theta) \quad E[S(X; \theta)] = 0$$

$$\int f(x; \theta) dx = 1 \quad \theta$$

$$\int \frac{\partial}{\partial \theta} f(x; \theta) dx = 0.$$

$$\frac{\partial}{\partial \theta} f = \frac{\partial}{\partial \theta} e^{\ln f} = e^{\ln f} \cdot \frac{\partial}{\partial \theta} \ln f = f \cdot S$$

$$\frac{\partial}{\partial \theta} f(x; \theta) = f(x; \theta) \cdot S(x; \theta),$$

$$E[S(X; \theta)] = 0$$

$$\int S(x; \theta) f(x; \theta) dx = 0 \quad \theta$$

$$\int \left(\frac{\partial^2}{\partial \theta^2} \ln f \cdot f + S^2 \cdot f \right) dx = 0,$$

$$E \left(\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right) + E(S^2) = 0.$$

$$E[S] = 0 \quad \text{Var}(S) = E(S^2)$$

$$I(\theta) = \text{Var} \left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right) = -E \left(\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right).$$

tip 5.43

$$f \quad f$$



$$\frac{\partial}{\partial \theta} (S \cdot f) = \underbrace{\frac{\partial S}{\partial \theta} \cdot f}_{Sf} + \underbrace{S \cdot \frac{\partial f}{\partial \theta}}_{S \cdot \frac{\partial f}{\partial \theta}}$$

- $Sf = \frac{\partial S}{\partial \theta} \cdot f = \frac{\partial^2}{\partial \theta^2} \ln f \cdot f \rightarrow E \left[\frac{\partial^2}{\partial \theta^2} \ln f \right]$
- $S \cdot \frac{\partial f}{\partial \theta} = S^2 \cdot f \rightarrow E[S^2]$

tip 5.44

$$I\theta = \text{var} \quad E(S) = 0 \quad 0 \quad I = -\ln S \quad 1/s \quad f \cdot S = f/d\theta$$



$$W = \theta \quad W$$

$$(1) \quad W$$

$$(2) \quad E_\theta W = \theta, \forall \theta \in \Theta$$

$$(3) \quad (1), (2) \quad W^*$$

$$\text{Var}_\theta(W) \leq \text{Var}_\theta(W^*), \quad \forall \theta \in \Theta,$$

$$W = \theta \quad X \sim N(\mu, \sigma^2) \quad (\mu, \sigma^2) \in \Theta = \{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 > 0\} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \mu$$

$$(1) \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1)$$

$$(2) \quad X \sim N(\mu, \sigma^2)$$

$$E_\mu \bar{X} = \frac{1}{n} \sum_{i=1}^n EX_i = \frac{1}{n} \cdot n\mu = \mu, \quad \forall (\mu, \sigma^2) \in \Theta,$$

$$\bar{X} = \mu \quad (2)$$

$$(3) \quad W^* = \sum_{i=1}^n c_i X_i \quad (I)(2)$$

$$EW^* = \sum_{i=1}^n c_i \mu = \mu \cdot \sum_{i=1}^n c_i = \mu,$$

$$\sum_{i=1}^n c_i = 1 \quad X_1, \dots, X_n$$

$$\text{Var}_\theta(W^*) = \sigma^2 \sum_{i=1}^n c_i^2.$$

Cauchy-Schwarz

$$\sum_{i=1}^n c_i^2 \geq \frac{(\sum_{i=1}^n c_i)^2}{n} = \frac{1}{n},$$

$$c_1 = \dots = c_n = \frac{1}{n} \quad W^* = \bar{X}$$

$$\text{Var}_\theta(W^*) \geq \frac{\sigma^2}{n} = \text{Var}_\theta(\bar{X}), \quad \forall \theta \in \Theta.$$

$$(3) \quad \bar{X} \quad \mu$$

$$X_i \quad EX_i = \mu \quad \text{Var}(X_i) = \sigma^2 < \infty \quad \text{Gauss-Markov} \quad \bar{X} \quad \mu \quad \text{BLUE}$$

tip 5.45

$$\text{Var}(W^*) = \text{Var}\left(\sum_{i=1}^n c_i X_i\right) = \sum_{i=1}^n c_i^2 \text{Var}(X_i)$$

$$\text{Var}(X_i) = \sigma^2 \quad \sigma^2 \sum_{i=1}^n c_i^2$$



[Gauss-Markov] X_i

1.

2. $EX_i = \mu$

3. $\text{Var}(X_i) = \sigma^2 < \infty$

$\bar{X} \quad \mu \quad \text{BLUE Best Linear Unbiased Estimator}$

tip 5.46

umvue

tip 5.47

(I)

$$W^* \quad W^* = \sum_{i=1}^n c_i X_i$$

$$EW^* = \sum_{i=1}^n c_i E(X_i),$$

$$E(X_i) = \mu \quad EW^* = \mu \cdot \sum_{i=1}^n c_i \quad \mu \cdot \sum_{i=1}^n c_i = 1$$

$$\text{Var}_\theta(W^*) \quad X_1, \dots, X_n \quad 0$$

$$\text{Var}_\theta(W^*) = \sigma^2 \sum_{i=1}^n c_i^2.$$

tip 5.48

5.1 (Cauchy-Schwarz)

$$a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$$

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

$$\frac{a_1}{b_1} = \dots = \frac{a_n}{b_n} \quad a_i = b_i$$

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$



11 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim P(\lambda)$

(1) $\bar{X} \sim \lambda$ UMVUE

(2) $\tilde{\lambda}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X} \sim \lambda^2$ UMVUE

(I) $P(\lambda)$

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

$$T = \sum_{i=1}^n X_i$$

$$\bar{X} = T/n$$

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \lambda,$$

$\bar{X} \sim \lambda$ Lehmann-Scheffé UMVUE $\bar{X} \sim \lambda$ UMVUE

tip 5.49

$$n \ln \lambda = - \sum_{i=1}^n \ln \lambda = - \sum_{i=1}^n \ln \frac{1}{\lambda} = - \ln \left(\frac{1}{\lambda} \right)^n = - \ln \left(\frac{1}{\lambda} \right)^T$$

tip 5.50 ()

/PMF

$$f(x; \theta) = h(x) \cdot \exp\{Q(\theta) \cdot t(x) - A(\theta)\}$$

$$T = \sum t(X_i)$$

$$P(\lambda)$$

$$P(X = x; \lambda) = \underbrace{\frac{1}{x!}}_{h(x)} \cdot \exp \left\{ \underbrace{\ln \lambda}_{Q(\theta)} \cdot \underbrace{x}_{t(x)} - \underbrace{\lambda}_{A(\theta)} \right\}$$

$$T = \sum X_i$$



$$f(x; \theta) = h(x) \cdot \exp\{Q(\theta) \cdot t(x) - A(\theta)\}$$

	$t(x)$	T
$P(\lambda)$	x	$\sum X_i$
$N(\mu, \sigma^2)$	x	$\sum X_i$
$\text{Exp}(\lambda)$	x	$\sum X_i$
$N(\mu, \sigma^2)$	(x, x^2)	$(\sum X_i, \sum X_i^2)$

$$(2) \quad P(\lambda) \quad E(X_i) = \lambda \quad \text{Var}(X_i) = \lambda \quad E(X_i^2) = \lambda^2 + \lambda$$

$$E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \lambda^2 + \lambda,$$

$$E(\bar{X}) = \lambda,$$

$$E(\tilde{\lambda}^2) = (\lambda^2 + \lambda) - \lambda = \lambda^2.$$

$$\tilde{\lambda}^2 = \lambda^2$$

$$UMVUE \quad T = \sum_{i=1}^n X_i \sim P(n\lambda) \quad \tilde{\lambda}^2 = T$$

$$\tilde{\lambda}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}.$$

$$\sum_{i=1}^n X_i^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n\bar{X}^2 = (n-1)s_n^2 + \frac{T^2}{n},$$

$$s_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \tilde{\lambda}^2 = T \quad \sum_{i=1}^n X_i^2 \quad \tilde{\lambda}^2 = T$$

$$T = \lambda^2 \quad UMVUE \quad g(T) \quad E[g(T)] = \lambda^2$$

$$g(T) = \frac{T(T-1)}{n^2} = \frac{(\sum_{i=1}^n X_i)(\sum_{i=1}^n X_i - 1)}{n^2}$$

$$E[g(T)] = \frac{E[T(T-1)]}{n^2} = \frac{n^2 \lambda^2}{n^2} = \lambda^2,$$

$$g(T) \quad g(T) = \lambda^2 \quad UMVUE$$

$$\tilde{\lambda}^2 = \lambda^2 \quad UMVUE$$

tip 5.51

$$\lambda = \lambda^2 \quad T = \sum X_i \quad T/n = \lambda \quad \lambda^2 = T \quad g(T) = E[g(T)] = \lambda^2$$

tip 5.52 (fisher)

Fisher

$$\ell(\theta) = \sum_{i=1}^n \ln f(x_i; \theta),$$

θ

$$\frac{\partial^2 \ell}{\partial \theta^2} = \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \ln f(x_i; \theta).$$

$X_1, \dots, X_n$

$$-E \left[\frac{\partial^2 \ell}{\partial \theta^2} \right] = -\sum_{i=1}^n E \left[\frac{\partial^2}{\partial \theta^2} \ln f(X_i; \theta) \right] = nI(\theta).$$

$$n \quad \text{Fisher} \quad n \quad \text{C-R} \quad \frac{1}{nI(\theta)} \quad n$$

 **12** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, \sigma^2)$ $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ μ σ^2 UMVUE μ σ^2 C-R

tip 5.53 (fisher)

$$L(\theta) = \ln \frac{1}{f}$$

$$\frac{\partial}{\partial \theta} \ln f \quad \text{score function} \quad \theta \quad X \quad \text{''} \quad \text{''}$$

$$I(\theta)_{\text{fisher}} \quad 0$$

tip 5.54 ()

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$X \sim N(\mu, \sigma^2)$$

$$\ln f(x; \mu, \sigma^2) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}.$$

tip 5.55

$$\mu \quad \sigma \quad \mu \quad C-R \quad \mu \quad \ln \quad \mu \quad Fisher \quad 0$$

$$\mu \quad C-R \quad \sigma^2 \quad \mu$$

$$\frac{\partial}{\partial \mu} \ln f = \frac{x - \mu}{\sigma^2}.$$

Fisher

$$I(\mu) = E \left[\left(\frac{\partial}{\partial \mu} \ln f \right)^2 \right] = E \left[\frac{(X - \mu)^2}{\sigma^4} \right] = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

$$\mu \quad C-R$$

$$\frac{1}{nI(\mu)} = \frac{\sigma^2}{n}.$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad C-R \quad \bar{X} \quad \mu \quad UMVUE$$

$$\sigma^2 \quad C-R \quad \mu \quad \lambda = \sigma^2 \quad \lambda$$

$$\frac{\partial}{\partial \lambda} \ln f = -\frac{1}{2\lambda} + \frac{(x - \mu)^2}{2\lambda^2}.$$

Fisher

$$I(\lambda) = E \left[\left(-\frac{1}{2\lambda} + \frac{(X - \mu)^2}{2\lambda^2} \right)^2 \right].$$

$$(X - \mu)^2 \sim \lambda \chi^2(1) \quad E(X - \mu)^2 = \lambda \quad E(X - \mu)^4 = 3\lambda^2$$

$$I(\lambda) = \frac{1}{4\lambda^2} - \frac{E(X - \mu)^2}{2\lambda^3} + \frac{E(X - \mu)^4}{4\lambda^4} = \frac{1}{4\lambda^2} - \frac{1}{2\lambda^2} + \frac{3}{4\lambda^2} = \frac{1}{2\lambda^2} = \frac{1}{2\sigma^4}.$$

$$\sigma^2 \quad C-R$$

$$\frac{1}{nI(\sigma^2)} = \frac{2\sigma^4}{n}.$$

$$\text{Var}(s^2) = \frac{2\sigma^4}{n-1} > \frac{2\sigma^4}{n} \quad s^2 \quad C-R \quad \sigma^2 \quad UMVUE$$

tip 5.56

$$\chi^2(1) - \mu \quad \lambda \quad \sigma^2 \quad \lambda \quad \chi^2(1)$$

tip 5.57

$$UMVUE \quad \text{---} \quad C-R \quad \text{---} \quad UMVUE \quad UMVUE \quad 2\pi$$

$$\ln(2\pi\lambda) = \ln(2\pi) + \ln \lambda \quad \lambda \quad \ln(2\pi) \quad \frac{1}{\lambda}$$

 **14** 3.3.3 $\hat{g}(X_1, X_2, \dots, X_n) \quad g(\theta)$

$$E_{\theta} \hat{g}(X_1, X_2, \dots, X_n) = g(\theta) + b(\theta), \quad \forall \theta \in \Theta \subset \mathbf{R},$$

$$b(\theta)$$

$$\text{Var}_{\theta}(\hat{g}(X_1, X_2, \dots, X_n)) \geq [b'(\theta) + g'(\theta)]^2 / nI(\theta).$$

$$\hat{g} = \hat{g}(X_1, X_2, \dots, X_n)$$

$$E_{\theta} \hat{g} = g(\theta) + b(\theta).$$

θ 3.3.3

$$\frac{d}{d\theta} E_{\theta} \hat{g} = g'(\theta) + b'(\theta).$$

$$\frac{d}{d\theta} E_{\theta} \hat{g} = \frac{d}{d\theta} \int \hat{g} \cdot L(\theta; \mathbf{x}) d\mathbf{x} = \int \hat{g} \cdot \frac{\partial \ln L(\theta; \mathbf{x})}{\partial \theta} \cdot L(\theta; \mathbf{x}) d\mathbf{x} = E_{\theta} \left[\hat{g} \cdot \frac{\partial \ln L}{\partial \theta} \right],$$

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n f(x_i; \theta)$$

$$E_{\theta} \left[\frac{\partial \ln L}{\partial \theta} \right] = 0,$$

$$E_{\theta} \left[\hat{g} \cdot \frac{\partial \ln L}{\partial \theta} \right] = \text{Cov}_{\theta} \left(\hat{g}, \frac{\partial \ln L}{\partial \theta} \right).$$

$$\text{Cov}_{\theta} \left(\hat{g}, \frac{\partial \ln L}{\partial \theta} \right) = g'(\theta) + b'(\theta).$$

Cauchy-Schwarz

$$\left[\text{Cov}_{\theta} \left(\hat{g}, \frac{\partial \ln L}{\partial \theta} \right) \right]^2 \leq \text{Var}_{\theta}(\hat{g}) \cdot \text{Var}_{\theta} \left(\frac{\partial \ln L}{\partial \theta} \right).$$

Fisher i.i.d.

$$\text{Var}_{\theta} \left(\frac{\partial \ln L}{\partial \theta} \right) = nI(\theta).$$

$$[g'(\theta) + b'(\theta)]^2 \leq \text{Var}_{\theta}(\hat{g}) \cdot nI(\theta),$$

$$\text{Var}_{\theta}(\hat{g}) \geq \frac{[g'(\theta) + b'(\theta)]^2}{nI(\theta)}.$$

tip 5.58 ()

$$\ln L = \theta \cdot \frac{\partial \ln L}{\partial \theta} = \frac{1}{L} \cdot \frac{\partial L}{\partial \theta} \quad \frac{\partial L}{\partial \theta} = L \cdot \frac{\partial \ln L}{\partial \theta} \quad L \quad L$$



$I = I$

$$\text{Cov}(X, Y) = E[XY] - E[X] \cdot E[Y]$$

$$E[Y] = 0$$

$$\text{Cov}(X, Y) = E[XY]$$

tip 5.59

$$E_{\theta} \hat{g} = g(\theta) + b(\theta) \quad g(\theta) = b(\theta) \quad E_{\theta} \hat{g} = \text{Fisher} \quad f = f = L$$

0

tip 5.60 (-)

$$\|\mathbf{u} \cdot \mathbf{v}\| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

$$\left(\sum a_i b_i\right)^2 \leq \left(\sum a_i^2\right) \left(\sum b_i^2\right)$$

$$\left(E[(X - EX)(Y - EY)]\right)^2 \leq E[(X - EX)^2] \cdot E[(Y - EY)^2]$$

15 $\hat{g}_1(\mathbf{X})$ $\hat{g}_2(\mathbf{X})$ $g(\theta)$ $\hat{g}_1(\mathbf{X})$ $g(\theta)$ UMVUE $\hat{g}_1(\mathbf{X})$ $\hat{g}_2(\mathbf{X})$

$$\sqrt{\text{Var}(\hat{g}_1)\text{Var}(\hat{g}_2)} \geq 0 \quad \text{Cov}_\theta(\hat{g}_1, \hat{g}_2) \geq 0$$

$$\hat{g}_\alpha = (1 - \alpha)\hat{g}_1 + \alpha\hat{g}_2 \quad (1 - \alpha) + \alpha = 1 \quad \hat{g}_\alpha \quad g(\theta)$$

$$\hat{g}_1 \quad \text{UMVUE} \quad \alpha \in \mathbb{R}$$

$$\text{Var}(\hat{g}_1) \leq \text{Var}(\hat{g}_\alpha)$$

$$D_1 = \text{Var}(\hat{g}_1) \quad D_2 = \text{Var}(\hat{g}_2) \quad C = \text{Cov}(\hat{g}_1, \hat{g}_2)$$

$$\varphi(\alpha) := \alpha^2(D_1 - 2C + D_2) - 2\alpha(D_1 - C) \geq 0 \quad \forall \alpha \in \mathbb{R}$$

$$\alpha > 0 \quad \alpha$$

$$\alpha(D_1 - 2C + D_2) - 2(D_1 - C) \geq 0$$

$$\alpha \rightarrow 0^+$$

$$-2(D_1 - C) \geq 0 \implies C \geq D_1 \geq 0$$

$$\text{Cov}_\theta(\hat{g}_1, \hat{g}_2) \geq 0$$

tip 5.61

$$\rho(\hat{g}_1, \hat{g}_2) = \frac{\text{Cov}(\hat{g}_1, \hat{g}_2)}{\sqrt{\text{Var}(\hat{g}_1) \cdot \text{Var}(\hat{g}_2)}}$$

$$\rho \geq 0 \quad \geq 0 \quad \text{Cov}(\hat{g}_1, \hat{g}_2) \geq 0$$

tip 5.62

$$\text{Cov}(\hat{g}_1, \hat{g}_2)$$

$$\hat{g}_1 \quad \text{UMVUE} \quad \text{Cov} \quad \hat{g}_1 \quad \hat{g}_2 \quad \text{UMVUE}$$

$$\hat{g}_\alpha = (1 - \alpha)\hat{g}_1 + \alpha\hat{g}_2$$

$$(1 - \alpha) + \alpha = 1$$

16

5.1 3.2.12 0-1 $X_1, X_2, \dots, X_n$ i.i.d. (0-1) $B(1, \theta)$ θ Beta Beta(a, b)

$$\pi(\theta) = \begin{cases} \frac{1}{B(a, b)} \theta^{a-1} (1 - \theta)^{b-1}, & \theta \in (0, 1); \\ 0, & \theta \notin (0, 1), \end{cases}$$

$$a > 0, b > 0 \quad B(a, b) \quad \text{Beta} \quad \theta$$

$$0-1 \quad 3.2.12 \quad \theta$$

$$\hat{\theta} = \frac{\sum_{i=1}^n X_i + a}{n + a + b}.$$

(1)

$$S = \sum_{i=1}^n X_i \sim B(n, \theta) \quad ES = n\theta \quad \text{Var}(S) = n\theta(1 - \theta)$$

$$E\hat{\theta} - \theta = \frac{n\theta + a}{n + a + b} - \theta = \frac{a - (a + b)\theta}{n + a + b}.$$

$$\text{Var}(\hat{\theta}) = \frac{\text{Var}(S)}{(n+a+b)^2} = \frac{n\theta(1-\theta)}{(n+a+b)^2}.$$

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (E\hat{\theta} - \theta)^2 = \frac{n\theta(1-\theta) + [a - (a+b)\theta]^2}{(n+a+b)^2}.$$

tip 5.63

$$n\theta \quad S \quad I.$$

tip 5.64

$$S$$



$$(\hat{\theta} - \theta) = (\hat{\theta} - E\hat{\theta}) + (E\hat{\theta} - \theta) \quad 0$$

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (E\hat{\theta} - \theta)^2.$$

$$S \sim B(n, \theta) \quad ES = n\theta \quad \text{Var}(S) = n\theta(1-\theta)$$

$$E(cS + d) = cES + d, \quad \text{Var}(cS) = c^2\text{Var}(S)$$

$$\hat{\theta} = \frac{S+a}{n+a+b} \quad \text{MSE}$$

5.6.1 3.4



$$1 \quad X_n \xrightarrow{d} X \quad Y_n \xrightarrow{a.s.} a > 0$$

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{a}.$$

$$Y_n \xrightarrow{a.s.} a \quad Y_n \xrightarrow{P} a$$

$$\frac{1}{Y_n} \xrightarrow{P} \frac{1}{a}.$$

$$\text{Slutsky} \quad X_n \xrightarrow{d} X \quad Z_n \xrightarrow{P} c \quad X_n Z_n \xrightarrow{d} cX \quad Z_n = \frac{1}{Y_n} \quad c = \frac{1}{a}$$

$$\frac{X_n}{Y_n} = X_n \cdot \frac{1}{Y_n} \xrightarrow{d} \frac{1}{a} \cdot X = \frac{X}{a}.$$

tip 5.65*Slutsky**Slutsky*

$$1 \quad \{\hat{\theta}_n\} \quad \lim_{n \rightarrow \infty} E_{\theta}(\hat{\theta}_n - \theta)^2 = 0 \quad \forall \theta \in \Theta \quad \hat{\theta}_n \rightarrow \theta$$

$$\text{Chebyshev} \quad \varepsilon > 0$$

$$P(|\hat{\theta}_n - \theta| \geq \varepsilon) \leq \frac{E_{\theta}(\hat{\theta}_n - \theta)^2}{\varepsilon^2}.$$

$$\lim_{n \rightarrow \infty} E_{\theta}(\hat{\theta}_n - \theta)^2 = 0$$

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0,$$

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \hat{\theta}_n \rightarrow \theta$$

tip 5.66*MSE*

$$2 \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim U(0, \theta) \quad \hat{\theta}_n = X_{(n)} \rightarrow \theta$$

$X_{(n)}$

$$f_{X_{(n)}}(x) = \frac{nx^{n-1}}{\theta^n}, \quad 0 < x < \theta.$$

$$E(X_{(n)}) = \int_0^\theta x \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{n+1}\theta,$$

$$E(X_{(n)}^2) = \int_0^\theta x^2 \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{n+2}\theta^2.$$

$$E_\theta(X_{(n)} - \theta)^2 = E(X_{(n)}^2) - 2\theta E(X_{(n)}) + \theta^2 = \frac{n}{n+2}\theta^2 - \frac{2n}{n+1}\theta^2 + \theta^2 = \frac{2\theta^2}{(n+1)(n+2)} \rightarrow 0.$$

 $X_{(n)} \quad \theta$ **tip 5.67**

$$\text{MSE} = E[(X_{(n)} - \theta)^2] \rightarrow 0$$



4 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim P(\lambda)$ $\bar{X} \quad s_n^2 \quad \lambda$
 $P(\lambda) \quad E(X_1) = \lambda \quad \text{Var}(X_1) = \lambda \quad E(X_1^2) = \lambda^2 + \lambda$
 $\bar{X} \quad \bar{X} \xrightarrow{P} E(X_1) = \lambda \quad \bar{X} \quad \lambda$
 s_n^2

$$s_n^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2.$$

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} E(X_1^2) = \lambda^2 + \lambda \quad \bar{X}^2 \xrightarrow{P} \lambda^2$$

$$s_n^2 \xrightarrow{P} (\lambda^2 + \lambda) - \lambda^2 = \lambda,$$

 $s_n^2 \quad \lambda$ **tip 5.68**

5 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(0, \theta)$

$$T(X_1, X_2, \dots, X_n) = \sqrt[n]{X_1 X_2 \cdots X_n}$$

$$g(\theta) = \frac{\theta}{e} \quad Y_i = \ln X_i \quad X_i \sim U(0, \theta)$$

$$E(Y_i) = \int_0^\theta \ln x \cdot \frac{1}{\theta} dx = \ln \theta - 1,$$

$$\text{Var}(Y_i) = E(Y_i^2) - (EY_i)^2 < \infty.$$

$$\frac{1}{n} \sum_{i=1}^n \ln X_i \xrightarrow{P} \ln \theta - 1 = \ln \frac{\theta}{e}.$$

$$T = \exp\left(\frac{1}{n} \sum_{i=1}^n \ln X_i\right) = \sqrt[n]{X_1 X_2 \cdots X_n} \xrightarrow{P} \frac{\theta}{e} = g(\theta),$$

 $T \quad g(\theta)$

tip 5.69

$$T = \ln \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) \xrightarrow{P} E[Y_1]$$

$$E[Y_1] = \ln \theta - 1$$

6 $X_1, X_2, \dots, X_n$

$$T(X_1, X_2, \dots, X_n) = \frac{2}{n(n+1)} \sum_{i=1}^n i X_i$$

$$\mu = E(X_1) = T$$

$$E(T) = \frac{2}{n(n+1)} \sum_{i=1}^n i \cdot E(X_i) = \frac{2\mu}{n(n+1)} \sum_{i=1}^n i = \frac{2\mu}{n(n+1)} \cdot \frac{n(n+1)}{2} = \mu.$$

$$T \quad \text{Var}(X_1) = \sigma^2 < \infty$$

$$\text{Var}(T) = \frac{4}{n^2(n+1)^2} \sum_{i=1}^n i^2 \sigma^2 = \frac{4\sigma^2}{n^2(n+1)^2} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{2\sigma^2(2n+1)}{3n(n+1)} \rightarrow 0.$$

$$\text{Chebyshev} \quad T \xrightarrow{P} \mu \quad T$$

tip 5.70

Chebyshev " " " "

$$P(|T - \mu| \geq \varepsilon) \leq \frac{E(T - \mu)^2}{\varepsilon^2} = \frac{\text{MSE}}{\varepsilon^2}$$

$$\rightarrow 0 \quad \text{MSE} = \text{Var}(T) + \text{Bias}^2(T) \rightarrow 0$$

- $E(T) = \mu \Rightarrow \text{Bias} = 0$
- $\text{Var}(T) \rightarrow 0$

$$T \quad \text{LLN} \quad + \text{Chebyshev}$$

8 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x, \theta) = \frac{1}{\theta} e^{-x/\theta} \quad x > 0 \quad \theta > 0$ MLE $N\left(\theta, \frac{\theta^2}{n}\right)$

$$\ell(\theta) = -n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n X_i.$$

$$\ell'(\theta) = 0 \quad \text{MLE} \quad \hat{\theta} = \bar{X}$$

$$E(X_1) = \theta \quad \text{Var}(X_1) = \theta^2$$

$$\sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0, \theta^2),$$

$$\hat{\theta} = \bar{X} \xrightarrow{d} N\left(\theta, \frac{\theta^2}{n}\right).$$

$$\theta \text{ MLE} \quad N\left(\theta, \frac{\theta^2}{n}\right)$$

tip 5.71

$$0 \text{ MLE} \quad 0 \text{ } \theta^2$$

5.7

9 3.3.11 $\lambda^2 \quad \bar{X}^2 \quad \tilde{\lambda}^2$
 $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} P(\lambda) \quad 3.3.11 \quad \lambda^2$

$$\tilde{\lambda}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}.$$

$\bar{X}^2 \quad \text{Delta} \quad g(x) = x^2 \quad g'(x) = 2x \quad \text{Poisson} \quad \text{Var}(X) = \lambda$
 $\sqrt{n}(\bar{X}^2 - \lambda^2) \xrightarrow{d} N(0, 4\lambda^3),$

$\bar{X}^2 \quad \frac{4\lambda^3}{n}$
 $\tilde{\lambda}^2 \quad Y_i = X_i^2 - X_i \quad \tilde{\lambda}^2 = \bar{Y} \quad \text{CLT}$
 $\sqrt{n}(\tilde{\lambda}^2 - \lambda^2) \xrightarrow{d} N(0, \text{Var}(Y)).$

Poisson

$$E[X^2] = \lambda^2 + \lambda, \quad E[X^3] = \lambda^3 + 3\lambda^2 + \lambda, \quad E[X^4] = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda,$$

$$E[Y^2] = E[(X^2 - X)^2] = E[X^4 - 2X^3 + X^2] = (\lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda) - 2(\lambda^3 + 3\lambda^2 + \lambda) + (\lambda^2 + \lambda) = \lambda^4 + 4\lambda^3 + 2\lambda^2,$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = \lambda^4 + 4\lambda^3 + 2\lambda^2 - \lambda^4 = 4\lambda^3 + 2\lambda^2,$$

$$\tilde{\lambda}^2 \quad \frac{4\lambda^3 + 2\lambda^2}{n}$$

$$e(\bar{X}^2, \tilde{\lambda}^2) = \frac{4\lambda^3/n}{(4\lambda^3 + 2\lambda^2)/n} = \frac{4\lambda^3}{4\lambda^3 + 2\lambda^2} = \frac{2\lambda}{2\lambda + 1}.$$

tip 5.72

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{V_2}{V_1}$$

$$V_1 \Rightarrow e > 1 \Rightarrow \hat{\theta}_1 \quad V_1 \Rightarrow e < 1 \Rightarrow \hat{\theta}_1$$

(9)

$$e(\bar{X}^2, \tilde{\lambda}^2) = \frac{V(\tilde{\lambda}^2)}{V(\bar{X}^2)}$$

tip 5.73

• Delta $\hat{\theta} \quad g(\hat{\theta})$

$$= [g'(\theta)]^2 \cdot \sigma^2$$

• CLT $\bar{Y}$

$$= \text{Var}(Y)$$

10 $X_1, \dots, X_n$ i.i.d. $X_1 \sim f(x, \theta) \quad \hat{\theta} = \hat{\theta}(X_1, \dots, X_n) \quad \theta$

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{L} N(0, V(\theta)), \quad \forall \theta \in \Theta,$$

$$V(\theta) \quad \theta \quad V(\hat{\theta}) \quad V(\theta)$$

$$\begin{aligned}\sqrt{n}(\hat{\theta}_n - \theta) &\xrightarrow{L} N(0, V(\theta)) \\ \hat{\theta}_n - \theta &= \frac{1}{\sqrt{n}} \cdot \sqrt{n}(\hat{\theta}_n - \theta), \\ \sqrt{n}(\hat{\theta}_n - \theta) &\xrightarrow{L} N(0, V(\theta)) \quad \sqrt{n} \rightarrow \infty \\ \hat{\theta}_n - \theta &\xrightarrow{P} 0 \implies \hat{\theta}_n \xrightarrow{P} \theta. \\ V(\hat{\theta}) &\xrightarrow{P} V(\theta).\end{aligned}$$

tip 5.74

$$\hat{\theta} \rightarrow \theta$$

tip 5.75

$$\begin{aligned}\hat{\theta} - \theta &= \frac{1}{\sqrt{n}} \cdot \sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{P} 0 \\ \hat{\theta}_n &\xrightarrow{P} \theta \quad V(\hat{\theta}_n) \xrightarrow{P} V(\theta)\end{aligned}$$



$$\hat{\theta}_n \xrightarrow{L} c \iff \hat{\theta}_n \xrightarrow{P} c.$$

12 3.4.6 $h(\hat{\theta}) - h(\theta_0) = \frac{1}{2}(\hat{\theta} - \theta_0)^2(h''(\theta_0) + R)$

$$n(h(\hat{\theta}) - h(\theta_0)) \xrightarrow{L} \frac{1}{2}A(\theta_0)h''(\theta_0)\chi^2(1),$$

$$\chi^2(1) \quad 1 \quad \chi^2$$

3.4.6 MLE $\hat{\theta}$

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, A(\theta_0)).$$

Taylor

$$h(\hat{\theta}) - h(\theta_0) = \frac{1}{2}(\hat{\theta} - \theta_0)^2(h''(\theta_0) + R),$$

$$R \xrightarrow{P} 0 \quad \hat{\theta} \xrightarrow{P} \theta_0 \quad h'' \quad n$$

$$n(h(\hat{\theta}) - h(\theta_0)) = \frac{1}{2} \left[\sqrt{n}(\hat{\theta} - \theta_0) \right]^2 (h''(\theta_0) + R).$$

$$\left[\sqrt{n}(\hat{\theta} - \theta_0) \right]^2 \xrightarrow{d} A(\theta_0) \cdot Z^2,$$

$$Z \sim N(0, 1) \quad Z^2 \sim \chi^2(1)$$

$$\left[\sqrt{n}(\hat{\theta} - \theta_0) \right]^2 \xrightarrow{d} A(\theta_0)\chi^2(1).$$

$$R \xrightarrow{P} 0 \quad h''(\theta_0) + R \xrightarrow{P} h''(\theta_0) \quad \text{Slutsky}$$

$$n(h(\hat{\theta}) - h(\theta_0)) \xrightarrow{d} \frac{1}{2}A(\theta_0)h''(\theta_0)\chi^2(1).$$

tip 5.76

$A(\theta_0)$ MLE Fisher

$$A(\theta_0) = \frac{1}{I(\theta_0)}$$

tip 5.77

fisher MLE mle fisher

tip 5.78

R Taylor $R = h''(\xi) - h''(\theta_0)$ $\xi \hat{\theta} \theta_0$
 $\hat{\theta} \xrightarrow{p} \theta_0$ MLE $\Rightarrow \xi \xrightarrow{p} \theta_0 \Rightarrow h''$ $h''(\xi) \xrightarrow{p} h''(\theta_0) \Rightarrow R = h''(\xi) - h''(\theta_0) \xrightarrow{p} 0$



$$\begin{aligned} \sqrt{n}(\hat{\theta} - \theta_0) &\xrightarrow{d} N(0, A(\theta_0)) & \sqrt{A(\theta_0)} \cdot Z \quad Z \sim N(0, 1) \\ \left[\sqrt{n}(\hat{\theta} - \theta_0) \right]^2 &\xrightarrow{d} \left[\sqrt{A(\theta_0)} \cdot Z \right]^2 = A(\theta_0) \cdot Z^2 \\ X \sim N(0, \sigma^2) &\Rightarrow X = \sigma Z \Rightarrow X^2 = \sigma^2 Z^2 \quad Z^2 \sim \chi^2(1) & 1 \quad A(\theta_0) \quad Z^2 \sim \chi^2(1) \end{aligned}$$

5.7.1 3.5

5.1 (Beta)

X

$$f(x; a, b) = \frac{x^{a-1}(1-x)^{b-1}}{B(a, b)}, \quad 0 < x < 1,$$

$$a > 0 \quad b > 0 \quad B(a, b) = \int_0^1 t^{a-1}(1-t)^{b-1} dt \quad X \quad (a, b) \quad \text{Beta} \quad X \sim \text{Beta}(a, b)$$



$$\text{Beta}(1, n) \quad n \quad U(0, 1) \quad Y_{(1)} \quad Y_i \stackrel{i.i.d.}{\sim} U(0, 1)$$

$$P(Y_{(1)} > y) = P(Y_i > y) = (1 - y)^n,$$

y

$$f_{Y_{(1)}}(y) = n(1 - y)^{n-1}, \quad 0 < y < 1,$$

$$\text{Beta}(1, n) \quad U(\theta, \theta + 1) \quad Y_i = X_i - \theta \sim U(0, 1)$$

$$X_{(1)} - \theta = Y_{(1)} \sim \text{Beta}(1, n),$$

$$\theta \quad X_{(1)} - \theta \quad n \quad U(0, 1) \quad Y_{(n)} \sim \text{Beta}(n, 1) \quad f(y) = ny^{n-1} \quad 0 < y < 1$$



$$1 \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim U(\theta, \theta + 1) \quad \theta \quad 100(1 - \alpha)\%$$

$$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} U(\theta, \theta + 1) \quad \theta \quad 100(1 - \alpha)\% \\ U(\theta, \theta + 1) \quad \text{,,} \quad \text{,,} \quad \text{,,} \quad \text{,,}$$

$$T = (X_{(1)}, X_{(n)}),$$

$$X_{(1)} = \min_i X_i \quad X_{(n)} = \max_i X_i \\ X_i - \theta \stackrel{i.i.d.}{\sim} U(0, 1)$$

$$G = X_{(1)} - \theta \sim \text{Beta}(1, n),$$

$$f(u) = n(1 - u)^{n-1} \quad 0 < u < 1 \quad \theta \quad G$$

$$S \quad S = [a, b] \quad P(a \leq X_{(1)} - \theta \leq b) = 1 - \alpha$$

$$P(X_{(1)} - \theta < a) = \frac{\alpha}{2}, \quad P(X_{(1)} - \theta > b) = \frac{\alpha}{2}.$$

$$\text{Beta}(1, n) \quad F(u) = 1 - (1 - u)^n$$

$$a = 1 - \left(1 - \frac{\alpha}{2}\right)^{1/n}, \quad b = 1 - \left(\frac{\alpha}{2}\right)^{1/n}.$$

$$a \leq X_{(1)} - \theta \leq b - \theta$$

$$X_{(1)} - b \leq \theta \leq X_{(1)} - a.$$

$$\theta = 100(1 - \alpha)\%$$

$$\left[X_{(1)} - 1 + \left(\frac{\alpha}{2}\right)^{1/n}, X_{(1)} - 1 + \left(1 - \frac{\alpha}{2}\right)^{1/n} \right].$$

tip 5.79

$-\theta$

$$Y_i = X_i - \theta \sim U(0, 1) \quad \theta \quad \text{pivot}$$

$$Y_i = X_{(1)} - \theta$$

$$X_{(1)} - \theta \quad X_i - \theta \quad \dots$$

 **2** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, \sigma^2)$ $\mu = 100(1 - \alpha)\%$ μ

$$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2) \quad \mu = 100(1 - \alpha)\%$$

μ

$$T = \sum_{i=1}^n (X_i - \mu)^2.$$

$$\frac{X_i - \mu}{\sigma} \stackrel{i.i.d.}{\sim} N(0, 1)$$

$$G = \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi^2(n),$$

$$\sigma^2 \quad G$$

$$S^2 \quad \chi^2(n) \quad \chi_{\alpha/2}^2(n) \quad \chi_{1-\alpha/2}^2(n)$$

$$P\left(\chi_{1-\alpha/2}^2(n) \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \leq \chi_{\alpha/2}^2(n)\right) = 1 - \alpha.$$

σ^2

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\alpha/2}^2(n)} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\alpha/2}^2(n)}.$$

$$\sigma^2 = 100(1 - \alpha)\%$$

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\alpha/2}^2(n)}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\alpha/2}^2(n)} \right].$$

$$\mu \quad \mu \quad \bar{X} \quad \mu$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1),$$

$$n \quad n-1 \quad \chi^2 \quad \mu \quad \sigma^2$$

tip 5.80

$$n \quad (X_1 - \bar{X}), \dots, (X_n - \bar{X})$$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

$$n-1 \quad n$$

$$n \quad 0 \quad n-1$$



$$T \mid \theta \rightarrow G(T, \theta) \begin{cases} \theta \\ \theta \end{cases}$$

- $T \mid \theta \sim \theta \mid T$
- $G \mid T \sim \theta \mid T$

3 $X_1, X_2, \dots, X_n$ i.i.d. X_1

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & x > 0; \\ 0, & . \end{cases}$$

- (1) $2n\bar{X}/\theta \sim \chi^2(2n)$
- (2) $\theta = 100(1 - \alpha)\%$
- (3) $T_{0.95} = P\{X_1 > T_{0.95}\} = 0.95 \quad T_{0.95} = 100(1 - \alpha)\%$

$$(1) \quad X_i \sim \Gamma(1, \theta) \quad \sum_{i=1}^n X_i \sim \Gamma(n, \theta)$$

$$\frac{2}{\theta} \sum_{i=1}^n X_i = \frac{2n\bar{X}}{\theta} \sim \Gamma(n, 2) = \chi^2(2n).$$

$$(2) \quad (1)$$

$$G = \frac{2n\bar{X}}{\theta} \sim \chi^2(2n),$$

$$\theta$$

$$P\left(\frac{2n\bar{X}}{\theta} \leq \chi_{\alpha}^2(2n)\right) = 1 - \alpha.$$

$$\theta$$

$$\theta \geq \frac{2n\bar{X}}{\chi_{\alpha}^2(2n)}.$$

$$\theta = 100(1 - \alpha)\%$$

$$\theta_1 = \frac{2n\bar{X}}{\chi_{\alpha}^2(2n)}.$$

$$(3)$$

$$S(x) = P\{X > x\} = 1 - F(x),$$

" x " CDF

$$P\{X_1 > T_{0.95}\} = e^{-T_{0.95}/\theta} = 0.95,$$

$$T_{0.95} = -\theta \ln 0.95.$$

$$T_{0.95} \mid \theta \quad (2) \quad \theta_1 = T_{0.95} = 100(1 - \alpha)\%$$

$$-\theta_1 \ln 0.95 = -\frac{2n\bar{X} \ln 0.95}{\chi_{\alpha}^2(2n)}.$$

tip 5.81

$$e^{-x/\beta}$$

$$x^{\alpha-1}$$



χ^2 Gamma Gamma

$$f(x) = \frac{1}{\beta^{\alpha} \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

$$\alpha = 1 \quad \beta = \theta \quad f(x) = \frac{1}{\theta} e^{-x/\theta}$$

$$X_i \sim \text{Exp}(1/\theta) \iff X_i \sim \Gamma(1, \theta).$$

$$\alpha = n/2 \quad \beta = 2 \quad f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2} \quad \chi^2(n)$$

$$\chi^2(n) = \Gamma\left(\frac{n}{2}, 2\right).$$

$$\text{Exp}(1/\theta) = \Gamma(1, \theta), \quad \chi^2(n) = \Gamma\left(\frac{n}{2}, 2\right).$$



•

$$f(x; \theta) = h(x) \exp(\eta(\theta)T(x) - A(\theta)),$$

Gamma Beta Poisson

•

$$\sim x^{-\alpha} \quad \text{Cauchy} \quad t \quad \text{Pareto}$$

•

$$U(a, b) \quad \text{Beta} \quad \text{Beta}(\alpha, \beta)$$

$$\subset \supset, \quad \cap = \emptyset.$$

5.8



$$4 \quad 0.1 \quad p \quad 100 \quad 2 \quad 95\%$$

$$X_i \sim B(1, p) \quad \hat{p} = \bar{X} = \frac{2}{100} = 0.02 \quad n = 100$$

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{p(1-p)}} \xrightarrow{d} N(0, 1)$$

$$\hat{p} \quad p$$

$$U = \frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1-\hat{p})}} \approx N(0, 1)$$

$$95\% \quad P(U \leq z_{0.05}) = 0.95 \quad z_{0.05} = 1.645$$

$$\sqrt{n}(\hat{p} - p) \leq z_{0.05} \sqrt{\hat{p}(1-\hat{p})}$$

$$p \geq \hat{p} - \frac{z_{0.05} \sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}$$

$$\bar{p} = \hat{p} + \frac{z_{0.05} \sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} = 0.02 + \frac{1.645 \times \sqrt{0.02 \times 0.98}}{\sqrt{100}}$$

$$= 0.02 + \frac{1.645 \times 0.1400}{10} = 0.02 + 0.0230 = 0.0430$$

$$95\% \quad \bar{p} = 0.0430$$

tip 5.82

$$\begin{array}{ccccccc} \text{clt} & & \hat{p} & \text{phat} & \text{Var}\left(\frac{1}{n} \sum X_i\right) & X_i & \frac{1}{n} \\ (\sqrt{n})^2 = n & X_i & p(1-p) & N(0, p(1-p)) & xi & 0 & 1 \end{array} \quad \begin{array}{c} CLT \\ \sqrt{n} \end{array}$$

$$CLT \quad X_i \quad 0-1 \quad CLT \quad \hat{p} = \frac{1}{n} \sum X_i \quad CLT$$

$$\begin{aligned} & \bullet \hat{p} - p = 0 \\ & \bullet \times \sqrt{n} = 0 \\ & \hat{p} = \frac{p(1-p)}{n} \quad \sqrt{n} \quad p(1-p) \quad n \quad n \rightarrow \infty \quad N(0, p(1-p)) \\ 5 \quad & X_1, X_2, \dots, X_m \quad Y_1, Y_2, \dots, Y_n \quad N(\mu_1, \sigma_1^2) \quad N(\mu_2, \sigma_2^2) \\ & (1) \quad \sigma_1^2, \sigma_2^2 \quad \mu_2 - \mu_1 \quad 1 - \alpha \\ & (2) \quad \sigma_1^2 = \sigma_2^2 \quad 90\% \quad \mu_2 - \mu_1 \quad \frac{2}{5}\sigma_1 \quad n_1 = n_2 \\ & (3) \quad \mu_1, \mu_2, \sigma_1^2, \sigma_2^2 \quad \sigma_2^2/\sigma_1^2 \quad 1 - \alpha \end{aligned}$$

$$\bar{Y} - \bar{X} \sim N\left(\mu_2 - \mu_1, \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}\right)$$

$$U = \frac{(\bar{Y} - \bar{X}) - (\mu_2 - \mu_1)}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} \sim N(0, 1)$$

$$P(|U| \leq z_{\alpha/2}) = 1 - \alpha \quad \mu_2 - \mu_1 \quad 1 - \alpha$$

$$\left(\bar{Y} - \bar{X} - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}, \bar{Y} - \bar{X} + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}} \right)$$

$$1 - \alpha$$

$$P(|U| \leq z_{\alpha/2}) = 1 - \alpha$$

$$\frac{\alpha}{2} \quad z_{\alpha/2} \quad z_{\alpha} \quad z_{\alpha}$$

$$\bullet \quad (2) \quad n_1 = n_2 = n \quad \sigma_1^2 = \sigma_2^2 = \sigma^2 \quad (1)$$

$$2z_{\alpha/2} \sqrt{\frac{2\sigma^2}{n}}$$

$$90\% \quad \alpha = 0.1 \quad z_{0.05} = 1.645 \quad \frac{2}{5}\sigma$$

$$2 \times 1.645 \times \sqrt{\frac{2\sigma^2}{n}} = \frac{2}{5}\sigma$$

$$1.645 \sqrt{\frac{2}{n}} = \frac{1}{5}$$

$$\frac{2}{n} = \left(\frac{1}{5 \times 1.645} \right)^2 = \frac{1}{67.65}$$

$$n = 135.3$$

$$n_1 = n_2 \quad 136$$

$$\bullet \quad (3) \quad \frac{(m-1)S_1^2}{\sigma_1^2} \sim \chi^2(m-1), \quad \frac{(n-1)S_2^2}{\sigma_2^2} \sim \chi^2(n-1)$$

$$F = \frac{S_2^2/\sigma_2^2}{S_1^2/\sigma_1^2} = \frac{S_2^2}{S_1^2} \cdot \frac{\sigma_1^2}{\sigma_2^2} \sim F(n-1, m-1)$$

$$P(F_{\alpha/2}(n-1, m-1) \leq F \leq F_{1-\alpha/2}(n-1, m-1)) = 1 - \alpha \quad \sigma_2^2/\sigma_1^2 \quad 1 - \alpha$$

$$\left(\frac{S_2^2}{S_1^2} \cdot \frac{1}{F_{1-\alpha/2}(n-1, m-1)}, \frac{S_2^2}{S_1^2} \cdot \frac{1}{F_{\alpha/2}(n-1, m-1)} \right)$$

tip 5.83 $\mu_2 - \mu_1$ σ_2^2/σ_1^2 $\mu_1 \mu_2$ $\mu_1 \mu_2$ F $n-1 \quad m-1$ **5.8.1 4.1**

 1 $X_1, X_2, \dots, X_n \quad X_1 \sim N(\mu, 1) \quad H_0 : \mu = 0 \quad H_1 : \mu > 0$

$$\phi(\mathbf{X}) = \phi(X_1, X_2, \dots, X_n) = \begin{cases} 1, & \bar{X}_n > 1.3 \\ 0, & \bar{X}_n \leq 1.3 \end{cases}$$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad n = 10$$

(1)

(2) $\mu = 1.0 \quad \mu = 1.5$

(3) $\alpha = 0.05 \quad \bar{X}_n$

$X_i \sim N(\mu, 1) \quad n = 10$

$$\bar{X}_{10} \sim N\left(\mu, \frac{1}{10}\right), \quad \frac{\bar{X}_{10} - \mu}{1/\sqrt{10}} \sim N(0, 1).$$

(I) $H_0 : \mu = 0$

$$\alpha = P_{\mu=0}(\bar{X}_{10} > 1.3) = P\left(Z > \frac{1.3}{1/\sqrt{10}}\right) = 1 - \Phi(4.11) \approx 0.$$

0

tip 5.84 z_α 1.3 $\bar{X}$ $\bar{X} \quad H_0$

$$\bar{X}_{10} \sim N\left(0, \frac{1}{10}\right)$$

(2)

$$\beta(\mu) = P_\mu(\bar{X}_{10} \leq 1.3) = \Phi\left(\sqrt{10}(1.3 - \mu)\right).$$

- $\mu = 1.0 \quad \beta(1.0) = \Phi(\sqrt{10} \times 0.3) = \Phi(0.949) \approx 0.8289$
- $\mu = 1.5 \quad \beta(1.5) = \Phi(\sqrt{10} \times (-0.2)) = \Phi(-0.632) \approx 0.2638$

tip 5.85 $\mu \quad X \quad x$

(3) $c \quad 0.05$

$$P_{\mu=0}(\bar{X}_{10} > c) = 0.05 \implies 1 - \Phi(\sqrt{10}c) = 0.05 \implies \sqrt{10}c = 1.645 \implies c = \frac{1.645}{\sqrt{10}} \approx 0.520.$$

$\alpha = 0.05$

$$\phi(\mathbf{X}) = \begin{cases} 1, & \bar{X}_{10} > 0.520, \\ 0, & \bar{X}_{10} \leq 0.520. \end{cases}$$

tip 5.86

$$H_1 \quad 0.05 \quad \Phi \quad CDF \quad c \quad 1 - \Phi$$

$$P_{\mu=0}(\bar{X}_{10} > c) = 1 - \Phi(\sqrt{10}c) = 0.05$$

5.2 ()

 $\phi(\mathbf{X})$

- 1 H_0
- 0 H_0



tip 5.87

$$h_0 \quad h_0 \quad h_0 \quad \theta \quad H_0 \quad \theta \quad h_1 \quad 1-$$

- $H_0 \quad H_0$
- $H_1 \quad H_0$

2 $X_1, X_2, \dots, X_{10} \sim U(0, \theta)$

$$\phi(\mathbf{X}) = \begin{cases} 1, & X_{(10)} > 0.95 \\ 0, & X_{(10)} \leq 0.95 \end{cases}$$

$$H_0 : \theta = 1 \quad H_1 : \theta > 1 \quad \mathbf{X} = (X_1, \dots, X_{10})^T \quad X_{(10)} = \max(X_1, X_2, \dots, X_{10})$$

(1)

(2)

$$(3) \quad X_{(10)} \quad \alpha = 0.05$$

$$X_i \sim U(0, \theta) \quad CDF \quad F(x) = \frac{x}{\theta} \quad 0 \leq x \leq \theta \quad X_{(10)} = \max(X_1, \dots, X_{10}) \quad CDF$$

$$F_{(10)}(x) = P(X_{(10)} \leq x) = \left(\frac{x}{\theta}\right)^{10}, \quad 0 \leq x \leq \theta.$$

$$(1) \quad H_0 : \theta = 1$$

$$\alpha = P_{\theta=1}(X_{(10)} > 0.95) = 1 - P_{\theta=1}(X_{(10)} \leq 0.95) = 1 - (0.95)^{10} \approx 1 - 0.5987 = 0.4013.$$

$$(2) \quad \theta \quad H_0$$

$$\beta(\theta) = P_{\theta}(X_{(10)} > 0.95) = 1 - \left(\frac{0.95}{\theta}\right)^{10}, \quad \theta \geq 0.95.$$

$$\theta < 0.95 \quad X_{(10)} \leq \theta < 0.95 \quad \beta(\theta) = 0$$

$$(3) \quad c \quad 0.05$$

$$P_{\theta=1}(X_{(10)} > c) = 0.05 \implies 1 - c^{10} = 0.05 \implies c^{10} = 0.95 \implies c = 0.95^{1/10} \approx 0.9949.$$

$$\alpha = 0.05$$

$$\phi(\mathbf{X}) = \begin{cases} 1, & X_{(10)} > 0.95^{1/10} \approx 0.9949, \\ 0, & X_{(10)} \leq 0.95^{1/10} \approx 0.9949. \end{cases}$$

tip 5.88

Cumulative Distribution Function

$$H_0 : \theta = 1 \quad H_1 : \theta > 1$$

- $= H_0 \quad \beta(1)$
 - $= 1 - \beta(\theta) \quad \theta > 1 \quad \beta$
- $$\beta(\theta)$$

3 $B(n, p) \quad P\{X \leq x\} = \sum_{i=0}^x \binom{n}{i} p^i (1-p)^{n-i} \quad p$

$$m = \lfloor x \rfloor - p$$

$$\frac{\partial F}{\partial p} = \sum_{i=0}^m C_n^i [ip^{i-1}(1-p)^{n-i} - (n-i)p^i(1-p)^{n-i-1}].$$

$$p^{i-1}(1-p)^{n-i-1}$$

$$= \sum_{i=0}^m C_n^i p^{i-1}(1-p)^{n-i-1}(i-np).$$

$$i C_n^i = n C_{n-1}^{i-1}$$

$$C_n^i p^{i-1}(1-p)^{n-i-1}(i-np) = n C_{n-1}^{i-1} p^{i-1}(1-p)^{n-i} - n C_{n-1}^i p^i(1-p)^{n-i-1}.$$

$$T_i = C_{n-1}^i p^i(1-p)^{n-1-i} - n(T_{i-1} - T_i)$$

$$\frac{\partial F}{\partial p} = n \sum_{i=0}^m (T_{i-1} - T_i).$$

$$= n(T_{-1} - T_m) = n(0 - T_m) = -n C_{n-1}^m p^m(1-p)^{n-1-m}.$$

$$n \geq 1, 0 < p < 1, 0 \leq m \leq n-1, \quad \frac{\partial F}{\partial p} < 0 \quad F(x; p) \text{ is decreasing in } p$$

tip 5.89

$$\begin{aligned} & 0 \quad 0 \quad (i-np) \\ \bullet \quad & i \cdot C_n^i \rightarrow n \cdot C_{n-1}^{i-1} \quad T_{i-1} \\ \bullet \quad & np - n \cdot C_n^i \cdot p \xrightarrow{\text{Pascal}} n(C_{n-1}^{i-1} + C_{n-1}^i) \cdot p - T_i \\ & n(T_{i-1} - T_i) \\ & i \quad n \quad -np \end{aligned}$$



$$T_{-1} = C_{n-1}^{-1} \cdot p^{-1}(1-p)^n = 0 \quad C_{n-1}^{-1} = 0 \quad n-1 \quad -1 \quad 0$$

5.8.2 4.2

$$5.2 \quad X \sim N(\mu, 9) \quad 8$$

$$12.5 \quad 13.8 \quad 14.1 \quad 14.1 \quad 14.8 \quad 15.0 \quad 15.4 \quad 16.0$$

$$\alpha = 0.05 \quad \mu = 14$$



$$1 \quad 4.2.1 \quad \sigma^2 \quad \alpha = 0.10$$

$$H_0 : \mu = 14; \quad H_1 : \mu \neq 14$$

t

$$\bar{X} = \frac{12.5 + 13.8 + 14.1 + 14.1 + 14.8 + 15.0 + 15.4 + 16.0}{8} = \frac{115.7}{8} = 14.4625$$

$$S^2 = \frac{1}{7} \sum_{i=1}^8 (X_i - \bar{X})^2 = \frac{7.9989}{7} \approx 1.1427, \quad S \approx 1.0690$$

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{14.4625 - 14}{1.0690/\sqrt{8}} = \frac{0.4625}{0.3780} \approx 1.223$$

$$\alpha = 0.10 \quad n-1 = 7$$

$$t_{0.05}(7) = 1.895$$

$$|t| = 1.223 < 1.895 \quad H_0 \quad 0.10 \quad \mu = 14$$

tip 5.90

 T 

$$P(T > t_{0.10}(n)) = 0.10$$

$$0.90 = 90\%$$

$$\underbrace{0.10} = \quad = 1 -$$

$$\alpha = 0.10 \implies t_{0.10} \quad 90\%$$

$$\alpha = 0.05 \implies t_{0.05} \quad 95\%$$

$$\alpha = 0.025 \implies t_{0.025} \quad 97.5\%$$



$$2 \quad X_1, X_2, \dots, X_{10} \sim N(\mu, \sigma^2)$$

$$(1) \quad H_0 : \mu \leq 1 \quad H_1 : \mu > 1 \quad \alpha = 0.1$$

$$H_1 : \mu > 1 \quad t$$

$$T = \frac{\bar{X} - 1}{S/\sqrt{10}} \sim t(9) \quad (H_0 : \mu = 1)$$

$$t_\alpha(n-1) = t_{0.1}(9) = 1.383$$

$$\phi(\mathbf{X}) = \begin{cases} 1, & T > 1.383 \\ 0, & T \leq 1.383 \end{cases}$$

$$T > 1.383 \quad H_0 : \mu > 1$$

tip 5.91

 I

5.2 ()

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2) \quad H_0 : \mu = \mu_0$$

$$\rightarrow Z$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

$$\rightarrow t$$

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$$

$$S \quad \sigma \quad n \quad t$$

$$H_0 : \sigma^2 = \sigma_0^2$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$$

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1-1, n_2-1)$$



$$2 \quad X_1, X_2, \dots, X_{10} \sim N(\mu, \sigma^2)$$

$$(2) \quad H_0 : \sigma^2 \leq 1 \quad H_1 : \sigma^2 > 1 \quad \alpha = 0.1$$

$$(3) \quad \alpha = 0.05 \quad (1) (2)$$

$$(2) \quad \mu$$

$$\chi^2 = 9S^2 \sim \chi^2(9) \quad (\sigma^2 = 1)$$

$$\alpha = 0.1 \quad \chi_{0.1}^2(9) = 14.684$$

$$\varphi(X) = \begin{cases} 1, & 9S^2 > 14.684 \\ 0, & 9S^2 \leq 14.684 \end{cases}$$

$$\beta(\sigma^2) = P\left(\chi^2(9) > \frac{14.684}{\sigma^2}\right)$$

$$(3) \quad \alpha = 0.05 \quad \chi_{0.05}^2(9) = 16.919 \quad (2)$$

$$\varphi(X) = \begin{cases} 1, & 9S^2 > 16.919 \\ 0, & 9S^2 \leq 16.919 \end{cases}$$

$$\beta(\sigma^2) = P\left(\chi^2(9) > \frac{16.919}{\sigma^2}\right)$$

$$\alpha = 0.1 \quad \sigma^2 > 1 \quad \alpha$$

$$\text{3} \quad X_1, X_2, \dots, X_m \sim N(\mu_1, \sigma_1^2) \quad Y_1, Y_2, \dots, Y_n \sim N(\mu_2, \sigma_2^2) \quad \sigma_1^2, \sigma_2^2 \quad H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

$$(2) \quad \sigma_1^2 = k\sigma_2^2 \quad k$$

$$(2) \quad \sigma_1^2 = k\sigma_2^2$$

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}\right) = N\left(\mu_1 - \mu_2, \sigma_2^2\left(\frac{k}{m} + \frac{1}{n}\right)\right)$$

$$\frac{(m-1)S_1^2}{\sigma_1^2} + \frac{(n-1)S_2^2}{\sigma_2^2} = \frac{(m-1)S_1^2}{k\sigma_2^2} + \frac{(n-1)S_2^2}{\sigma_2^2} \sim \chi^2(m+n-2)$$

$$S_p^2 = \frac{\frac{(m-1)S_1^2}{k} + (n-1)S_2^2}{m+n-2}$$

$$\frac{(m+n-2)S_p^2}{\sigma_2^2} \sim \chi^2(m+n-2) \quad H_0 : \mu_1 = \mu_2 \quad t$$

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{k}{m} + \frac{1}{n}}} \sim t(m+n-2)$$

$$\varphi = \begin{cases} 1, & |T| > t_{\alpha/2}(m+n-2) \\ 0, & |T| \leq t_{\alpha/2}(m+n-2) \end{cases}$$

$$\text{4} \quad 3$$

$$(1) \quad H_0 : \sigma_1^2 \leq \sigma_2^2 \quad H_1 : \sigma_1^2 > \sigma_2^2$$

$$(2) \quad H_0 : \sigma_1^2 = k\sigma_2^2 \quad H_1 : \sigma_1^2 \neq k\sigma_2^2 \quad k$$

$$(1) \quad \mu_1, \mu_2$$

$$F = \frac{S_1^2}{S_2^2} \sim F(m-1, n-1) \quad (\sigma_1^2 = \sigma_2^2)$$

$$H_1 : \sigma_1^2 > \sigma_2^2$$

$$\varphi = \begin{cases} 1, & F > F_\alpha(m-1, n-1) \\ 0, & F \leq F_\alpha(m-1, n-1) \end{cases}$$

$$(2) \quad H_0 : \sigma_1^2 = k\sigma_2^2$$

$$F = \frac{S_1^2/k}{S_2^2} = \frac{S_1^2}{kS_2^2} \sim F(m-1, n-1)$$

$$\varphi = \begin{cases} 1, & F > F_{\alpha/2}(m-1, n-1) \quad F < F_{1-\alpha/2}(m-1, n-1) \\ 0, & \text{otherwise} \end{cases}$$



6

A B

A

A 10

10.5 10.3 10.4 9.6 9.7 9.8 9.9 10.1 10.2 9.7

B 10

10.2 10.7 10.8 10.9 10.4 10.6 11.0 11.1 10.9 11.2

$$(1) \quad B$$

$$(2) \quad A \quad B$$



$$" \quad " \quad \mu_B = \mu_A \quad " \quad " \quad \sigma_B^2 = \sigma_A^2 \mu_A, \mu_B, \sigma_A^2, \sigma_B^2$$

$$(1) \quad H_0 : \mu_B = \mu_A \quad H_1 : \mu_B \neq \mu_A \quad \sigma_A^2, \sigma_B^2 \quad t$$

$$\bar{X} = \frac{1}{10}(10.5 + 10.3 + 10.4 + 9.6 + 9.7 + 9.8 + 9.9 + 10.1 + 10.2 + 9.7) = 10.02$$

$$\bar{Y} = \frac{1}{10}(10.2 + 10.7 + 10.8 + 10.9 + 10.4 + 10.6 + 11.0 + 11.1 + 10.9 + 11.2) = 10.78$$

$$S_A^2 = \frac{1}{9} \sum_{i=1}^{10} (X_i - \bar{X})^2 = \frac{1}{9}(0.48^2 + 0.28^2 + 0.38^2 + 0.42^2 + 0.32^2 + 0.22^2 + 0.12^2 + 0.08^2 + 0.18^2 + 0.32^2)$$

$$0.2304 + 0.0784 + 0.1444 + 0.1764 + 0.1024 + 0.0484 + 0.0144 + 0.0064 + 0.0324 + 0.1024 = 0.9360$$

$$S_A^2 = \frac{0.9360}{9} \approx 0.1040, \quad S_A \approx 0.3225$$

$$S_B^2 = \frac{1}{9} \sum_{i=1}^{10} (Y_i - \bar{Y})^2$$

$$0.58^2 + 0.08^2 + 0.02^2 + 0.12^2 + 0.38^2 + 0.18^2 + 0.22^2 + 0.32^2 + 0.12^2 + 0.42^2 = 0.3364 + 0.0064 + 0.0004 + 0.0144 + 0.1444 + 0.0324 + 0.0484 + 0.1024 + 0.0144 + 0.1764 = 0.8760$$

$$S_B^2 = \frac{0.8760}{9} \approx 0.0973, \quad S_B \approx 0.3120$$

$$S_p^2 = \frac{9S_A^2 + 9S_B^2}{18} = \frac{0.9360 + 0.8760}{18} = \frac{1.8120}{18} \approx 0.1007, \quad S_p \approx 0.3173$$

$$T = \frac{\bar{Y} - \bar{X}}{S_p \sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{10.78 - 10.02}{0.3173 \times \sqrt{0.2}} = \frac{0.76}{0.3173 \times 0.4472} \approx \frac{0.76}{0.1419} \approx 5.356$$

$$\alpha = 0.05 \quad t_{0.025}(18) = 2.101 \quad |T| = 5.356 > 2.101 \quad H_0$$

B

$$(2) \quad H_0 : \sigma_B^2 = \sigma_A^2 \quad H_1 : \sigma_B^2 \neq \sigma_A^2 \quad F$$

$$F = \frac{S_A^2}{S_B^2} = \frac{0.1040}{0.0973} \approx 1.069$$

$$\alpha = 0.05 \quad F_{0.025}(9, 9) = 4.03 \quad F_{0.975}(9, 9) = \frac{1}{4.03} \approx 0.248$$

$$0.248 < F = 1.069 < 4.03 \quad H_0$$

$$0.05$$

5.9 10

5.9.1 4.3

$$X \sim B(n, p) \quad n = 36 \quad \hat{p} = \frac{3}{36} = \frac{1}{12} \approx 0.0833$$

$$H_0 : p \leq 0.03, \quad H_1 : p > 0.03$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{\frac{1}{12} - 0.03}{\sqrt{\frac{0.03 \times 0.97}{36}}} \approx 1.875$$

$$\alpha = 0.05 \quad z_{0.05} = 1.645 \quad Z \approx 1.875 > 1.645 \quad H_0 \quad 3\%$$

$$p \quad 0.95$$

$$p \leq \hat{p} + z_{0.05} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \frac{1}{12} + 1.645 \times \sqrt{\frac{\frac{1}{12} \times \frac{11}{12}}{36}} \approx 0.160$$

16.0%

tip 5.92

 $n \quad p \quad p$

tip 5.93

$$\bar{X} \quad n \quad X_i \quad n$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \implies \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\bar{X} \quad \frac{\sqrt{n}}{\sigma^2} \quad \sqrt{n}$$

$$\bar{X} \implies \frac{(n-1)S^2}{\sigma^2} \quad \sqrt{n}$$

tip 5.94

 n

$$N(0, 1) \quad \hat{p}$$

$$2 \quad X_1, X_2, \dots, X_n \quad P(\lambda) \quad H_0 : \lambda = \lambda_0 \quad H_1 : \lambda \neq \lambda_0 \quad 5, 7, 8, 6, 9, 10 \quad \alpha = 0.05 \quad \lambda = 3$$

$$X_i \sim P(\lambda) \quad \lambda \quad \lambda \quad \text{Poisson}$$

$$\bar{X} \quad \frac{\lambda}{n} \quad \sqrt{\frac{\lambda}{n}}$$

$$X_i \sim P(\lambda) \quad \bar{X} \quad \lambda$$

$$Z = \frac{\bar{X} - \lambda_0}{\sqrt{\lambda_0/n}} \stackrel{H_0}{\approx} N(0, 1)$$

$$|Z| > z_{\alpha/2} = z_{0.025} = 1.96$$

$$\bar{x} = \frac{5 + 7 + 8 + 6 + 9 + 10}{6} = 7.5$$

$$Z = \frac{7.5 - 3}{\sqrt{3/6}} = \frac{4.5}{\sqrt{0.5}} \approx 6.36$$

$$|Z| \approx 6.36 > 1.96 \quad H_0 \quad \alpha = 0.05 \quad \lambda = 3$$

tip 5.95

01

5.10 11

5.10.1 4.4

 1 $X \sim \lambda$

$$f(x, \lambda) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots$$

$$\lambda > 0 \quad X_1, X_2, \dots, X_n$$

$$H_0 : \lambda \leq \lambda_0; \quad H_1 : \lambda > \lambda_0$$

1. UMP

2. UMP $\alpha = 0.05 \quad n = 10 \quad \lambda_0 = 0.1$

$$\lambda_1 < \lambda_2$$

$$\frac{\prod_{i=1}^n f(X_i; \lambda_2)}{\prod_{i=1}^n f(X_i; \lambda_1)} = \frac{\prod_{i=1}^n \frac{\lambda_2^{X_i}}{X_i!} e^{-\lambda_2}}{\prod_{i=1}^n \frac{\lambda_1^{X_i}}{X_i!} e^{-\lambda_1}} = \left(\frac{\lambda_2}{\lambda_1} \right)^{\sum_{i=1}^n X_i} e^{-n(\lambda_2 - \lambda_1)}$$

$$e^{-n(\lambda_2 - \lambda_1)} \quad \lambda \quad \lambda_2 > \lambda_1 \quad \frac{\lambda_2}{\lambda_1} > 1 \quad T(\mathbf{X}) = \sum_{i=1}^n X_i \quad \text{MLR} \quad \text{MLR} \quad \text{UMP} \quad T(\mathbf{X}) = \sum_{i=1}^n X_i > c$$

tip 5.96

H_0

$$\alpha = \underbrace{P_{H_0}(\quad)}_{\underbrace{\int_{H_0} f_{H_0}(x) dx}}$$

α

α

α

$$\alpha = 0.05 \quad n = 10 \quad \lambda_0 = 0.1$$

$$E_{\lambda_0}[\phi(\mathbf{X})] = \alpha$$

$$H_0 \quad \sum_{i=1}^n X_i \sim P(n\lambda_0) = P(1) \quad c$$

$$P_{\lambda_0} \left(\sum_{i=1}^n X_i > c \right) \leq 0.05$$

$$\sum X_i \sim P(1)$$

c	$P(\sum X_i \leq c)$	$P(\sum X_i > c)$
2	0.9197	0.0803
3	0.9810	0.0190

$$c = 3 \quad r \cdot E_{\lambda_0}[\phi(\mathbf{X})] = \alpha$$

$$P_{\lambda_0} \left(\sum X_i > 3 \right) + r \cdot P_{\lambda_0} \left(\sum X_i = 3 \right) = 0.05$$

$$0.0190 + r \times (0.9810 - 0.9197) = 0.05$$

$$r = \frac{0.05 - 0.0190}{0.0613} \approx 0.506$$

$$\phi(\mathbf{X}) = \begin{cases} 1, & \sum_{i=1}^n X_i > 3 \\ 0.506, & \sum_{i=1}^n X_i = 3 \\ 0, & \sum_{i=1}^n X_i < 3 \end{cases}$$

tip 5.97

$$\phi(X) = 0 \quad \phi(X) = 1 \quad E_{H_0}[\phi(X)] = \alpha \quad 1 \quad 0 \quad (0, 1)$$

tip 5.98 (-)

$$\sum X_i = 3 \quad r \quad 1 - r \quad \alpha$$

$$2 \quad n \quad X_i \quad i \quad X_1, X_2, \dots, X_n \quad \theta \quad n$$

$$1. \quad 10 \quad 0.01$$

$$2. \quad 15 \quad 0.01$$

n UMP

tip 5.99

$$H_0 \quad H_1 \quad H_0$$

$$H_0 : \theta \leq \theta_0 \quad H_1 : \theta > \theta_0$$

$$\bullet \quad 1 \quad \theta = 10 \quad 0.01 \quad \beta_\phi(10) \leq 0.01$$

$$\bullet \quad 2 \quad \theta = 15 \quad 0.01 \quad \beta_\phi(15) \geq 0.99$$

$$\text{UMP} \quad \text{MLR} \quad T = \sum_{i=1}^n X_i \sim P(n\theta) \quad \sum_{i=1}^n X_i > c$$

$$n \quad \sum X_i \approx N(n\theta, n\theta)$$

$$\beta_\phi(\theta) = P_\theta \left(\sum X_i > c \right) = P \left(Z > \frac{c - n\theta}{\sqrt{n\theta}} \right)$$

$$\frac{c - 10n}{\sqrt{10n}} = 2.326, \quad \frac{c - 15n}{\sqrt{15n}} = -2.326$$

c

$$\frac{c - 10n}{\sqrt{10n}} - \frac{c - 15n}{\sqrt{15n}} = 4.652$$

$$\frac{1}{\sqrt{10}} \approx 0.3162 \quad \frac{1}{\sqrt{15}} \approx 0.2582$$

$$\sqrt{n} (5 \times 0.3162 + 5 \times 0.2582) = 4.652$$

$$\sqrt{n} \times 2.872 = 4.652$$

$$\sqrt{n} \approx 1.620 \implies n \approx 25$$

$$n = 25$$



$$n = 10 \quad \lambda_0 = 0.1 \quad n\lambda_0 = 1 \quad n \quad \theta = 10 \quad \theta = 15 \quad n\theta$$

$$\sum X_i \sim P(n\theta) \approx N(n\theta, n\theta)$$



$$1. \quad P(\lambda)$$

$$\frac{\prod f(X_i; \lambda_2)}{\prod f(X_i; \lambda_1)} = \left(\frac{\lambda_2}{\lambda_1}\right)^{\sum X_i} e^{-n(\lambda_2 - \lambda_1)}$$

$$2. \quad \lambda_2 > \lambda_1 \quad \frac{\lambda_2}{\lambda_1} > 1 \quad \sum X_i$$

$$N(\mu, \sigma^2) \quad \sigma^2$$

$$\frac{\prod f(X_i; \mu_2)}{\prod f(X_i; \mu_1)} = \exp\left(\frac{(\mu_2 - \mu_1) \sum X_i}{\sigma^2} - \frac{n(\mu_2^2 - \mu_1^2)}{2\sigma^2}\right)$$

$$3. \quad \bar{X} = \sum X_i$$

$$f(x; \theta) = \theta e^{-\theta x}$$

$$\frac{\prod f(X_i; \theta_2)}{\prod f(X_i; \theta_1)} = \left(\frac{\theta_2}{\theta_1}\right)^n e^{-(\theta_2 - \theta_1) \sum X_i}$$

$$\theta_2 > \theta_1 \quad \sum X_i$$

$$e^{Q(\theta) \cdot T(\mathbf{X})} \quad Q(\theta) \quad MLR$$

tip 5.100

$$\mu \quad " \quad " \quad \sigma^2 \quad " \quad " \quad \sigma^2$$



$$3. \quad n \quad X_1, X_2, \dots, X_n \quad X_1 \sim N(\mu, \sigma^2) \quad \mu_0 \pm \Delta$$

$$1.$$

$$2. \quad \text{UMP}$$

$$3. \quad \chi^2 \quad n \quad 0.01 \quad 1.1\Delta \quad 0.9$$

tip 5.101

$$S^2 \quad (n-1) \quad \sigma^2$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\mu_0 \pm \Delta \quad |\mu - \mu_0| \leq \Delta \quad |\mu - \mu_0| > \Delta$$

$$H_0 : |\mu - \mu_0| \leq \Delta \quad H_1 : |\mu - \mu_0| > \Delta$$

$$H_0 : \sigma^2 \leq \Delta^2 \quad H_1 : \sigma^2 > \Delta^2$$

$$\sigma^2 \quad MLR \quad T(\mathbf{X}) = \sum_{i=1}^n (X_i - \mu_0)^2 \quad UMP$$

$$\sum_{i=1}^n (X_i - \mu_0)^2 > c$$

$$H_0$$

$$\frac{\sum_{i=1}^n (X_i - \mu_0)^2}{\Delta^2} \sim \chi^2(n)$$

$$E_{\sigma^2=\Delta^2}[\phi(\mathbf{X})] = \alpha = 0.01$$

$$c = \Delta^2 \chi_{0.01}^2(n)$$

$$\phi(\mathbf{X}) = \begin{cases} 1, & \sum_{i=1}^n (X_i - \mu_0)^2 > \Delta^2 \chi_{0.01}^2(n) \\ 0, & \sum_{i=1}^n (X_i - \mu_0)^2 \leq \Delta^2 \chi_{0.01}^2(n) \end{cases}$$

$$1.1\Delta \quad \sigma^2 = (1.1\Delta)^2 = 1.21\Delta^2 \quad 0.9$$

$$\beta_\phi(1.21\Delta^2) = P_{\sigma^2=1.21\Delta^2} \left(\sum_{i=1}^n (X_i - \mu_0)^2 > \Delta^2 \chi_{0.01}^2(n) \right) \geq 0.9$$

$$\frac{\sum_{i=1}^n (X_i - \mu_0)^2}{1.21\Delta^2} \sim \chi^2(n)$$

$$P \left(\chi^2(n) > \frac{\chi_{0.01}^2(n)}{1.21} \right) \geq 0.9$$

$$\frac{\chi_{0.01}^2(n)}{1.21} \leq \chi_{0.9}^2(n)$$

 χ^2

$$\frac{\chi_{0.01}^2(n)}{\chi_{0.9}^2(n)} \leq 1.21$$

$$n \quad n = 152 \quad 1.21 \quad n = 152$$

tip 5.102

$$\mu_0 \quad X \quad n$$

 **5** $X_1, X_2, \dots, X_n$

$$f(x; \theta) = \theta e^{-\theta x}, \quad x > 0, \quad \theta > 0.$$

1. UMP

2. UMP

$$(1) \quad \theta_1 < \theta_2$$

$$\frac{\prod_{i=1}^n f(X_i; \theta_2)}{\prod_{i=1}^n f(X_i; \theta_1)} = \left(\frac{\theta_2}{\theta_1} \right)^n e^{-(\theta_2 - \theta_1) \sum_{i=1}^n X_i}$$

$$\theta_2 > \theta_1 \quad T(\mathbf{X}) = \sum_{i=1}^n X_i \quad \text{MLR} \quad \text{MLR} \quad \text{UMP}$$

$$(a) \quad H_0 : \theta \leq \theta_0, \quad H_1 : \theta > \theta_0$$

$$(b) \quad H_0 : \theta \geq \theta_0, \quad H_1 : \theta < \theta_0$$

$$H_0 : \theta = \theta_0 \quad H_1 : \theta \neq \theta_0 \quad \text{UMP}$$

$$(2) \quad \sum X_i$$

$$(a) \quad H_0 : \theta \leq \theta_0 \quad H_1 : \theta > \theta_0 \quad \theta \quad \sum X_i$$

$$\varphi(\mathbf{X}) = \begin{cases} 1, & \sum_{i=1}^n X_i < c_1 \\ 0, & \sum_{i=1}^n X_i \geq c_1 \end{cases}$$

$$c_1 \quad P_{\theta_0}(\sum X_i < c_1) = \alpha$$

$$(b) H_0 : \theta \geq \theta_0 \quad H_1 : \theta < \theta_0$$

$$\varphi(\mathbf{X}) = \begin{cases} 1, & \sum_{i=1}^n X_i > c_2 \\ 0, & \sum_{i=1}^n X_i \leq c_2 \end{cases}$$

$$c_2 \quad P_{\theta_0}(\sum X_i > c_2) = \alpha \quad \sum_{i=1}^n X_i \sim \Gamma(n, \theta)$$

tip 5.103

$$\theta_1 > \theta_0 \quad N-P \quad T > c \quad \theta_2 < \theta_0 \quad N-P \quad T < c' \quad \phi \quad \theta \neq \theta_0 \quad UMP$$

 **6** n X_i i $X_1, \dots, X_n$ Weibull

$$f(x, \lambda) = \lambda c x^{c-1} e^{-\lambda x^c}, \quad x > 0$$

$$c \quad \lambda > 0$$

1.

$$H_0 : \frac{1}{\lambda} \leq \frac{1}{\lambda_0}, \quad H_1 : \frac{1}{\lambda} > \frac{1}{\lambda_0}$$

UMP

2. (1) UMP

$$3. \quad \lambda_0 = \frac{1}{12} \quad n \quad 0.01 \quad \lambda = \frac{1}{15} \quad 0.95$$

X_i^c

$$(I) \quad H_0 : \frac{1}{\lambda} \leq \frac{1}{\lambda_0} \quad H_0 : \lambda \geq \lambda_0 \quad H_1 : \lambda < \lambda_0 \quad \lambda_1 < \lambda_2$$

$$\frac{\prod_{i=1}^n f(X_i; \lambda_2)}{\prod_{i=1}^n f(X_i; \lambda_1)} = \left(\frac{\lambda_2}{\lambda_1} \right)^n e^{-(\lambda_2 - \lambda_1) \sum_{i=1}^n X_i^c}$$

$$\lambda_2 > \lambda_1 \quad T(\mathbf{X}) = \sum_{i=1}^n X_i^c \quad \text{Weibull} \quad \text{MLR} \quad \text{UMP}$$

$$(2) \quad \sum X_i^c \quad \lambda \quad \frac{1}{\lambda} \quad \sum X_i^c$$

$$\varphi(\mathbf{X}) = \begin{cases} 1, & \sum_{i=1}^n X_i^c > c_0 \\ 0, & \sum_{i=1}^n X_i^c \leq c_0 \end{cases}$$

$$X_i^c \sim \text{Exp}(\lambda) \quad \sum_{i=1}^n X_i^c \sim \Gamma(n, \lambda) \quad c_0$$

$$P_{\lambda_0} \left(\sum_{i=1}^n X_i^c > c_0 \right) = \alpha$$

$$2\lambda_0 \sum X_i^c \sim \chi^2(2n)$$

$$c_0 = \frac{\chi_{\alpha}^2(2n)}{2\lambda_0}$$

$$(3) \quad \lambda_0 = \frac{1}{12} \quad \lambda = \frac{1}{15} \quad \alpha = 0.01 \quad \beta_{\varphi} \left(\frac{1}{15} \right) \geq 0.95$$

$$\beta_{\varphi} \left(\frac{1}{15} \right) = P_{\lambda = \frac{1}{15}} \left(\sum X_i^c > \frac{\chi_{0.01}^2(2n)}{2\lambda_0} \right) \geq 0.95$$

$$\lambda = \frac{1}{15} \quad 2\lambda \sum X_i^c = \frac{2}{15} \sum X_i^c \sim \chi^2(2n)$$

$$P \left(\chi^2(2n) > \frac{2\lambda}{2\lambda_0} \chi_{0.01}^2(2n) \right) = P \left(\chi^2(2n) > \frac{\lambda}{\lambda_0} \chi_{0.01}^2(2n) \right) \geq 0.95$$

$$\frac{\lambda}{\lambda_0} = \frac{1/15}{1/12} = \frac{12}{15} = 0.8$$

$$P(\chi^2(2n) > 0.8 \chi_{0.01}^2(2n)) \geq 0.95$$

$$0.8 \chi_{0.01}^2(2n) \leq \chi_{0.95}^2(2n)$$

$$\frac{\chi_{0.95}^2(2n)}{\chi_{0.01}^2(2n)} \geq 0.8$$

$$\chi^2 \quad n \quad \chi^2(m) \approx m \left(1 - \frac{2}{9m} + z \sqrt{\frac{2}{9m}} \right)^3 \quad n$$

$$n = 76$$

 7

$$\{f(x; \theta) : \theta \in \Theta\}$$

$$(a) \quad H_0 : \theta = \theta_0, \quad H_1 : \theta > \theta_0,$$

$$(b) \quad H_0 : \theta = \theta_0, \quad H_1 : \theta < \theta_0,$$

1. UMP

2. (1)

(1) UMP

$$\begin{array}{ccccccc} \text{MLR} & \frac{\prod f(X_i; \theta_2)}{\prod f(X_i; \theta_1)} & T(\mathbf{X}) & H_0 & \theta = \theta_0 & H_1 & \text{MLR} & \text{UMP} \\ (2) & (a) H_0 : \theta = \theta_0 & H_1 : \theta > \theta_0 & & & & & \end{array}$$

$$\varphi_a(\mathbf{X}) = \begin{cases} 1, & T(\mathbf{X}) > c_1 \\ \gamma_1, & T(\mathbf{X}) = c_1 \\ 0, & T(\mathbf{X}) < c_1 \end{cases}$$

$$\begin{array}{l} c_1 \quad \gamma_1 \quad E_{\theta_0}[\varphi_a(\mathbf{X})] = \alpha \\ (b) H_0 : \theta = \theta_0 \quad H_1 : \theta < \theta_0 \end{array}$$

$$\varphi_b(\mathbf{X}) = \begin{cases} 1, & T(\mathbf{X}) < c_2 \\ \gamma_2, & T(\mathbf{X}) = c_2 \\ 0, & T(\mathbf{X}) > c_2 \end{cases}$$

$$c_2 \quad \gamma_2 \quad E_{\theta_0}[\varphi_b(\mathbf{X})] = \alpha$$

5.11 12



1

$$\begin{array}{cccccccccccc} 0.473 & 0.224 & 0.827 & 0.223 & 0.867 & 0.779 & 0.554 & 0.529 & 0.705 & 0.656 & & U(0, 1) \\ U(0, 1) & F(x) = x & x \in [0, 1] & & & & & & & & & \end{array}$$

$$0.223, 0.224, 0.473, 0.529, 0.554, 0.656, 0.705, 0.779, 0.827, 0.867$$

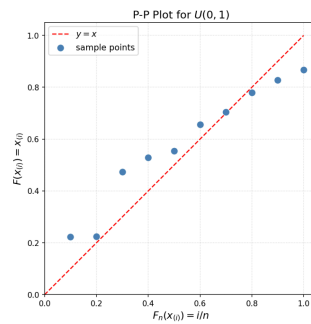
$$F_n(x_{(i)}) = \frac{i}{n} = \frac{i}{10}, \quad F(x_{(i)}) = x_{(i)}$$

i	$x_{(i)}$	$F_n(x_{(i)}) = i/10$	$F(x_{(i)}) = x_{(i)}$
1	0.223	0.1	0.223
2	0.224	0.2	0.224
3	0.473	0.3	0.473
4	0.529	0.4	0.529
5	0.554	0.5	0.554
6	0.656	0.6	0.656
7	0.705	0.7	0.705
8	0.779	0.8	0.779
9	0.827	0.9	0.827
10	0.867	1.0	0.867

P-P

$$F(x_{(i)}) = x_{(i)} \quad F_n(x_{(i)}) = i/10 \quad U(0, 1) \quad y = x$$

$$P-P \quad y = x \quad U(0, 1)$$

**2**

0.54 0.32 0.68 0.71 1.04 0.94 1.34 1.10 0.44 1.01

$$\hat{\mu} = \bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i = 0.8120, \quad \hat{\sigma} = S = 0.3252$$

0.32, 0.44, 0.54, 0.68, 0.71, 0.94, 1.01, 1.04, 1.10, 1.34

$$F_n(x_{(i)}) = \frac{i}{10}, \quad F(x_{(i)}) = \Phi\left(\frac{x_{(i)} - \hat{\mu}}{\hat{\sigma}}\right)$$

i	$x_{(i)}$	$F_n(x_{(i)}) = i/10$	$F(x_{(i)})$
1	0.32	0.1	0.0651
2	0.44	0.2	0.1263
3	0.54	0.3	0.2014
4	0.68	0.4	0.3424
5	0.71	0.5	0.3769
6	0.94	0.6	0.6531
7	1.01	0.7	0.7287
8	1.04	0.8	0.7584
9	1.10	0.9	0.8121
10	1.34	1.0	0.9478

P-P

$$F_n(x_{(i)}) \quad F(x_{(i)}) \quad y = x$$

P-P $y = x$

 **extra** 1 Kolmogorov

30 1 2 3 4 5 6 3 6 4 7 4 6

$H_0 : p_1 = \dots = p_6 = \frac{1}{6}$ H_0 $F_0(x) = \lfloor x \rfloor / 6$
 $n = 30$

k	n_k	$F_n(k)$	$F_0(k)$	$ F_n(k) - F_0(k) $
1	3	0.1000	0.1667	0.0667
2	6	0.3000	0.3333	0.0333
3	4	0.4333	0.5000	0.0667
4	7	0.6667	0.6667	0.0000
5	4	0.8000	0.8333	0.0333
6	6	1.0000	1.0000	0.0000

$D_{30} = 0.0667$

$$\sqrt{n} \cdot D_n = \sqrt{30} \times 0.0667 \approx 0.365$$

$\alpha = 0.05$ $\lambda_{0.05} = 1.358$ $0.365 < 1.358$ H_0

tip 5.104

f_n f_0 D_n n k

 **3** 9, 6, 13, 22, 18, 37, 36, 52, 41, 36 Weibull

Weibull

Weibull

$$F(x) = 1 - \exp \left[- \left(\frac{x}{\lambda} \right)^k \right], \quad x > 0$$

$k > 0$ $\lambda > 0$

$$\hat{k} = 1.9155, \quad \hat{\lambda} = 30.4547$$

6, 9, 13, 18, 22, 36, 36, 37, 41, 52

$$F_n(x_{(i)}) = \frac{i}{10}, \quad F(x_{(i)}) = 1 - \exp \left[- \left(\frac{x_{(i)}}{\hat{\lambda}} \right)^{\hat{k}} \right]$$

P-P

$F_n(x_{(i)})$ $F(x_{(i)})$ $y = x$

P-P $y = x$ **Weibull**

5.12 13

 **5.2.1**

9

i	$x_{(i)}$	$F_n(x_{(i)}) = i/10$	$F(x_{(i)})$
1	6	0.1	0.0436
2	9	0.2	0.0923
3	13	0.3	0.1778
4	18	0.4	0.3059
5	22	0.5	0.4151
6	36	0.6	0.7478
7	36	0.7	0.7478
8	37	0.8	0.7659
9	41	0.9	0.8292
10	52	1.0	0.9384

	0	0.5	1	2	3	4	5	6	7	8
$/10^{-6}$	104.6	104.1	104.4	105.0	106.0	106.8	107.7	108.7	110.6	112.1

$$n = 10 \quad \sum x_i^2 = 204.25 \quad \sum x_i y_i = 3973.35 \quad \bar{x} = 3.65 \quad \bar{y} = 107.0$$

$$S_{xx} = 204.25 - 10 \times 3.65^2 = 71.025, \quad S_{xy} = 3973.35 - 10 \times 3.65 \times 107.0 = 67.85$$

$$\hat{\beta}_1 = \frac{67.85}{71.025} \approx 0.9553, \quad \hat{\beta}_0 = 107.0 - 0.9553 \times 3.65 \approx 103.513$$

$$\hat{y} = 103.513 + 0.9553x$$

$$9 \quad \hat{y} = 103.513 + 0.9553 \times 9 \approx 112.1 \times 10^{-6}$$

 **5.2.2** $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(0, \sigma^2)$ $u \in \mathbb{R}^n$ $u^T u = 1$

$$\frac{X^T(I_n - uu^T)X}{\sigma^2}$$

$$n-1 \quad \chi^2$$

$$X^T = (X_1, X_2, \dots, X_n),$$

$$I_n \quad n$$

$$P = I_n - uu^T \quad P$$

$$P^T = I_n - (uu^T)^T = I_n - uu^T = P$$

$$P^2 = (I_n - uu^T)^2 = I_n - 2uu^T + \underbrace{u(u^T u)u^T}_{=1} = I_n - uu^T = P.$$

$$\text{rank}(uu^T) = 1 \quad \text{rank}(P) = n-1$$

$$X \sim N(\mathbf{0}, \sigma^2 I_n) \quad P \quad n-1$$

$$\frac{X^T P X}{\sigma^2} = \frac{X^T (I_n - uu^T) X}{\sigma^2} \sim \chi^2(n-1). \quad \square$$

 **6** $1 \quad 9 \quad 90\%$

$$I \quad n = 10 \quad \bar{x} = 3.65 \quad \bar{y} = 107.0 \quad S_{xx} = 71.025 \quad S_{xy} = 67.85 \quad \hat{\beta}_1 = 0.9553 \quad \hat{\beta}_0 = 103.513 \quad S_{yy} = \sum y_i^2 - n\bar{y}^2 = 114558.32 - 114490 = 68.32$$

$$\alpha = 0.05 \quad H_0 : \beta_1 = 0$$

$$SS_e = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 68.32 - \frac{67.85^2}{71.025} \approx 3.503, \quad S = \sqrt{\frac{SS_e}{n-2}} \approx 0.6617$$

$$T = \frac{\hat{\beta}_1}{S/\sqrt{S_{xx}}} = \frac{0.9553}{0.6617/\sqrt{71.025}} \approx 12.17$$

$$|T| = 12.17 > t_{0.025}(8) = 2.306 \quad H_0$$

$$90\% \quad x^* = 9$$

$$\hat{y}(9) = 103.513 + 0.9553 \times 9 \approx 112.11$$

$$\sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} = \sqrt{1 + 0.1 + 0.4031} \approx 1.226$$

$$\Delta = t_{0.05}(8) \cdot S \cdot 1.226 = 1.860 \times 0.6617 \times 1.226 \approx 1.509$$

$$9 \quad 90\% \quad (110.60, 113.62) \times 10^{-6}$$

$$7 \quad 18 \quad 5-8$$

(kg)	((dm) ³)	(kg)	((dm) ³)	(kg)	((dm) ³)	(kg)	((dm) ³)
17.1	16.7	15.8	15.2	15.0	14.5	16.7	16.6
10.5	10.4	15.1	14.8	16.0	15.8	16.5	15.9
13.8	13.5	12.1	11.9	17.8	17.6	15.1	15.1
11.9	11.6	18.4	18.3	15.7	15.7	15.1	14.5
10.4	10.2	17.1	16.7				

1.

2. 90% 14 kg

3. 90% 14 kg

4.

1. $n = 18 \quad \bar{x} \approx 15.006 \quad \bar{y} \approx 14.722 \quad S_{xx} \approx 96.39 \quad S_{xy} \approx 95.24$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \approx 0.988, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx -0.114$$

$$\hat{y} = -0.114 + 0.988x \quad \text{SSE} \approx 0.61 \quad \hat{\sigma} \approx 0.195$$

2. $x^* = 14 \quad \hat{y}^* \approx 13.72 \quad \text{SE}_{\hat{y}} = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} \approx 0.0501 \quad t_{0.05}(16) = 1.746 \quad 90\%$

$$13.72 \pm 1.746 \times 0.0501 \approx (13.63, 13.81) \text{ (dm)}^3$$

3. $\text{SE}_{\text{pred}} = \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} \approx 0.2014 \quad 90\%$

$$13.72 \pm 1.746 \times 0.2014 \approx (13.37, 14.07) \text{ (dm)}^3$$

4. $\hat{\beta}_1 \approx 1 \quad \text{dm}^3 \quad \text{kg} \quad 1 \text{ kg/dm}^3 \quad (2) \quad (3)$

5.13 14

1

$$y_{ij} = \mu + \mu_i + e_{ij}, \quad (6.2.4)$$

$$\bar{\mu} = \frac{1}{k} \sum_{i=1}^k \mu'_i$$

$$(6.2.4) \quad \mu \quad y_{ij}$$

$$y_{ij} = \bar{\mu} + \mu_i + e_{ij},$$

$$\mu_i = \mu'_i - \bar{\mu}.$$

1. $\{\mu_i\}$
2. $H_0 : \mu'_1 = \mu'_2 = \cdots = \mu'_k \quad H_0 : \mu_1 = \mu_2 = \cdots = \mu_k = 0$

1. $\mu_i = \mu'_i - \bar{\mu} \quad i$

$$\sum_{i=1}^k \mu_i = \sum_{i=1}^k (\mu'_i - \bar{\mu}) = \sum_{i=1}^k \mu'_i - k\bar{\mu} = k\bar{\mu} - k\bar{\mu} = 0.$$

$$\{\mu_i\}$$

$$\sum_{i=1}^k \mu_i = 0.$$

2. $\mu_i = \mu'_i - \bar{\mu} \quad \mu'_i = \mu_i + \bar{\mu}$
 $\mu'_1 = \mu'_2 = \cdots = \mu'_k \quad c \quad \bar{\mu} = c$

$$\mu_i = \mu'_i - \bar{\mu} = c - c = 0, \quad i = 1, \dots, k.$$

$$\mu_1 = \mu_2 = \cdots = \mu_k = 0$$

$$\mu'_i = \mu_i + \bar{\mu} = \bar{\mu}, \quad i = 1, \dots, k,$$

$$\mu'_1 = \mu'_2 = \cdots = \mu'_k.$$

 2

1()	5	8	7	7	10	8
2	4	6	6	3	5	6
3	6	4	4	5	4	3
4	7	4	6	6	3	5
5	9	3	5	7	7	6

- 1.
- 2.

1. $n_i = 6$

$$\bar{y}_1 = \frac{5+8+7+7+10+8}{6} = \frac{45}{6} = 7.5, \quad s_1 = \sqrt{\frac{13.5}{5}} = \sqrt{2.7} \approx 1.643$$

$$\bar{y}_2 = \frac{4+6+6+3+5+6}{6} = \frac{30}{6} = 5.0, \quad s_2 = \sqrt{\frac{8}{5}} = \sqrt{1.6} \approx 1.265$$

$$\bar{y}_3 = \frac{6+4+4+5+4+3}{6} = \frac{26}{6} \approx 4.333, \quad s_3 = \sqrt{\frac{16/3}{5}} = \sqrt{\frac{16}{15}} \approx 1.033$$

$$\bar{y}_4 = \frac{7+4+6+6+3+5}{6} = \frac{31}{6} \approx 5.167, \quad s_4 = \sqrt{\frac{65/6}{5}} = \sqrt{\frac{13}{6}} \approx 1.472$$

$$\bar{y}_5 = \frac{9+3+5+7+7+6}{6} = \frac{37}{6} \approx 6.167, \quad s_5 = \sqrt{\frac{125/6}{5}} = \sqrt{\frac{25}{6}} \approx 2.041$$

2. $k = 5 \quad n_i = 6 \quad N = 30$

$$\bar{y} = \frac{45+30+26+31+37}{30} = \frac{169}{30} \approx 5.633.$$

$$SS_T = \sum_{i,j} y_{ij}^2 - \frac{(\sum_{i,j} y_{ij})^2}{N} = 1047 - \frac{169^2}{30} = \frac{2849}{30} \approx 94.967,$$

$$SS_A = \sum_{i=1}^5 \frac{(\sum_j y_{ij})^2}{n_i} - \frac{169^2}{30} = \frac{45^2 + 30^2 + 26^2 + 31^2 + 37^2}{6} - \frac{169^2}{30} = \frac{5931}{6} - \frac{28561}{30} = \frac{1094}{30} \approx 36.467,$$

$$SS_E = SS_T - SS_A = \frac{2849}{30} - \frac{1094}{30} = \frac{1755}{30} = 58.5.$$

$$f_A = k - 1 = 4 \quad f_E = N - k = 25$$

				F
A	$\frac{1094}{30} \approx 36.467$	4	$\frac{547}{60} \approx 9.117$	≈ 3.896
E	58.5	25	2.34	
	$\frac{2849}{30} \approx 94.967$	29		

$$F \quad F_{0.05}(4, 25) \approx 2.76$$

$$F = \frac{MS_A}{MS_E} = \frac{547/60}{117/50} = \frac{2735}{702} \approx 3.896 > F_{0.05}(4, 25) \approx 2.76,$$

$$\alpha = 0.05 \quad H_0$$

6

6.1

3 720

668	711	625	701	688	667	694	630	547	703	688	697	703
656	677	700	702	688	691	664	688	679	708	699	667	703

- 1.
- 2.
- 3.

	4	(mg/L)	20										
				133	137	148	149	152	167	174	179	189	192
				201	209	210	211	218	238	245	248	253	257

(1)
(2)

5

page44

3 $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim N(\mu, \sigma^2)$ μ σ^2

$\sum_{i=1}^n X_i$, $X_1 + 2\mu$, $\max_{1 \leq i \leq n} \{X_i\}$, $\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}$, $\sum_{i=1}^n (X_i - \mu)^2$.

4 $N(\mu, \sigma^2)$ μ, σ^2 μ, σ^2 n n

$x_1, x_2, \dots, x_n$.

 **6** $X_1, X_2, \dots, X_n$ i.i.d. X_1

$$f(x, \sigma) = \frac{1}{\pi \sigma} \frac{1}{1 + (x/\sigma)^2}, \quad -\infty < x < \infty,$$

$$\overline{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

page45

 1 (2.2.3)

 2 X

$$P\{X = k\} = p(1 - p)^{k-1}, \quad k = 1, 2, \dots,$$

$$X_1, X_2, \dots, X_n \quad X \quad X_{(1)}, X_{(n)}$$

 **3** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$ $X_{(1)} \quad X_{(n)}$

6.2

6.2.1 45,3,4,6

 **4** $X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim F(x) F(x) f(x)$

(1) $X_{(1)} X_{(n)}$

(2) $R = X_{(n)} - X_{(1)}$

 **5** 5

$$-2.8, -1, 1.5, 2.1, 3.4.$$

 **6** $N(\mu, \sigma^2)$ α $0 < \alpha < 1$ z_α α u_α

$$z_\alpha = \mu + \sigma u_\alpha.$$

6.2.2 46,1,2,3

 **1** χ^2 $X \sim \chi^2(n)$

$$E(X^k) = 2^k \left(\frac{n}{2} + k - 1 \right) \left(\frac{n}{2} + k - 2 \right) \cdots \frac{n}{2}, \quad k = 1, 2, \dots.$$

$$E(X) = n, \quad \text{Var}(X) = 2n.$$

 **2** t $X \sim t(n)$ $k < n$ $E(X^k)$

$$E(X^k) = \begin{cases} 0, & k \\ \frac{n^{k/2} \Gamma\left(\frac{k+1}{2}\right) \Gamma\left(\frac{n-k}{2}\right)}{\sqrt{\pi} \Gamma\left(\frac{n}{2}\right)}, & k \end{cases}$$

$$k \geq n \quad E(X^k)$$

 **3** χ^2 $X_1 \sim \chi^2(m)$ $X_2 \sim \chi^2(n)$ X_1, X_2 $X_1 + X_2$ X_1/X_2

 **5** $F \sim F(m, n)$

$$F_{1-\alpha}(m, n) F_\alpha(n, m) = 1, \quad \forall \alpha \in (0, 1).$$

 $X_1, X_2, \dots, X_n$ $A_{2 \times n}$

$$Y = AX,$$

$$Y \sim X$$

 χ^2 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(0, 1)$

$$\mathbf{X} = (X_1, \dots, X_n)^T,$$

$$A \sim n$$

$$\mathbf{X}^T A \mathbf{X} \sim \chi^2(r)$$

$$A$$

$$A^2 = A,$$

$$\text{rank}(A) = r.$$

6.3

$X_1, \dots, X_n$ i.i.d., $X_i \sim N(\mu, \sigma^2)$ $X_{n+1} \sim N(\mu, \sigma^2)$ $X_1, \dots, X_n$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$T = \sqrt{\frac{n}{n+1}} \frac{\bar{X} - X_{n+1}}{s} \sim t(n-1).$$

$X_1, \dots, X_n$ i.i.d. $X_1 \sim N(0, 1)$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

Var(s^2)

$X_1, \dots, X_m$ i.i.d. $X_1 \sim N(\mu_1, \sigma^2)$ $Y_1, \dots, Y_n$ i.i.d. $Y_1 \sim N(\mu_2, \sigma^2)$

$$s_1^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - \bar{X})^2, \quad s_2^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

1.

$$\frac{(m-1)s_1^2 + (n-1)s_2^2}{\sigma^2}$$

2. s_1^2

$$\frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2}$$

$X_1, \dots, X_n$ i.i.d. $X_1 \sim N(0, 1)$

$$Z = \frac{\sum_{i=1}^{n-1} X_i}{X_n}$$

$X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim P(\lambda)$ λ

$X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim B(k, p)$ p

$X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim U(\theta_1, \theta_2)$ θ_1, θ_2

$X_1, X_2, \dots, X_n$ i.i.d., $X_1 \sim f(x)$

$$f(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right),$$

 σ

6.4

$X_1, X_2, \dots, X_n$ i.i.d. θ

2. $f(x; \theta) = \frac{x}{\theta^2} e^{-x^2/\theta^2}, x > 0, \theta > 0.$

$X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right)$

$$T \in \left[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}\right]$$

 θ

10 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x; \alpha, \beta)$

$$f(x; \alpha, \beta) = \begin{cases} \beta \exp\{-\beta(x - \alpha)\}, & x \geq \alpha, \\ 0, & \end{cases}$$

$\beta > 0 \quad \alpha, \beta$

11 $X_1, X_2, \dots, X_n$ i.i.d. X

$$P\{X = k\} = \theta^{k-1}(1 - \theta), \quad k = 1, 2, \dots, r;$$

$$P\{X = r + 1\} = \theta^r.$$

$$M = \#\{i : X_i = r + 1, i = 1, 2, \dots, n\},$$

θ MLE

$$\hat{\theta} = \frac{\sum_{i=1}^n X_i - n}{\sum_{i=1}^n X_i - M}.$$

12 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim f(x, \theta_1, \theta_2)$

$$f(x, \theta_1, \theta_2) = \begin{cases} \frac{\theta_1}{\theta_2} x^{\theta_1-1} \exp\left(-\frac{x^{\theta_1}}{\theta_2}\right), & x \geq 0, \\ 0, & \end{cases}$$

$\theta_1 \quad \theta_2$

18 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, \sigma^2) \quad \sigma^2 \quad \mu \quad N(\nu, \tau^2)$

1. μ

2. μ

20 Pareto $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(0, \theta) \quad \theta \quad \text{Pareto}(a, b)$

$$\pi(\theta) = \frac{ba^b}{\theta^{b+1}}, \quad a < \theta < \infty, a > 0, b > 0.$$

1. θ

2. θ

6.5

1 $X \sim P(\lambda) \quad 1 \quad g(\lambda) = e^{-2\lambda}$

2 $X \sim B(n, p) \quad g(p) = \frac{1}{1+p^2}$

3 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, 1) \quad g(\mu) = |\mu|$

4 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(\theta_1, \theta_2) \quad \theta_1, \theta_2$

5 $\hat{\theta} \quad \theta^* \quad \theta \quad \tilde{\theta} = a\hat{\theta} + b\theta^* \quad a, b \quad \tilde{\theta} \quad \theta$

6 $X \sim U(0, \theta) \quad X_1, X_2, \dots, X_n$ i.i.d. $X \quad \theta^2, \theta^{-1}$

6.6

1

$$I(\theta) = \text{Var} \left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right) = -E \left(\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right)$$

7 $W \quad \theta \quad W$

(1) W

(2) $E_\theta W = \theta, \forall \theta \in \Theta$

$$(3) \quad (1),(2) \quad W^*$$

$$\text{Var}_\theta(W) \leq \text{Var}_\theta(W^*), \quad \forall \theta \in \Theta,$$

$$W \sim \theta \quad X \sim N(\mu, \sigma^2) \quad (\mu, \sigma^2) \in \Theta = \{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 > 0\} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \mu$$

$$\text{11} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim P(\lambda)$$

$$(1) \quad \bar{X} \sim \lambda \quad \text{UMVUE}$$

$$(2) \quad \tilde{\lambda}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \quad \lambda^2 \quad \text{UMVUE}$$

$$\text{12} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim N(\mu, \sigma^2) \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \mu \quad \sigma^2 \quad \text{UMVUE}$$

$$\mu \quad \sigma^2 \quad \text{C-R}$$

$$\text{14} \quad 3.3.3 \quad \hat{g}(X_1, X_2, \dots, X_n) \quad g(\theta)$$

$$E_\theta \hat{g}(X_1, X_2, \dots, X_n) = g(\theta) + b(\theta), \quad \forall \theta \in \Theta \subset \mathbf{R},$$

$$b(\theta)$$

$$\text{Var}_\theta(\hat{g}(X_1, X_2, \dots, X_n)) \geq [b'(\theta) + g'(\theta)]^2 / nI(\theta).$$

$$\text{15} \quad \hat{g}_1(\mathbf{X}) \quad \hat{g}_2(\mathbf{X}) \quad g(\theta) \quad \hat{g}_1(\mathbf{X}) \quad g(\theta) \quad \text{UMVUE} \quad \hat{g}_1(\mathbf{X}) \quad \hat{g}_2(\mathbf{X})$$

$$\text{16} \quad 0-1 \quad 3.2.12 \quad \theta$$

$$\hat{\theta} = \frac{\sum_{i=1}^n X_i + a}{n + a + b}.$$

$$(1)$$

6.6.1 3.4

$$\text{1} \quad X_n \xrightarrow{d} X \quad Y_n \xrightarrow{a.s.} a > 0$$

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{a}.$$

$$\text{1} \quad \{\hat{\theta}_n\} \quad \lim_{n \rightarrow \infty} E_\theta(\hat{\theta}_n - \theta)^2 = 0 \quad \forall \theta \in \Theta \quad \hat{\theta}_n \quad \theta$$

$$\text{2} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim U(0, \theta) \quad \hat{\theta}_n = X_{(n)} \quad \theta$$

$$\text{4} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim P(\lambda) \quad \bar{X} \quad s_n^2 \quad \lambda$$

$$\text{5} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim U(0, \theta)$$

$$T(X_1, X_2, \dots, X_n) = \sqrt[n]{X_1 X_2 \cdots X_n}$$

$$g(\theta) = \frac{\theta}{e}$$

$$\text{6} \quad X_1, X_2, \dots, X_n$$

$$T(X_1, X_2, \dots, X_n) = \frac{2}{n(n+1)} \sum_{i=1}^n i X_i$$

$$\text{8} \quad X_1, X_2, \dots, X_n \text{ i.i.d. } X_1 \sim f(x, \theta) = \frac{1}{\theta} e^{-x/\theta} \quad x > 0 \quad \theta \quad \text{MLE} \quad N\left(\theta, \frac{\theta^2}{n}\right)$$

6.7

$$\text{9} \quad 3.3.11 \quad \lambda^2 \quad \bar{X}^2 \quad \tilde{\lambda}^2$$

$$\text{10} \quad X_1, \dots, X_n \text{ i.i.d. } X_1 \sim f(x, \theta) \quad \hat{\theta} = \hat{\theta}(X_1, \dots, X_n) \quad \theta$$

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{L} N(0, V(\theta)), \quad \forall \theta \in \Theta,$$

$$V(\theta) \quad \theta \quad V(\hat{\theta}) \quad V(\theta)$$

12 3.4.6 $h(\theta) \quad h'(\theta_0) = 0 \quad h''(\theta_0) \neq 0 \quad h(\hat{\theta}) - h(\theta_0) = \frac{1}{2}(\hat{\theta} - \theta_0)^2(h''(\theta_0) + R)$

$$n(h(\hat{\theta}) - h(\theta_0)) \xrightarrow{L} \frac{1}{2}A(\theta_0)h''(\theta_0)\chi^2(1),$$

$$\chi^2(1) \quad 1 \quad \chi^2$$

6.7.1 3.5

1 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim U(\theta, \theta + 1) \quad \theta \quad 100(1 - \alpha)\%$

2 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(\mu, \sigma^2) \quad \mu \quad \sigma^2 \quad 100(1 - \alpha)\% \quad \mu$

3 $X_1, X_2, \dots, X_n$ i.i.d. X_1

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & x > 0; \\ 0, & . \end{cases}$$

(1) $2n\bar{X}/\theta \sim \chi^2(2n)$

(2) $\theta \quad 100(1 - \alpha)\%$

(3) $T_{0.95} \quad P\{X_1 > T_{0.95}\} = 0.95 \quad T_{0.95} \quad 100(1 - \alpha)\%$

6.8

4 0-1 $p \quad 100 \quad 2 \quad 95\%$

5 $X_1, X_2, \dots, X_m \quad Y_1, Y_2, \dots, Y_n \quad N(\mu_1, \sigma_1^2) \quad N(\mu_2, \sigma_2^2)$

(1) $\sigma_1^2, \sigma_2^2 \quad \mu_2 - \mu_1 \quad 1 - \alpha$

(2) $\sigma_1^2 = \sigma_2^2 \quad 90\% \quad \mu_2 - \mu_1 \quad \frac{2}{5}\sigma_1 \quad n_1 = n_2$

(3) $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2 \quad \sigma_2^2/\sigma_1^2 \quad 1 - \alpha$

6.8.1 4.1

1 $X_1, X_2, \dots, X_n \quad X_1 \sim N(\mu, 1) \quad H_0 : \mu = 0 \quad H_1 : \mu > 0$

$$\phi(\mathbf{X}) = \phi(X_1, X_2, \dots, X_n) = \begin{cases} 1, & \bar{X}_n > 1.3 \\ 0, & \bar{X}_n \leq 1.3 \end{cases}$$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad n = 10$$

(1)

(2) $\mu = 1.0 \quad \mu = 1.5$

(3) $\alpha = 0.05 \quad \bar{X}_n$

2 $X_1, X_2, \dots, X_{10} \quad U(0, \theta)$

$$\phi(\mathbf{X}) = \begin{cases} 1, & X_{(10)} > 0.95 \\ 0, & X_{(10)} \leq 0.95 \end{cases}$$

$$H_0 : \theta = 1 \quad H_1 : \theta > 1 \quad \mathbf{X} = (X_1, \dots, X_{10})^T \quad X_{(10)} = \max(X_1, X_2, \dots, X_{10})$$

(1)

(2)

(3) $X_{(10)} \quad \alpha = 0.05$

3 $B(n, p) \quad P\{X \leq x\} = \sum_{i=0}^x \binom{n}{i} p^i (1-p)^{n-i} \quad p$

6.8.2 4.2

1 4.2.1 σ^2 $\alpha = 0.10$

$$H_0 : \mu = 14; \quad H_1 : \mu \neq 14$$

2 $X_1, X_2, \dots, X_{10} \sim N(\mu, \sigma^2)$

$$(1) \quad H_0 : \mu \leq 1 \quad H_1 : \mu > 1 \quad \alpha = 0.1$$

2 $X_1, X_2, \dots, X_{10} \sim N(\mu, \sigma^2)$

$$(2) \quad H_0 : \sigma^2 \leq 1 \quad H_1 : \sigma^2 > 1 \quad \alpha = 0.1$$

$$(3) \quad \alpha = 0.05 \quad (1) \quad (2)$$

3 $X_1, X_2, \dots, X_m \sim N(\mu_1, \sigma_1^2) \quad Y_1, Y_2, \dots, Y_n \sim N(\mu_2, \sigma_2^2) \quad \sigma_1^2, \sigma_2^2 \quad H_0 : \mu_1 = \mu_2$

$$H_1 : \mu_1 \neq \mu_2$$

$$(2) \quad \sigma_1^2 = k\sigma_2^2 \quad k$$

4 3

$$(1) \quad H_0 : \sigma_1^2 \leq \sigma_2^2 \quad H_1 : \sigma_1^2 > \sigma_2^2$$

$$(2) \quad H_0 : \sigma_1^2 = k\sigma_2^2 \quad H_1 : \sigma_1^2 \neq k\sigma_2^2 \quad k$$

6 $A \quad B \quad A \quad A \quad 10$

10.5 10.3 10.4 9.6 9.7 9.8 9.9 10.1 10.2 9.7

$B \quad 10$

10.2 10.7 10.8 10.9 10.4 10.6 11.0 11.1 10.9 11.2

$$(1) \quad B$$

$$(2) \quad A \quad B$$

6.9 10

6.9.1 4.3

1 36 3 3%

2 $X_1, X_2, \dots, X_n \sim P(\lambda) \quad H_0 : \lambda = \lambda_0 \quad H_1 : \lambda \neq \lambda_0 \quad 5, 7, 8, 6, 9, 10 \quad \alpha = 0.05 \quad \lambda = 3$

6.10 11

6.10.1 4.4

1 $X \sim \lambda$

$$f(x, \lambda) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots$$

$$\lambda > 0 \quad X_1, X_2, \dots, X_n$$

$$H_0 : \lambda \leq \lambda_0; \quad H_1 : \lambda > \lambda_0$$

1. UMP

2. UMP $\alpha = 0.05 \quad n = 10 \quad \lambda_0 = 0.1$

2 $n \quad X_i \quad i \quad X_1, X_2, \dots, X_n \quad \theta \quad n$

1. 10 0.01

2. 15 0.01

 **3** n UMP $X_1, X_2, \dots, X_n$ $X_1 \sim N(\mu, \sigma^2)$ $\mu_0 \pm \Delta$

1.

2. UMP

3. χ^2 n 0.01 1.1 Δ 0.9

 **5** $X_1, X_2, \dots, X_n$

$$f(x; \theta) = \theta e^{-\theta x}, \quad x > 0, \quad \theta > 0.$$

1. UMP

2. UMP

 **6** n X_i i $X_1, \dots, X_n$ Weibull

$$f(x, \lambda) = \lambda c x^{c-1} e^{-\lambda x^c}, \quad x > 0$$

c $\lambda > 0$

1.

$$H_0 : \frac{1}{\lambda} \leq \frac{1}{\lambda_0}, \quad H_1 : \frac{1}{\lambda} > \frac{1}{\lambda_0}$$

UMP

2. (1) UMP

3. $\lambda_0 = \frac{1}{12}$ n 0.01 $\lambda = \frac{1}{15}$ 0.95
 X_i^c

 **7**

$$\{f(x; \theta) : \theta \in \Theta\}$$

$$(a) \quad H_0 : \theta = \theta_0, \quad H_1 : \theta > \theta_0,$$

$$(b) \quad H_0 : \theta = \theta_0, \quad H_1 : \theta < \theta_0,$$

1. UMP

2. (1)

6.11 12

 **1** 0.473 0.224 0.827 0.223 0.867 0.779 0.554 0.529 0.705 0.656 $U(0, 1)$

 **2** 0.54 0.32 0.68 0.71 1.04 0.94 1.34 1.10 0.44 1.01

 **extra** 1 Kolmogorov

30 1 2 3 4 5 6 3 6 4 7 4 6

 **3** 9, 6, 13, 22, 18, 37, 36, 52, 41, 36 Weibull

6.12 13

 **5.2.1**

9

	0	0.5	1	2	3	4	5	6	7	8
$/10^{-6}$	104.6	104.1	104.4	105.0	106.0	106.8	107.7	108.7	110.6	112.1

5.2.2 $X_1, X_2, \dots, X_n$ i.i.d. $X_1 \sim N(0, \sigma^2)$ $u \in \mathbb{R}^n$ $u^T u = 1$

$$\frac{X^T(I_n - uu^T)X}{\sigma^2}$$

$$n-1 \quad \chi^2$$

$$X^T = (X_1, X_2, \dots, X_n),$$

I_n n
6 1 9 90%
7 18 5-8

(kg)	((dm) ³)	(kg)	((dm) ³)	(kg)	((dm) ³)	(kg)	((dm) ³)
17.1	16.7	15.8	15.2	15.0	14.5	16.7	16.6
10.5	10.4	15.1	14.8	16.0	15.8	16.5	15.9
13.8	13.5	12.1	11.9	17.8	17.6	15.1	15.1
11.9	11.6	18.4	18.3	15.7	15.7	15.1	14.5
10.4	10.2	17.1	16.7				

- 1.
- 2. 90% 14 kg
- 3. 90% 14 kg
- 4.

6.13 14

1

$$y_{ij} = \mu + \mu_i + e_{ij}, \tag{6.2.4}$$

$$\bar{\mu} = \frac{1}{k} \sum_{i=1}^k \mu'_i$$

$$(6.2.4) \quad \mu \quad y_{ij}$$

$$y_{ij} = \bar{\mu} + \mu_i + e_{ij},$$

$$\mu_i = \mu'_i - \bar{\mu}.$$

- 1. $\{\mu_i\}$
- 2. $H_0 : \mu'_1 = \mu'_2 = \dots = \mu'_k \quad H_0 : \mu_1 = \mu_2 = \dots = \mu_k = 0$

2

1()	5	8	7	7	10	8
2	4	6	6	3	5	6
3	6	4	4	5	4	3
4	7	4	6	6	3	5
5	9	3	5	7	7	6

- 1.
- 2.



[] μ $X_1, X_2, \dots, X_n$ n
 n n " θ " $T(X_1, \dots, X_n)$
 T θ —

7.1 ()

$X_1, X_2, \dots, X_n$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$R = X_{(n)} - X_{(1)}$$



7.1

7.1

$n = 5$ —
 X_i $F(x)$ $\bar{X}$ $\bar{X}$ " "

$$T = \min\{X_1, X_2, X_3, X_4, X_5\} = X_{(1)}.$$

—

$$T = \max\{X_1, X_2, X_3, X_4, X_5\} = X_{(n)}.$$

- $\bar{X}$ " "
- $X_{(1)}, \dots, X_{(n)}$ " "
 " "" "" r "" "

tip 7.1

" " " "

7.1 (r)

$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} F(x)$ $f(x)$ $X_{(r)}$

$$f_{(r)}(x) = rC_n^r F^{r-1}(x) [1 - F(x)]^{n-r} f(x)$$



- []
- $f(x)$ r x
 - $F^{r-1}(x)$ $r-1$ x
 - $[1 - F(x)]^{n-r}$ $n-r$ x
 - $rC_n^r = \frac{n!}{(r-1)!(n-r)!}$



[]

- $U(\theta_1, \theta_2)$ $[\theta_1, \theta_2]$ $\rightarrow X_{(1)} X_{(n)}$
- $U(0, \theta)$ $[0, \theta]$ $\theta \rightarrow X_{(n)}$
- $f(x) = e^{-(x-\theta)}$ $x \geq \theta$ $\rightarrow X_{(1)}$

- $N(\mu, \sigma^2) \rightarrow \bar{X}, s^2$
- $P(\lambda) \rightarrow \bar{X}$

tip 7.2

$X_1, X_2, \dots, X_8$ i.i.d. $\sim U(0, \theta)$ $1, 2, 3, 4, 5, 6, 7, 8$ $X_{(8)} = \max\{X_1, \dots, X_8\} = 8$ $X_{(8)} = \theta$ $X_{(n)}$
 $\theta = \bar{X} = \mu$

7.2 $X_{(n)} = X_1, \dots, X_8$
 $\theta = 10$

$$P(X_i \leq 5) = \frac{5}{10} = 0.5$$

$$X_i = 50\% \leq 5$$

$$P(X_{(8)} \leq 5) = \left(\frac{5}{10}\right)^8 = \frac{1}{256} \approx 0.4\%$$

$$8 \leq 5 \quad 0.4\%$$

$$8 \leq 5 \quad 5 = X_{(n)} = X_i = X_i$$

tip 7.3

$X_1, \dots, X_8 = X_{(8)} = 5 = 5 = U(0, \theta) = 5 = \frac{5}{\theta}$



$$F_{(n)}(x) = \left(\frac{x}{\theta}\right)^n$$

$$f_{(n)}(x) = \frac{nx^{n-1}}{\theta^n}$$

$$E[X_{(n)}] = \int_0^\theta x \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{\theta^n} \int_0^\theta x^n dx = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n\theta}{n+1}$$

tip 7.4

$$0 = \theta$$

$$E[X_{(n)}] = \frac{n\theta + 0}{n+1}$$

$$\theta_1 = \theta_2 = \theta = \theta_2 = 0 = \theta_1$$

$$E[X_{(n)}] = \frac{n\theta_2 + \theta_1}{n+1}$$

$$E[X_{(1)}] = \theta_2 = \theta_1 = n = \theta_1 = \theta_2$$

$$E[X_{(1)}] = \frac{n\theta_1 + \theta_2}{n+1}$$



$$\rightarrow \bar{X}, s^2 \rightarrow X_{(1)}, X_{(n)}$$

$$I = I_n = [\theta_1, \theta_2] \quad n = n+1$$

$$\frac{\theta_2 - \theta_1}{n+1}$$

$$\bullet X_{(1)} \quad \theta_1$$

$$E[X_{(1)}] = \theta_1 + \frac{\theta_2 - \theta_1}{n+1} = \frac{n\theta_1 + \theta_2}{n+1}$$

$$\bullet X_{(n)} \quad \theta_2$$

$$E[X_{(n)}] = \theta_2 - \frac{\theta_2 - \theta_1}{n+1} = \frac{\theta_1 + n\theta_2}{n+1}$$

$$\begin{array}{ccccccc} X_{(1)} & \theta_1 & \theta_1 & n & \theta_2 & 1 & X_{(n)} \\ n & n+1 & & 1 & & n & \end{array}$$

7.2 t F

7.2 (χ^2)

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(0, 1)$$

$$\xi = \sum_{i=1}^n X_i^2$$

$$\xi \sim \chi^2(n) \quad \xi \sim \chi^2(n)$$



$$[\quad]$$

$$E(\xi) = n, \quad \text{Var}(\xi) = 2n.$$

tip 7.5

$$\xi \quad E X \quad 0 \quad I \quad n \quad 0I$$



$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 & E(X^2) &= \text{Var}(X) + [E(X)]^2 \\ [\text{Var}(\xi) = 2n] \end{aligned}$$

$$\text{Var}(\xi) = \sum_{i=1}^n \text{Var}(X_i^2)$$

$$X_i \sim N(0, 1)$$

$$\text{Var}(X_i^2) = E(X_i^4) - [E(X_i^2)]^2 = 3 - 1 = 2$$

$$E(X^4) = 3 \quad \text{Var}(\xi) = 2n$$

7.2 ()

$$\xi \sim \chi^2(m) \quad \eta \sim \chi^2(n) \quad \xi \perp \eta$$

$$\xi + \eta \sim \chi^2(m+n).$$



$$[\quad]$$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0 \quad \text{---} \quad \chi^2(n-1) \quad \chi^2(n)$$

7.3 (t - / /)

$$X \sim N(0, 1) \quad Y \sim \chi^2(n) \quad X \perp Y$$

$$\xi = \frac{X}{\sqrt{Y/n}}$$

$$\xi \sim t(n) \quad \xi \sim t(n)$$



tip 7.6

$$t$$



$$[t] \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad \sigma \quad S \quad t$$

tip 7.7

$$\sigma \quad Z \quad \sigma \quad S \quad S \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1) \quad t$$



$$[]$$

$$t(n) \xrightarrow{n \rightarrow \infty} N(0, 1)$$

t

7.4 (F)

$$X \sim \chi^2(m) \quad Y \sim \chi^2(n) \quad X \perp Y$$

$$\xi = \frac{X/m}{Y/n}$$

$$\xi \quad (m, n) \quad F \quad \xi \sim F(m, n)$$

**tip 7.8**

$$F \quad F$$



$$[]$$

$$\xi \sim F(m, n) \implies \frac{1}{\xi} \sim F(n, m)$$



$$[]$$

$$F_{1-\alpha}(m, n) = \frac{1}{F_{\alpha}(n, m)}$$

$$F_{\alpha} \quad F_{1-\alpha}$$

7.2.1 t

$$f(x) = \quad \times \quad \chi^2(n)$$

$$f(x) = \underbrace{\frac{1}{2^{n/2}\Gamma(n/2)}}_{\chi^2(n)} \times \underbrace{x^{\frac{n}{2}-1}e^{-x/2}}_{\chi^2(n)}, \quad x > 0.$$

$$t(n)$$

$$f(x) = \underbrace{\frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}}_{\chi^2(n)} \times \underbrace{\left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}}_{\chi^2(n)}.$$



$$N(0, 1): \quad \varphi(x) = \frac{1}{\sqrt{2\pi}} \cdot \underbrace{e^{-x^2/2}}_{\chi^2(n)}, \quad x \in \mathbb{R}.$$

$$\chi^2(1): \quad f(x) = \frac{1}{\sqrt{2\pi}} \cdot \underbrace{x^{-1/2}e^{-x/2}}_{\chi^2(n)}, \quad x > 0.$$

$$\chi^2(n): f(x) = \underbrace{\frac{1}{2^{n/2}\Gamma(\frac{n}{2})}} \cdot \underbrace{x^{\frac{n}{2}-1}e^{-x/2}}, \quad x > 0.$$

tip 7.9

$$\Gamma(x) = \int_0^\infty e^{-x} x^{x-1} dx \quad x > 0$$

tip 7.10

$$x = 2u \quad \frac{dx}{du} = 2 \quad \Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du \quad \text{---} \quad \int_0^\infty x^{z-1} e^{-x/2} \frac{dx}{2} = \int_0^\infty u^{z-1} e^{-u} du$$

tip 7.11

$$\begin{aligned} \Gamma(1/2) &= \int_0^\infty x^{-1/2} e^{-x} dx \quad x = u^2 \quad dx = 2u du \quad x^{-1/2} = u^{-1} \\ \Gamma\left(\frac{1}{2}\right) &= \int_0^\infty x^{-1/2} e^{-x} dx = \int_0^\infty u^{-1} e^{-u^2} \cdot (2u du) = \int_0^\infty 2e^{-u^2} du \\ e^{-u^2} \quad 2 \int_0^\infty e^{-u^2} du &= \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi} \\ \Gamma\left(\frac{1}{2}\right) &= \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi} \\ x = u^2 \quad Y = X^2 \quad N(0, 1) \quad \chi^2(1) &= \frac{1}{\sqrt{2\pi}} \end{aligned}$$

 Γ

- $\Gamma(z+1) = z\Gamma(z)$ " "
- $\Gamma(n) = (n-1)!$ " "
- $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ " " $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$

tip 7.12

$$\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du \quad \text{---} \quad \Gamma(z+1) = z\Gamma(z)$$

 Γ

$$\Gamma(z+1) = z\Gamma(z) \quad \text{---} \quad \Gamma(z+1) = \int_0^\infty u^z e^{-u} du = \left[-u^z e^{-u}\right]_0^\infty + z \int_0^\infty u^{z-1} e^{-u} du = z\Gamma(z).$$

$$u^z = z u^{z-1} e^{-u} - e^{-u} \quad 0 \quad \Gamma(z+1) = z\Gamma(z)$$

$$\text{Euler motivation } n! = n \cdot (n-1) \cdots 1 \quad \text{---} \quad \Gamma(n) = (n-1)!$$



$$\Gamma\left(\frac{n}{2}\right) = \frac{\sqrt{\pi}}{2^{n/2}} \chi^2(n/2) \quad \text{---} \quad \Gamma\left(\frac{n}{2}\right) = \frac{\sqrt{\pi}}{2^{n/2}} \chi^2(n/2)$$

7.3

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1), \quad \bar{X} \perp s^2$$

$$\sigma^2 \quad \mu$$

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$$

$$Y_1, \dots, Y_m \stackrel{i.i.d.}{\sim} N(\mu_2, \sigma_2^2)$$

$$F = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim F(n-1, m-1)$$

7.5 ()

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$n-1 \quad n \quad E(s^2) = \sigma^2$$



tip 7.13

$$n-1 \quad \mu \quad 0 \quad \sigma$$



$$[\quad]$$

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2/\sigma^2}{n-1}}} = \frac{N(0,1)}{\sqrt{\chi^2(n-1)/(n-1)}} \sim t(n-1)$$

$$\chi^2(n-1) \quad \sigma^2 \quad t$$

tip 7.14

$$n \quad \sigma \quad l \quad t \quad n-1$$



$$[\quad n-1] \quad n$$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

$$n-1 \quad n-1 \quad n-1$$



$$[\quad]$$

- $\sigma^2 = E[(X - \mu)^2] \quad \mu$
- $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \bar{X} \quad E \quad /n-1$

7.3.1

7.1 ()

$$X_1, \dots, X_n \quad X_i \sim N(\mu_i, \sigma^2) \quad A \quad n \times n \quad A^\top A = I \quad Y = AX \quad Y_1, \dots, Y_n$$

$$Y_i \sim N\left(\sum_k a_{ik} \mu_k, \sigma^2\right)$$



tip 7.15

$$X_i \quad i \quad \mathbf{X} = (X_1, X_2, \dots, X_n)^\top \quad n$$

tip 7.16

,

$$[\text{Strang}] \quad AX = A \quad X$$

$$AX = x_1 \text{col}_1(A) + x_2 \text{col}_2(A) + \cdots + x_n \text{col}_n(A)$$

$$AX = A$$

7.3

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

3

3

4

$$AX = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 4 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 8 \\ 4 \end{pmatrix} = \begin{pmatrix} 11 \\ 10 \end{pmatrix}$$



$$[\quad] \quad n$$

—

$$Y \quad \sigma^2$$

7.3 ()

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(0, 1)$$

$$\sum_{i=1}^n X_i^2 = Q_1 + Q_2 + \cdots + Q_k$$

$$Q_i = \mathbf{X}^\top A_i \mathbf{X} \quad \text{rank}(A_i) = n_i \quad Q_1, \dots, Q_k \sim \chi^2(n_i)$$

$$n_1 + n_2 + \cdots + n_k = n$$



$$[\quad] \quad n$$

7.4

7.6 ()

$$X, Y \stackrel{i.i.d.}{\sim} N(0, 1)$$

$$\xi = \frac{X}{Y}$$

$$\xi \quad \xi \sim C(0, 1)$$

$$f(x) = \frac{1}{\pi(1+x^2)}.$$



$$[\quad t \quad t(1) = C(0, 1)] \quad t \quad \xi = \frac{X}{\sqrt{Y/n}} \quad X \sim N(0, 1) \quad Y \sim \chi^2(n)$$

$$n=1 \quad Y \sim \chi^2(1) \quad Y = Z^2 \quad Z \sim N(0, 1)$$

$$\xi = \frac{X}{\sqrt{Z^2}} = \frac{X}{|Z|}.$$

$$X \geq 0 \quad Z = X/|Z| \quad X/Z$$

$$t(1) = C(0, 1).$$

tip 7.17

t

n = 1

n → ∞

"

"

n

"

"

n = 1



$$[\quad]$$

$$\bullet \quad f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

• $f(x) = \frac{1}{\pi(1+x^2)}$ $1/x^2$ " "

$$\int_{-\infty}^{\infty} |x| \cdot \frac{1}{\pi(1+x^2)} dx$$

$$\pm\infty \int \frac{1}{x} dx \implies E(X)$$

tip 7.18

$$44(6) \quad e^{-t^2/2} \quad t=0 \quad e^{-|t|} \quad t=0 \quad " \quad " \quad 0 \quad \iff \quad \text{---} \quad " \quad "$$



$$[\quad \bar{X} \quad]$$

	$\bar{X}$	n
$N(\mu, \sigma^2)$	$N(\mu, \sigma^2/n)$	$\rightarrow 0 \quad \bar{X} \rightarrow \mu \quad LLN$
$C(\mu, \gamma)$	$C(\mu, \gamma)$	LLN



LLN " " " "

$$[\quad] \quad a^2$$

$$aX \sim N(a\mu, a^2\sigma^2).$$

$$|a|$$

$$aC(0, 1) = C(0, a).$$

" " γ " "

tip 7.19

$$" \quad " \quad "t(1)" \rightarrow \rightarrow LLN$$

7.7 ($C(\mu, \gamma)$)

$$f(x) = \frac{\gamma}{\pi(\gamma^2 + (x - \mu)^2)}$$

$$\mu \quad \gamma > 0 \quad \mu = 0, \gamma = 1 \quad f(x) = \frac{1}{\pi(1+x^2)}$$



tip 7.20

$$C(\mu, \gamma)$$

• μ *location* " " — $x = \mu$ μ

• γ *scale* " / " σ σ^2

γ $x = \mu$ " " $x = \mu$ γ — γ " " *half-width at half-maximum* γ γ

8.1



θ " " $X_1, \dots, X_n$ " "

- $E(X^k) = g_k(\theta)$ θ θ
 - $\frac{1}{n} \sum_{i=1}^n X_i^k$ θ
- LLN $n \rightarrow \infty$

$$\frac{1}{n} \sum_{i=1}^n X_i^k \xrightarrow{P} E(X^k).$$

" " " " θ —

$$g_k(\theta) = \frac{1}{n} \sum_{i=1}^n X_i^k,$$

θ " " θ

tip 8.1

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad P(\lambda) \quad \lambda \quad \lambda \quad \text{—} \quad \lambda \quad \lambda \quad E(X) = \lambda$$



$\hat{\theta}$ *hat* θ *hat*

$$\begin{array}{c} \text{MLE} \quad \hat{\theta}_{\text{MLE}} \\ \hat{\theta}_M = \\ \hat{\theta}_B \quad E(\theta | \mathbf{X}) \end{array}$$

hat

8.2 MLE

8.1 ()

$X_1, \dots, X_n$ $f(x; \theta)$ $\theta \in \Theta$ $x_1, \dots, x_n$

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

$$\hat{\theta} \in \Theta$$

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta),$$

$\hat{\theta}$ θ *MLE*



tip 8.2

$$\begin{array}{lcl} f(x; \theta) & \theta & x \rightarrow \text{'' '' } x \quad \theta \rightarrow \text{'' '' } \text{---} \\ \hat{\theta} & L(\theta) & \theta \\ \text{'' ''} = & \theta & \theta \quad \hat{\theta} \end{array}$$

8.2.1 mle

8.2 ()

$$X \quad f(x; \theta)$$

$$\text{supp}(f) = \{x \in \mathbb{R} : f(x; \theta) > 0\}$$

$$\text{---} \quad x \quad f(x; \theta) = 0$$

$$\bullet N(\mu, \sigma^2) = \mathbb{R} \text{ ---}$$

$$\bullet U(0, \theta) = [0, \theta] \text{ --- } \theta$$



$$[MLE \quad \theta \quad]$$

$$\bullet \theta \quad \theta \quad N(\mu, \sigma^2) \text{ Exp}(\lambda)$$

$$f(x; \theta) \quad \theta$$

$$\ell(\theta) = \sum_{i=1}^n \ln f(x_i; \theta), \quad \ell'(\theta) = 0 \implies \hat{\theta}.$$

$$\ell''(\hat{\theta}) < 0$$

$$\bullet \theta \quad \theta \quad U(0, \theta) \quad x \geq \mu$$

$$L(\theta) \quad \mathbb{K}[\dots]$$

$$1. \quad \text{''}\theta \quad \text{''} \quad \theta$$

$$2. \quad L(\theta) \quad \theta$$

$$3. \quad \hat{\theta} \quad X_{(1)} \quad X_{(n)}$$



$$[MLE \quad] \quad \hat{\theta} \quad \theta \quad MLE \quad g(\hat{\theta}) \quad g(\theta) \quad MLE$$

$$\widehat{g(\theta)}_{MLE} = g(\hat{\theta}_{MLE}).$$

$$L(\theta) \quad \theta = \hat{\theta} \quad \theta \quad \eta = g(\theta) \quad \text{---} \quad \hat{\theta} \quad g(\hat{\theta})$$

$$X \sim \text{Exp}(\lambda) \quad MLE \quad \hat{\lambda} = 1/\bar{X} \quad 1/\lambda \quad MLE \quad 1/\hat{\lambda} = \bar{X}$$

8.3

8.3.1



$$\theta$$

$$\begin{array}{c} \text{---} \\ \text{---} \\ \pi(\theta) \quad \theta \\ f(X | \theta) \quad \theta \\ p(\theta | X) \end{array}$$



$$[MLE \text{ Bayes} \quad] \quad \text{---} \quad \theta \quad \text{''} \quad \text{''}$$

$$\bullet MLE \quad \text{''} \quad \text{''} \quad \theta$$

• Bayes () + →

Bayes " "

$$p \approx 0.5 \quad \approx 0.8 \quad \hat{p} \approx 0.7$$

8.1 $n = 10 \quad k = 7 \quad \theta \quad \pi(\theta) \sim \text{Beta}(2, 2) \quad \pi(\theta) \propto \theta(1 - \theta)$

$$f(X | \theta) = \binom{10}{7} \theta^7 (1 - \theta)^3 \propto \theta^7 (1 - \theta)^3.$$

$$p(\theta | X) \propto \theta^7 (1 - \theta)^3 \cdot \theta(1 - \theta) = \theta^8 (1 - \theta)^4,$$

Beta(9, 5) Bayes

$$\hat{\theta} = E[\theta | X] = \frac{9}{9 + 5} = \frac{9}{14} \approx 0.643.$$

$$7/10 = 0.7 \quad 0.5$$



$$p(\mu | \mathbf{X}) = \frac{f(\mathbf{X} | \mu) \cdot \pi(\mu)}{f(\mathbf{X})}$$

$$f(\mathbf{X}) = \int f(\mathbf{X} | \mu) \pi(\mu) d\mu \quad \mu$$

$$p(\mu | \mathbf{X}) \propto f(\mathbf{X} | \mu) \cdot \pi(\mu)$$

$$\int p(\mu | \mathbf{X}) d\mu = 1$$

$$p(\mu | \mathbf{X}) \propto \exp\left(-\frac{(\mu - \mu_1)^2}{2\tau_1^2}\right)$$

$$N(\mu_1, \tau_1^2) \propto \frac{1}{\sqrt{2\pi} \tau_1}$$

8.3.2 beta

8.3 (Beta)

Beta B

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt, \quad \alpha, \beta > 0.$$

Beta $B(2, 2) = \int_0^1 t(1-t) dt = \frac{1}{6}$

Gamma

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$



Beta

1. $[0, 1]$ Gamma

Gamma $(0, \infty)$

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

Beta $(0, 1)$

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

$x \quad 1-x \quad 1$ — “ ”

2.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{1}{(n+1)B(k+1, n-k+1)}$$

*Beta**Beta***tip 8.3** $B(\alpha, \beta)$ $\alpha \quad \beta$

α	$+1$
β	$+1$

- Beta(2, 2) " **1 1** "
- Beta(1, 1) 0 0 — $U(0, 1)$
 $\alpha \quad \beta$



$$\text{Beta-Binomial} = \sum_{k=0}^n \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

$$\underbrace{\text{Beta}(\alpha, \beta)} \xrightarrow{n} \underbrace{\text{Beta}(\alpha + k, \beta + n - k)}$$

$$\theta^{\alpha-1}(1-\theta)^{\beta-1} \cdot \theta^k(1-\theta)^{n-k} = \theta^{(\alpha+k)-1}(1-\theta)^{(\beta+n-k)-1},$$

*Beta***tip 8.4**

				<i>Bayes</i>
$B(n, \theta)$	$\text{Beta}(\alpha, \beta)$	$\text{Beta}(\alpha + k, \beta + n - k)$		$\frac{\alpha + k}{\alpha + \beta + n}$
$P(\lambda)$	$\Gamma(\alpha, \beta)$	$\Gamma(\alpha + n\bar{x}, \beta + n)$		$\frac{\alpha + n\bar{x}}{\beta + n}$
$\text{Exp}(\lambda)$	$\Gamma(\alpha, \beta)$	$\Gamma(\alpha + n, \beta + n\bar{x})$		$\frac{\alpha + n}{\beta + n\bar{x}}$
$N(\theta, \sigma^2) \quad \sigma^2$	$N(\mu_0, \tau_0^2)$	$N(\mu_1, \tau_1^2)$		μ_1
$U(0, \theta)$	$\text{Pareto}(\alpha, \beta)$	$\text{Pareto}(\alpha + n, \max(\beta, x_{(n)}))$		$\frac{\alpha + n}{\alpha + n - 1} \max(\beta, x_{(n)})$

$$\bullet \quad = \quad +$$

•

$$\bullet \quad - \quad n \quad \max$$



Gamma $\Gamma(\alpha, \beta) \quad \pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda} \quad \alpha \quad \beta \quad \alpha/\beta$
Beta(α, β)

Beta(α, β)	$\Gamma(\alpha, \beta)$
α	+1
β	+1
$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha}{\beta}$
(0, 1)	(0, ∞)

Beta $\alpha + \beta$ 1 *Gamma* β
 $B(1, \theta) +$ *Beta*(a, b) *Binomial/Beta* *Beta*

$$p(\theta|\mathbf{X}) \propto \theta^{S+a-1}(1-\theta)^{n-S+b-1} \implies \theta|\mathbf{X} \sim \text{Beta}(S+a, n-S+b)$$

$$\text{Beta} \quad E[\text{Beta}(\alpha, \beta)] = \frac{\alpha}{\alpha + \beta}$$

$$\hat{\theta}_B = \frac{S+a}{n+a+b}, \quad S = \sum_{i=1}^n X_i$$

$$\hat{\theta}_B \quad \text{MSE} \quad S \sim B(n, \theta) \quad ES = n\theta \quad \text{Var}(S) = n\theta(1-\theta) \quad \text{MSE} \quad S$$

8.3.3



Pareto(a, b)

$$\pi(\theta) = \frac{ba^b}{\theta^{b+1}}, \quad \theta > a.$$

$$a \quad \theta$$

$$b$$

Pareto($\max(x_{(n)}, a), n+b$)

- $\max(x_{(n)}, a) \quad \theta$
- $n+b$

9.1

$X_1, X_2, \dots, X_n \sim \mu$
 $n = 3, 2, 4, 6$

$\bar{X}$	$(2+4+6)/3$	4
X_1		2
$(X_1 + X_3)/2$	$(2+6)/2$	4
5	5	5
$\max(X_i)$		6

” ” ——— ” $\mu =$ ”

$\bar{X}$ ” ” $\bar{X} = X_1$
 = ” ”

“ ”

μ
 “ ”

$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$ $E[X_i] = \mu$

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \cdot n\mu = \mu$$

μ “ $\mu = 3$ $\mu = 100$ μ
 ” ———

x1 x2

tip 9.1

$x1$ $x2$



[UMVUE]

A B

UMVUE

UMVUE

9.1UMVUE $X_1, X_2, X_3, X_4 \stackrel{\text{i.i.d.}}{\sim} N(\mu, 1)$ μ μ

$$\hat{\mu}_A = \bar{X} = \frac{X_1 + X_2 + X_3 + X_4}{4}, \quad \text{Var}(\hat{\mu}_A) = \frac{1}{4}$$

$$\hat{\mu}_B = \frac{X_1 + X_2}{2}, \quad \text{Var}(\hat{\mu}_B) = \frac{1}{2}$$

$$\hat{\mu}_C = X_1, \quad \text{Var}(\hat{\mu}_C) = 1$$

$$\mu \quad \bar{X} \quad \bar{X} \quad \mu \quad X_1 \quad \text{UMVUE}$$

- X_1 160 180 $N(\mu, \sigma^2)$ $E[X_1] = \mu$
- x_1 172 $\bar{x} = (172 + 168 + \dots)/n$
 X_1 $E[\cdot]$

tip 9.2

$$h(X) \quad E[h(X)] = \sum_k h(k) \cdot P(X = k) = g(\theta) \quad g(\theta)$$

$$X_1 \quad " \quad x_1 \quad " \quad "$$

tip 9.3

tip 9.4

9.2 umvue+CR

9.2.1 fisher

Fisher θ " "

$$\theta \quad \theta \quad \theta \quad \theta$$

$$S = \frac{\partial}{\partial \theta} \ln f(X; \theta) \quad \theta \quad \theta$$

$$\bullet S \rightarrow \theta \rightarrow$$

$$\bullet S \rightarrow \theta \rightarrow$$

$$f(X; \theta) \quad X \quad X \quad \theta$$

$$f(X; \theta) = " \quad \theta \quad X \quad "$$

$$\theta$$

$$\bullet X \quad S = \frac{\partial}{\partial \theta} \ln f(X; \theta) \quad f(X; \theta) \quad L(\theta)$$

$$\bullet X_1, \dots, X_n \quad L(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

$$S_n = \frac{\partial}{\partial \theta} \ln L(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \ln f(X_i; \theta) = \sum_{i=1}^n S(X_i; \theta)$$

$$L(\theta)$$

tip 9.5

$$\ln \quad \theta$$

Likelihood function Likelihood " / " " θ " " " likelihood—

$$\bullet f(x | \theta) \quad \theta \quad " \quad \theta \quad "$$

- $f(x; \theta)$ θ

$$S(X; \theta) = X$$

$$S(X; \theta) = \frac{\partial}{\partial \theta} \ln f(X; \theta)$$

$$X \quad S \quad S$$

$$E[S] = 0 \quad f(x; \theta) \quad x \quad 1 \quad \theta \quad 0$$

$$"S \quad 0 \quad "$$

- $\text{Var}(S) \rightarrow S \rightarrow \theta \rightarrow$

- $\text{Var}(S) \rightarrow S \rightarrow 0 \rightarrow \theta \rightarrow$

$$E[S] = 0$$

$$\frac{\partial}{\partial \theta} \int f(x; \theta) dx = \int \frac{\partial}{\partial \theta} f(x; \theta) dx$$

$$\int f dx = 1 \quad 0 \quad E[S] \quad E[S] = 0$$

$$\theta U(0, \theta) \quad \theta$$

$$\frac{\partial}{\partial \theta} \int_0^\theta \frac{1}{\theta} dx = \frac{\partial}{\partial \theta} 1 = 0$$

$$\int_0^\theta \frac{\partial}{\partial \theta} \frac{1}{\theta} dx = \int_0^\theta -\frac{1}{\theta^2} dx = -\frac{1}{\theta}$$

$$E[S] = -\frac{1}{\theta} \neq 0 \quad C-R \quad U(0, \theta)$$

9.2.2 CR

C-R

$$\hat{\theta}$$

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

$$" \quad " \quad A \quad B \quad A \quad B \quad A$$

$$C-R \quad \frac{1}{nI(\theta)}$$

- $\text{Var}(\hat{\theta}) = \frac{1}{nI(\theta)} \rightarrow$

- $\text{Var}(\hat{\theta}) > \frac{1}{nI(\theta)} \rightarrow$

$$E[\hat{\theta}] = \theta$$

$$\int \hat{\theta} \cdot \frac{\partial f}{\partial \theta} dx = 1$$

$$\frac{\partial f}{\partial \theta} = f \cdot S$$

$$E[\hat{\theta} \cdot S] = 1$$

$$E[S] = 0$$

$$\text{Cov}(\hat{\theta}, S) = E[\hat{\theta} \cdot S] - E[\hat{\theta}] \cdot E[S] = 1$$

-

$$\text{Cov}(\hat{\theta}, S)^2 \leq \text{Var}(\hat{\theta}) \cdot \text{Var}(S) \implies 1 \leq \text{Var}(\hat{\theta}) \cdot I(\theta) \implies \text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

$$n \quad n \quad \frac{1}{nI(\theta)}$$

tip 9.6

$$\theta \quad \frac{\partial f}{\partial \theta} \quad S \quad E[S] = 0 \quad \text{Var}(S) = I(\theta)$$



[]

$$\frac{d}{d\theta} \int \hat{g} \cdot L(\theta; \mathbf{x}) d\mathbf{x} = \int \hat{g} \cdot \frac{\partial L(\theta; \mathbf{x})}{\partial \theta} d\mathbf{x}$$

$$1. \quad \theta \longrightarrow \theta$$

$$2. \quad f(x; \theta) \quad \theta$$

3.

9.3 umvue



Uniformly Minimum Variance Unbiased Estimator

$$\theta \quad \text{---} \quad X_1 \quad \bar{X}$$

" " $\theta \quad \theta \in \Theta$ "Uniformly"

UMVUE

UMVUE

tip 9.7

uniformity uniform m v ue

9.3.1



$$[\quad] \quad N(\mu, 1) \quad n = 2 \quad T = X_1$$

$$X_1 = 3, X_2 = 5$$

$$T = X_1 = 3 \quad \mu \quad \mu \approx 3 \quad X_2 = 5 \quad N(\mu, 1) \quad \mu \quad \text{---} \quad X_2 = 5 \quad \mu \quad 4 \quad T = X_1 \quad X_2$$

$$\mu \quad \text{" } \theta \quad \text{"}$$

$$T = \bar{X} = 4 \quad \bar{X} = 4 \quad X_1 = 3, X_2 = 5 \quad \text{---} X_1 \quad | \quad X_2 \quad | \text{---} \quad \mu \quad \mu$$

$$\bar{X} = 4 \quad (3, 5) \quad (4, 4) \quad \text{"}\mu \quad \text{"}$$

$$T \quad \text{"} \quad T \quad \theta \quad \text{"}$$

" " --- Cauchy " "

$$\mathbb{Q} \quad 1, 1.4, 1.41, 1.414, \dots \quad \text{Cauchy} \quad \sqrt{2} \quad \mathbb{Q}$$

$$\mathbb{R} \quad \text{Cauchy} \quad \mathbb{R}$$

$$T \quad g(T) \quad \text{"} \quad \text{"} \quad E[g(T)] = 0 \quad \forall \theta \quad g \quad \sqrt{2} \quad \mathbb{Q} \quad \text{"} \quad \text{"}$$

$$T \quad E[g(T)] = 0 \quad g \quad \text{"} \quad \text{"}$$

$$g_1(T) \quad g_2(T) \quad \theta$$

$$E_\theta[g_1(T) - g_2(T)] = 0 \quad \forall \theta$$

$$g_1(T) = g_2(T) \text{---} T \quad \text{UMVUE}$$

tip 9.8

$$T \quad \text{umvue}$$

9.3.2

9.1 (Fisher-Neyman)

$$T(\mathbf{X}) \quad \theta \quad \Longleftrightarrow$$

$$f(\mathbf{x}; \theta) = g(T(\mathbf{x}), \theta) \cdot h(\mathbf{x})$$

$$g \quad T \quad h \quad \theta$$



$$f(\mathbf{x}; \theta) = \prod_{i=1}^n f(x_i; \theta) \quad \theta$$

$$f(\mathbf{x}; \theta) = g(T(\mathbf{x}), \theta) \cdot h(\mathbf{x})$$

$$T(\mathbf{X}) \quad \theta$$

- $g(T(\mathbf{x}), \theta) \quad \theta \quad T(\mathbf{x}) \quad \theta \quad T(\mathbf{x})$
- $h(\mathbf{x}) \quad \theta \quad \theta \quad \text{---} \quad \theta \quad h(\mathbf{x}) \quad \theta$

tip 9.9

- θ
- $T(\mathbf{X}) \quad \theta \quad h(\mathbf{x}) \quad \theta$

$$9.2 \quad \sigma^2 \quad X_i \sim N(\mu, \sigma^2)$$

$$\prod_{i=1}^n f(x_i; \mu) = \underbrace{\exp\left\{\frac{\mu \sum x_i}{\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right\}}_{g(T, \mu), \quad T = \sum x_i} \cdot \underbrace{\left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left\{-\frac{\sum x_i^2}{2\sigma^2}\right\}}_{h(\mathbf{x}), \quad \mu}$$

$$T = \sum X_i \quad \bar{X} \quad \mu$$

$$[\text{Rao-Blackwell}] \quad T \quad \delta(\mathbf{X})$$

$$\varphi(T) = E[\delta(\mathbf{X}) \mid T]$$

$$\text{Var}(\varphi(T)) \leq \text{Var}(\delta(\mathbf{X}))$$

tip 9.10

$$T \quad \theta \quad \delta(\mathbf{X}) \quad X_1 \quad E[\delta | T] \quad \varphi(T) \quad \theta$$

Rao-Blackwell “ ” “ ” “ ”



[Lehmann-Scheffé] T $\varphi(T) = g(\theta)$ $\varphi(T) = g(\theta)$ *UMVUE*

$$T \quad \varphi_1(T) \quad \varphi_2(T)$$

$$E[\varphi_1(T) - \varphi_2(T)] = 0 \quad \forall \theta$$

$$\Rightarrow \quad \varphi_1(T) = \varphi_2(T) \text{ a.s.}$$

9.3.3

[]

- $N(\mu, \sigma^2)$
- $P(\lambda)$
- $B(n, p)$
- $\text{Exp}(\lambda)$
- *Gamma* $\Gamma(\alpha, \beta)$
- $B(1, p)$

$$U(0, \theta) \quad \theta$$



$$f(x; \theta) = h(x) \cdot \exp\{Q(\theta) \cdot t(x) - A(\theta)\}$$

$$f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!} = \underbrace{\frac{1}{x!}}_{h(x)} \cdot \exp\left\{ \underbrace{\ln \lambda}_{Q(\lambda)} \cdot \underbrace{x}_{t(x)} - \underbrace{\lambda}_{A(\lambda)} \right\}$$

$$N(\mu, \sigma^2) \quad \sigma^2$$

$$f(x; \mu) = \underbrace{\frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma^2}}}_{h(x)} \cdot \exp\left\{ \underbrace{\frac{\mu}{\sigma^2}}_{Q(\mu)} \cdot \underbrace{x}_{t(x)} - \underbrace{\frac{\mu^2}{2\sigma^2}}_{A(\mu)} \right\}$$



$$e^{Q(\theta) \cdot t(x)} \quad T = \sum t(X_i)$$

$$f_T(s; \theta) = \tilde{h}(s) \cdot e^{Q(\theta) \cdot s}$$

$$E_\theta[\varphi(T)] = 0$$

$$\int \varphi(s) \cdot \tilde{h}(s) \cdot e^{Q(\theta) \cdot s} ds = 0 \quad \forall \theta$$

$$\eta = Q(\theta) \quad \varphi(s) \tilde{h}(s)$$

$$\Rightarrow \quad \varphi(T) = 0 \text{ a.s.}$$

$$e^{Q(\theta) \cdot s}$$

tip 9.11

+ *umvue*

9.4

9.4.1



$$g(\theta) \quad \text{---} \quad \theta \quad \theta$$

- $\theta \quad g(\theta) = \theta \quad \hat{g}_n \rightarrow \theta$
- $\theta^2 \quad g(\theta) = \theta^2 \quad \hat{g}_n \rightarrow \theta^2$
- $e^\theta \quad g(\theta) = e^\theta \quad \hat{g}_n \rightarrow e^\theta$

tip 9.12

1. $g(\theta) \quad \theta$
2. $\hat{g}_n(X_1, X_2, \dots, X_n) \quad n \quad g(\theta)$



$$\hat{g}_n \quad g(\theta)$$

tip 9.13

ε “ ”



[MSE] MSE Mean Squared Error

$$E[(\hat{g}_n - g(\theta))^2] = \underbrace{[E\hat{g}_n - g(\theta)]^2}_2 + \underbrace{\text{Var}(\hat{g}_n)}_2$$

$$= E[\hat{g}_n] - g(\theta)$$

-
-

tip 9.14

$$E[(\hat{g}_n - E\hat{g}_n)^2] \quad E\hat{g}_n$$

$$(\hat{g}_n - \theta^*) = \underbrace{(\hat{g}_n - E\hat{g}_n)}_2 + \underbrace{(E\hat{g}_n - \theta^*)}_2$$

$$E[\hat{g}_n - E\hat{g}_n] = 0 \quad 2$$



MSE $\rightarrow 0$

MSE $\rightarrow 0$

$$\text{MSE}(\hat{g}_n) = E[(\hat{g}_n - g(\theta))^2] \rightarrow 0 \implies \hat{g}_n \xrightarrow{P} g(\theta)$$

$$P(|\hat{g}_n - g(\theta)| > \varepsilon) \leq \frac{\text{MSE}(\hat{g}_n)}{\varepsilon^2} \rightarrow 0$$

MSE

$$\text{MSE} = \quad^2 + \quad = [E\hat{g}_n - g(\theta)]^2 + \text{Var}(\hat{g}_n)$$

MSE $\rightarrow 0 \quad 0$

- $\quad^2 \rightarrow 0$
- $\rightarrow 0$

$$= 0 \rightarrow 0$$



$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} E[X]$$

Slutsky

$$T_n \xrightarrow{P} c \implies g(T_n) \xrightarrow{P} g(c) \quad (g \text{ continuous})$$



[Slutsky +] —

Slutsky

MSE

tip 9.15

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} E[X]$$

$$T_n \xrightarrow{P} c$$

$$T_n \xrightarrow{P} c \implies g(T_n) \xrightarrow{P} g(c) \quad (g \text{ continuous})$$

$$k \quad \frac{1}{n} \sum_{i=1}^n X_i^k \xrightarrow{P} E[X^k]$$



Slutsky

$$X_n \xrightarrow{d} X, Y_n \xrightarrow{P} c$$

$$1. X_n + Y_n \xrightarrow{d} X + c$$

$$2. X_n \cdot Y_n \xrightarrow{d} c \cdot X$$

$$3. X_n / Y_n \xrightarrow{d} X / c \quad (c \neq 0)$$

“ ”

$$Y_n \xrightarrow{a.s.} a \implies \frac{1}{Y_n} \xrightarrow{P} \frac{1}{a} \quad \text{— “ ” Slutsky}$$

tip 9.16

Slutsky

d- distribution



- $X_n \xrightarrow{d} X$ X “ ”
- $Y_n \xrightarrow{P} c$ $Y_n \xrightarrow{P} c \implies |Y_n - c| \xrightarrow{P} 0$ $X_n \cdot Y_n \approx X_n \cdot c$
- c $X_n - Y_n$

9.5 MLE



[MLE]

$$X_1, \dots, X_n$$

$$L(\theta) = \prod_{i=1}^n f(X_i, \theta)$$

$$L(\theta) \quad \theta$$

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(X_i, \theta)$$

$$\ell'(\theta) = 0 \implies \hat{\theta}_{\text{MLE}}$$

" " θ MLE

tip 9.17

Maximum Likelihood Estimation

- **Max**
- **Likelihood** $L(\theta) = \prod_{i=1}^n f(X_i, \theta) \quad \theta$
- **Estimation** $\hat{\theta}$

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(X_i, \theta)$$

$$\ell'(\theta) = 0 \quad \hat{\theta}$$

$A \quad \theta \quad A$

$\theta \quad " \theta \quad " \text{MLE} \quad L(\theta) = P(\quad | \theta) \quad \theta \quad P(\theta | \quad)$



MLE $\hat{\theta}$

- 1 MLE
- 2
- $\hat{\theta}$

$$\hat{\theta} \approx N\left(\theta, \frac{1}{nI(\theta)}\right) \quad (n \rightarrow \infty)$$

- $\frac{1}{nI(\theta)}$
-
- C-R MLE
- MLE $\hat{\theta}$
- MLE

tip 9.18

MLE 0

9.5.1

9.1 ()

$$\hat{\theta}_n \mid \theta \sim \sigma^2(\theta)$$

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2(\theta))$$

$$\hat{\theta}_n \mid \theta \sim \sigma^2(\theta)$$

$$\hat{\theta}_n \xrightarrow{d} N\left(\theta, \frac{\sigma^2(\theta)}{n}\right)$$

$$MLE \sigma^2(\theta) = \frac{1}{I(\theta)}$$

$$\hat{\theta} \xrightarrow{d} N\left(\theta, \frac{1}{nI(\theta)}\right)$$



tip 9.19

MLE

” ”

$$\bullet \quad \frac{\theta}{MLE} = \frac{1}{nI(\theta)} = C-R$$

MLE $\ell'(\hat{\theta}) = 0$ θ Taylor

$$0 = \ell'(\hat{\theta}) \approx \ell'(\theta) + (\hat{\theta} - \theta)\ell''(\theta)$$

$$\sqrt{n}(\hat{\theta} - \theta) \approx \frac{\frac{1}{\sqrt{n}}\ell'(\theta)}{-\frac{1}{n}\ell''(\theta)}$$

- $CLT \Rightarrow N(0, I(\theta)) \leftarrow Fisher$
- $LLN \Rightarrow I(\theta) \leftarrow Fisher$
- $Slutsky \Rightarrow N\left(0, \frac{1}{I(\theta)}\right)$

tip 9.20

n , n 0 $1/n$ $fisher$, $Slutsky$

tip 9.21

Fisher

$$I(\theta) = E\left[\left(\frac{\partial \ln f}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \ln f}{\partial \theta^2}\right]$$

$$0 =$$



- CLT n iid μ σ^2

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{d} N(0, \sigma^2)$$

$$\sqrt{n}$$

- **LLN** n iid μ

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

9.2 (Delta)

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2) \quad g'(\theta) \neq 0$$

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2 \sigma^2)$$



[Delta] ” ” $\sqrt{n}(\bar{X} - \lambda) \xrightarrow{d} N(0, \lambda)$ $\sqrt{n}(\bar{X}^2 - \lambda^2)$ $g(x) = x^2 - \lambda$ *Taylor*

$$g(\bar{X}) = g(\lambda) + g'(\lambda)(\bar{X} - \lambda) + \underbrace{O((\bar{X} - \lambda)^2)}$$

$$g(\bar{X}) - g(\lambda) \approx g'(\lambda) \cdot (\bar{X} - \lambda) \quad \sqrt{n}$$

$$\sqrt{n}(g(\bar{X}) - g(\lambda)) \approx g'(\lambda) \cdot \sqrt{n}(\bar{X} - \lambda)$$

tip 9.22

$$g'(\lambda)$$

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \approx g'(\theta) \cdot \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, [g'(\theta)]^2 \sigma^2)$$

[] *Central Limit Theorem*

$$\text{Var}\left(\sum X_i - n\mu\right) = \text{Var}\left(\sum X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2$$

- $\sum X_i - n\mu = n\sigma^2 \rightarrow \infty$
- $n \cdot \bar{X} - \mu = \frac{\sigma^2}{n} \rightarrow 0$ *LLN*
- $\sqrt{n} \frac{1}{\sqrt{n}} \sum (X_i - \mu) = \sigma^2$

tip 9.23

$$n \quad n\sigma^2 \quad \bar{X} \quad n \quad \sqrt{n} \quad \frac{1}{\sqrt{n}}$$

$$\text{Var}(\sqrt{n}(\bar{X} - \mu)) = n \cdot \text{Var}(\bar{X}) = n \cdot \frac{\sigma^2}{n} = \sigma^2$$

- $z_{\alpha/2}$
- σ
- $\sqrt{n}$

tip 10.3

$$z_{\alpha/2} \quad \sqrt{n}$$



[] 100

$$\begin{array}{llll} 1 & \rightarrow & [168.1, 172.5] & \leftarrow \mu \\ 2 & \rightarrow & [169.0, 173.4] & \leftarrow \mu \\ 3 & \rightarrow & [171.8, 176.2] & \leftarrow \\ \vdots & & \vdots & \\ 100 & \rightarrow & \dots & \end{array}$$

$$\begin{array}{ccccccc} 95\% & & 100 & & 95 & & \mu \\ & & & & 95 & \text{---} & 95\% \end{array}$$



[] 99%

tip 10.4

[] θ
 θ

—

$$P(\theta_1(\mathbf{X}) \leq \theta \leq \theta_2(\mathbf{X})) \geq 1 - \alpha$$

— θ

$$P(\theta \leq \theta_2(\mathbf{X})) \geq 1 - \alpha$$

— θ

$$P(\theta \geq \theta_1(\mathbf{X})) \geq 1 - \alpha$$

$$[\text{---}\theta\text{---}] \quad \theta$$

$$(\text{---}\theta] \quad \theta$$

$$[\theta\text{---}) \quad \theta$$

$$\alpha/2 \quad z_{\alpha/2} \quad z_{\alpha} \quad \text{---}$$

tip 10.5

$$\underbrace{[\theta_1(\mathbf{X}), \theta_2(\mathbf{X})]}_{\theta_1 \quad \theta_2} \xrightarrow{(\sigma^2)} \underbrace{\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]}_{\theta_1 \quad \theta_2}$$

- $\mathbf{X} = (X_1, X_2, \dots, X_n)$
- $\theta_1(\mathbf{X})$
- $\theta_2(\mathbf{X})$

10.2 Pivotal quantity method

- $P(f(X) < \theta < p(X)) = 1 - \alpha$ $f(X)$ $p(X)$ θ
- θ ——— pivot ——— θ $\frac{\bar{X} - \theta}{\sigma/\sqrt{n}}$ pivot
- θ θ 0
- $z_{\alpha/2}$ $-z_{\alpha/2}$ θ θ

tip 10.6

- μ σ^2 $N(0, 1)$ σ^2 1 t
-

1. μ $\sigma^2 \rightarrow Z$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

2. μ $\sigma^2 \rightarrow t$

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

3. σ^2 $\mu \rightarrow \chi^2$

$$\chi^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi^2(n)$$

4. σ^2 $\mu \rightarrow \chi^2$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

5. $\mu_1 - \mu_2$ $\sigma_1^2 = \sigma_2^2 \rightarrow t$

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{1/m + 1/n}} \sim t(m+n-2)$$

6. σ_1^2/σ_2^2 $\mu_1, \mu_2 \rightarrow F$

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(m-1, n-1)$$

tip 10.7

$$\mu_1 - \mu_2 \quad 2 \quad 5 \quad 3$$

F

$$U \sim \chi^2(m) \quad V \sim \chi^2(n)$$

$$F = \frac{U/m}{V/n} \sim F(m, n)$$

tip 10.8

$$t \quad t \quad \bar{X} - \bar{Y}$$

$$S_p^2 = \frac{(m-1)S_1^2 + (n-1)S_2^2}{m+n-2}$$

t

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{1/m + 1/n}} \sim t(m+n-2)$$

n
CLT n

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

σ S

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \approx N(0, 1)$$

1. σ^2 Z T
 θ $X_1 \sim \frac{1}{\theta} e^{-x/\theta}$

pivot $G = \frac{2n\bar{X}}{\theta} \sim \chi^2(2n)$

$$\frac{2n\bar{X}}{\chi_{\alpha/2}^2(2n)} \leq \theta \leq \frac{2n\bar{X}}{\chi_{1-\alpha/2}^2(2n)}$$

2. θ $X_1 \sim U(\theta, \theta + 1)$

pivot $G = X_{(1)} - \theta \sim \text{Beta}(1, n)$

$$a = 1 - \left(1 - \frac{\alpha}{2}\right)^{1/n}, \quad b = 1 - \left(\frac{\alpha}{2}\right)^{1/n}$$

$$X_{(1)} - b \leq \theta \leq X_{(1)} - a$$

11 4

1.	4.1	3	
2.	4.2	5	
3.	4.3	2	
4.	NP	→ UMP 4.4	6 ← *
5.	12	4 ← 3	*

11.1

×	
H_0	$H_0 \rightarrow = \alpha$
H_1	$H_0 \rightarrow = \beta$
$\alpha \downarrow \Rightarrow \beta \uparrow$	
$\beta(\theta) = P_\theta(H_0)$	
$\theta \in \Theta_0$	$\beta(\theta) = \alpha$
$\theta \in \Theta_1$	$1 - \beta(\theta) = \beta$
p	H_0
$p < \alpha$	$\rightarrow H_0 p$
p	α

tip 11.1 ()

$$-\mu_0 \quad N(0,1) \quad \bar{X} \quad \mu_0 \quad \bar{X} \quad \mu_0 \quad \sigma/\sqrt{n} \quad x$$

tip 11.2

$$\sigma \quad H_0 \quad \sigma \quad \sigma \quad S \quad N(0,1) \quad t$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \quad T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

$$\bar{X} \quad Z$$



$$\bar{X} \quad N(0,1) \quad x \quad y \quad H_1 \quad \alpha \quad \bar{X} \quad x \quad H_1 \quad \alpha$$

$$\varphi(X) = \begin{cases} 1 & T \in W \\ 0 & T \notin W \end{cases}$$

$$W \quad \varphi = 1 \quad H_0 \quad \varphi = 0 \quad " \quad "$$

tip 11.3 (p -)

$$p \quad H_0 \quad \bar{X} \quad 8 \quad [1, +\infty) \quad 8 \quad 8 \quad 1.1 \quad 8 \quad H_0$$

$$p \quad H_0$$



$$p \quad \alpha \quad c \quad p \quad \alpha \quad \alpha \quad c \quad p$$

tip 11.4

(1)	H_0	H_1	p	H_0
1.	H_0			
2.	H_1	c	H_0	H_1



$$\beta(\theta) = P_{\theta}(\quad)$$

$$\theta \quad \theta$$

$$= \int_c^{+\infty} f_{H_0}(x) \, dx = \alpha$$

$$= \int_{-\infty}^c f_{H_1}(x) \, dx = \beta$$

tip 11.5

$$H_0 \quad H_1 \quad f$$

11.2

11.1 ()

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2) \quad H_0 : \mu = \mu_0$$

$$\rightarrow Z$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

$$\rightarrow t$$

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$$

$$S \quad \sigma \quad n \quad t$$

$$H_0 : \sigma^2 = \sigma_0^2$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$$

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1-1, n_2-1)$$



CI pivot

$$\mu \quad 0 \quad 1 \quad x$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \quad T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \quad \chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

tip 11.6

•

•

$$\theta \quad \theta_0$$

•

—

$$\alpha$$

$$\begin{aligned}
 & \bullet \quad \frac{\alpha}{2} \quad z_{\alpha/2} \\
 & \bullet \quad z_{\alpha} \\
 & \bullet \quad -z_{\alpha} \\
 & H_1 \\
 & \theta_0 \quad CLT \\
 & \frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\sqrt{V(\theta_0)}} \approx N(0, 1)
 \end{aligned}$$

$$N(0, 1) \quad x \quad z_{\alpha} \quad z_{\alpha/2} \quad H_0$$

$$\begin{aligned}
 & \bullet \quad \hat{\theta} \quad \hat{p} = \bar{X} \quad \bar{X} \\
 & \bullet \quad \theta_0 \quad H_0 \quad p_0 = 0.1 \\
 & \bullet \quad V(\theta_0) \quad H_0
 \end{aligned}$$

CLT

$$CLT \quad \sqrt{n}$$

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

$$N(0, \sigma^2) \quad 1 \quad \sigma$$

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

$$\begin{aligned}
 & \bullet \quad \sqrt{V(\theta_0)} \quad \sigma \quad x \\
 & \bullet \quad \hat{\theta} = \bar{X} \\
 & \bullet \quad V(\theta_0) = \sigma^2 \quad 1/n \\
 & \sqrt{n} \quad \sqrt{V(\theta_0)} \quad \sigma \quad \sigma/\sqrt{n} \\
 & \bullet \quad \theta_0 \quad H_0 \quad \theta_0 \quad H_0 \\
 & \bullet \quad \hat{\theta} \quad \theta_0 \quad \hat{\theta} \xrightarrow{P} \theta_0
 \end{aligned}$$

11.3 NP-UMP

NP

$$[\quad] \quad H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta = \theta_1 \quad \text{power} \quad \text{---}$$

$$P_{\theta_0}(X \in W) = \alpha, \quad \min P_{\theta_1}(X \notin W) \iff \max P_{\theta_1}(X \in W)$$

“ ” α “ ”

$$H_1 \quad c \quad H_1$$

tip 11.7

$$H_0 \quad W \quad = \alpha$$

$$W_1 = \{X \geq c_1\} \quad P_{\theta_0}(X \in W_1) = \alpha$$

$$W_2 = \{X \leq c_2\} \quad P_{\theta_0}(X \in W_2) = \alpha$$

$$W_3 = \{ \quad \} \quad P_{\theta_0}(X \in W_3) = \alpha$$

$$W \quad H_0 \quad x \quad H_1$$

$$P_{\theta_1}(X \in W_1) \neq P_{\theta_1}(X \in W_2) \neq P_{\theta_1}(X \in W_3)$$

$$H_1 \quad x \quad H_1 \quad W$$

$$H_0 \quad H_1 \quad NP \quad \Lambda(x) \geq k \quad W \quad H_1$$



$$\begin{aligned}
 & x \quad W \\
 & x \quad W \\
 & \bullet \quad \alpha \rightarrow f(x; \theta_0) \quad f(x; \theta_0) \quad \theta_0 \quad x \\
 & \bullet \quad \rightarrow f(x; \theta_1) \quad f(x; \theta_1) \quad \theta_1 \quad x \\
 & x \\
 & \Lambda(x) = \frac{f(x; \theta_1)}{f(x; \theta_0)} = - = \frac{H_1}{H_0} \\
 & x \quad W \quad \alpha \\
 & W = \{\Lambda(x) \geq k\}
 \end{aligned}$$

NP —

tip 11.8

$$\begin{aligned}
 & \Lambda(x) \geq k \\
 & \bullet f(x; \theta_0) \quad x \quad H_0 \quad \rightarrow \quad W \quad \alpha \\
 & \bullet f(x; \theta_1) \quad x \quad H_1 \quad \rightarrow \quad H_1 \quad W \quad P_{\theta_1}(X \in W) \\
 & = 1 - P_{\theta_1}(X \in W)
 \end{aligned}$$



[UMP] **UMP = Uniformly Most Powerful Test**

$$\begin{aligned}
 & NP \quad \text{vs} \quad H_1 \quad H_1 \\
 & H_1 : \theta > \theta_0 \quad (H_1 \quad)
 \end{aligned}$$

$$\begin{aligned}
 & \theta_1 \quad NP \quad \theta_1 \\
 & \mathbf{UMP} \quad W \quad H_1 \quad \theta_1 \\
 & \beta_W(\theta_1) \geq \beta_{W'}(\theta_1) \quad \theta_1 \in H_1
 \end{aligned}$$

UMP

$$\begin{aligned}
 & \bullet NP \rightarrow \text{vs} \\
 & \bullet UMP \rightarrow \text{vs} \quad \theta_1
 \end{aligned}$$

UMP UMP



[UMP] UMP

$$\begin{aligned}
 & NP \quad \theta_1 \quad \{\Lambda(x) \geq k\} \quad \Lambda(x) = \frac{f(x; \theta_1)}{f(x; \theta_0)} \quad T(X) \\
 & \Lambda(x) \geq k \iff T(X) \geq c
 \end{aligned}$$

$$\{T(X) \geq c\} \quad \theta_1 \quad \theta_1 > \theta_0$$

tip 11.9

$$\begin{aligned}
 & k \quad x \quad W = \{\Lambda(x) \geq k\} \quad H_0 \quad \alpha \\
 & P_{\theta_0}(X \in W) = \int_W f(x; \theta_0) dx = \alpha \\
 & k \quad \alpha \quad x \quad y \quad dx
 \end{aligned}$$



[]

$$\ln \Lambda(x) = \frac{\mu_1 - \mu_0}{\sigma^2} \cdot \bar{X} -$$

$$\begin{aligned}
 & \bar{X} \quad \frac{\mu_1 - \mu_0}{\sigma^2} \quad \text{—} \\
 & \theta_1 \quad H_1 : \theta > \theta_0 \quad \theta_1 - \theta_0 > 0 \quad k\bar{X} - b \quad k > 0 \quad \bar{X} \quad \Lambda(x) \quad \theta_1 \\
 & \{\bar{X} \geq c\} \quad \theta_1 > \theta_0 \quad \Rightarrow \mathbf{UMP}
 \end{aligned}$$



$I \quad J$

$$\Lambda(x) = \exp \left\{ \frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i - \right\} \geq k$$

$\ln \quad c'$

$$\frac{\mu_1 - \mu_0}{\sigma^2} \sum x_i \geq c'$$

$\mu_1 > \mu_0 \quad > 0 \quad c$

$$\Lambda(x) \geq k \iff \bar{X} \geq c$$

$\theta_1 \quad c \quad c \quad P_{\theta_0}(\bar{X} \geq c) = \alpha \quad \theta_1$

tip 11.10 (NP UMP)

$NP \quad x \quad k$

$$W = \{\Lambda(x) \geq k\}$$

$k \quad x \quad H_0 \quad \alpha$

$\Lambda(x) \quad \theta_1$

$T(X)$

$$\Lambda(x) \geq k \iff T(X) \geq c, \quad c \quad P_{\theta_0}(T(X) \geq c) = \alpha$$

$T(X)$



$I \quad J \quad \{\Lambda(x) \geq k\} \quad \Lambda(x) = \frac{f(x; \theta_1)}{f(x; \theta_0)} \quad \theta_1 \quad x$

$T(X)$

$$\Lambda(x) \geq k \iff T(X) \geq c$$

$c \quad P_{\theta_0}(T(X) \geq c) = \alpha \quad \theta_1 \quad k$

- $NP \quad \{\Lambda(x) \geq k\}$
- $T(X)$
- $T(X) \geq c$

11.4



$F_0 \quad F_n$

$H_0 : \mu = \mu_0$

$H_0 : F = F_0, \quad H_1 : F \neq F_0$

$F_0 \quad F_n$

$(\bar{X} \text{ vs } \mu_0) \quad (F_n \text{ vs } F_0)$

$F_n \quad F_0$

- k
- **K-S** $\sup_x |F_n(x) - F_0(x)|$
- **Q-Q**

H_0

tip 11.11 (P-P)

- $x_{(i)}$
- $x \quad F_n(x_{(i)}) = i/n \quad 10 \quad 55 \quad 10 \quad 10\%$
 - $y \quad F(x_{(i)}) \quad 55 \quad 0.045$
 - $\left(\frac{i}{n}, F(x_{(i)})\right) \quad y = x$



$$[F_n(x_{(i)}) = i/n]$$

- $n \quad 10 \quad n = 10$
- $i \quad i \quad 55 \quad 1 \quad i = 1 \quad 70 \quad 5 \quad i = 5$

$$F_n(x_{(i)}) = i/n \quad i \quad n$$



$$[\quad]$$

$$CDF$$

$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$p \quad x \quad F^{-1}(p)$$

$$F^{-1}\left(\frac{i}{n}\right) = \hat{\mu} + \hat{\sigma} \cdot \Phi^{-1}\left(\frac{i}{n}\right)$$

- $\Phi^{-1}(p) \quad p \quad \Phi^{-1}(0.1) = -1.28 \quad 10\% \quad -1.28$
- $\hat{\sigma} \cdot \Phi^{-1}(p) \quad 0 \quad 1 \quad \hat{\sigma}$
- $\hat{\mu} + \hat{\sigma} \cdot \Phi^{-1}(p) \quad \hat{\mu}$



$$[Q-Q \text{ Quantile-Quantile Plot} - \quad]$$

- $x \quad F^{-1}(i/n) \quad i/n$
- $y \quad x_{(i)} \quad i$
- $y = x$

11.1 (Kolmogorov-Smirnov)

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} F(x) \quad F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x) \quad H_0 : F(x) = F_0(x) \quad F_0$$

$$D_n = \sup_{x \in \mathbb{R}} |F_n(x) - F_0(x)|$$

$$H_0 \quad \sqrt{n} D_n \xrightarrow{d} K(\lambda) \quad K(\lambda) = \sum_{k=-\infty}^{+\infty} (-1)^k e^{-2k^2 \lambda^2} \quad \alpha \quad \sqrt{n} D_n > \lambda_\alpha$$



12

12.1



$$(x_i, Y_i)$$

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2) \quad x \quad Y \quad \varepsilon$$

$$H_0: \beta_1 = 0 \quad x \quad Y$$

- $H_0: x \quad Y$

- $H_0: \hat{\beta}_1$

$$\rightarrow OLS \rightarrow \beta_1 \neq 0 \rightarrow$$

tip 12.1

$$y = kx + b \quad k \quad x \quad y \quad b \quad 0$$



$$\hat{\beta}_1 = S_{xy}/S_{xx} = \widehat{\text{Cov}}(x, Y)/\widehat{\text{Var}}(x) = \frac{\sigma_Y}{\sigma_x} \quad (\bar{x}, \bar{Y}) \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

tip 12.2

$$(Y - kx - b)^2 \quad 0$$



$$x \quad \hat{\beta}_1 \quad H_0: 0 \quad \hat{\beta}_1 \neq 0 \quad \text{---}$$

$$\hat{\beta}_1$$

$$\hat{\beta}_1$$

$$T = \frac{\hat{\beta}_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t(n-2) \quad (H_0)$$

$$T \quad H_0$$

- $S_{xx} = \sum (x_i - \bar{x})^2 \quad x \quad x \quad \text{---} \quad x$

- $\hat{\sigma} = \sqrt{SS_e/(n-2)} \quad \text{---}$

- $T = \frac{\hat{\beta}_1}{\hat{\sigma}/\sqrt{S_{xx}}} \quad x \quad T = \quad \Rightarrow \quad H_0$

$$T = \frac{\hat{\beta}_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t(n-2) \quad (H_0)$$

tip 12.3

$$T = \hat{\beta}_1/(\hat{\sigma}/\sqrt{S_{xx}}) \quad \sigma \quad \sigma$$

$$\hat{\sigma}^2 = \frac{SS_e}{n-2}, \quad SS_e = \sum (Y_i - \hat{Y}_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

$$S_{yy} \quad Y \quad S_{xy}^2/S_{xx}$$

$$SS_e \rightarrow \hat{\sigma} \rightarrow T \rightarrow t(n-2) \rightarrow H_0$$

- $\hat{Y} = 43.5 + 7.5x \quad x^* = 3$

1. $x = 3$

2. $x = 3$

- $E(Y|x^*) = \beta_0 + \beta_1 x^* \quad \hat{\beta}_0, \hat{\beta}_1$
 - $Y^* = \beta_0 + \beta_1 x^* + \varepsilon^* \quad + \quad \varepsilon^*$
- $$\hat{Y}^* \pm t_{\alpha/2}(n-2) \cdot \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$
- $$\hat{Y}^* \pm t_{\alpha/2}(n-2) \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$
- ”1” ε^*



$$\hat{Y}^* \pm t_{\alpha/2}(n-2) \cdot SE$$

$$t(n-2) \quad SE$$

12.2



$$t \quad k \quad \binom{k}{2} \quad \alpha = 0.05$$

$$1 - (1 - 0.05)^{\binom{k}{2}}$$

$$k = 4 \quad 26\%$$

ANOVA Analysis of Variance

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

$$F \quad \alpha$$

” ” ” ”

$$SS_T = SS_A + SS_E$$

12.1 ()

$$y_{ij} = \mu + \mu_i + e_{ij}, \quad e_{ij} \sim N(0, \sigma^2) \text{ iid}$$

$$k \quad i \quad n_i \quad N = \sum n_i \quad \sum \mu_i = 0$$



- μ ” ”
 - μ_i i ” ”
 - e_{ij}
 - $\sum \mu_i = 0$ — μ μ_i
 - $H_0 : \mu_1 = \dots = \mu_k = 0$
 - $SS_T = SS_A + SS_E$
 - $SS_T = \sum_i \sum_j (y_{ij} - \bar{y})^2$
 - $SS_A = \sum_i n_i (\bar{y}_i - \bar{y})^2$ SS_A
 - $SS_E = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2$
- F



$$F = \frac{MS_A}{MS_E} = \frac{SS_A/(k-1)}{SS_E/(N-k)}$$

$$\bullet H_0 \quad MS_A \quad MS_E \quad \sigma^2 \quad 1$$

$$\bullet H_1 \quad MS_A \quad F \gg 1$$

$$F > F_\alpha(k-1, N-k) \quad H_0$$

tip 12.4

$$SS_T \quad N - 1$$

$$\underbrace{SS_T}_{N-1} = \underbrace{SS_A}_{k-1} + \underbrace{SS_E}_{N-k}$$

$$H_0 \quad SS_A \quad SS_E$$

$$\frac{SS_A}{\sigma^2} \sim \chi^2(k-1), \quad \frac{SS_E}{\sigma^2} \sim \chi^2(N-k)$$

$$F \quad F \quad F \quad \text{---}$$

$$F = \frac{MS_A}{MS_E} = \frac{SS_A/(k-1)}{SS_E/(N-k)} \sim F(k-1, N-k)$$